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Averting the Next Credibility Crisis in Psychological Science: Within-Person Methods for Personalized Diagnostics and Intervention

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Abstract: Personalizing assessments, predictions, and treatments of individuals is currently a defining trend in psychological research and applied fields, including personalized learning, personalized medicine, and personalized advertisement. For instance, the recent pandemic has reminded parents and educators of how challenging yet crucial it is to get the right learning task to the right student at the right time. Increasingly, psychologists and social scientists are realizing that the between-person methods that we have long relied upon to describe, predict, and treat individuals may fail to live up to these tasks (e.g., Molenaar, 2004). Consequently, there is a risk of a credibility loss, possibly similar to the one seen during the replicability crisis (Ioannides, 2005), because we have only started to understand how many of the conclusions that we tend to draw based on between-person methods are based on a misunderstanding of what these methods can tell us and what they cannot. An imminent methodological revolution will likely lead to a change of even well-established psychological theories (Barbot et al., 2020). Fortunately, methodological solutions for personalized descriptions and predictions, such as many within-person analyses, are available and undergo rapid development, although they are not yet embraced in all areas of psychology, and some come with their own limitations. This article first discusses the extent of the theory-method gap, consisting of theories about within-person patterns being studied with between-person methods in psychology, and the potential loss of trust that might follow from this theory-method gap. Second, this article addresses advantages and limitations of available within-person methods. Third, this article discusses how within-person methods may help improving the individual descriptions and predictions that are needed in many applied fields that aim for tailored individual solutions, including personalized learning and personalized medicine.

Keywords: within-person methods, theory-method gap, personalized learning, personalized medicine

Psychology aspires to understand what determines the behavior and mental states of, and differences between, *individuals*, in contrast to sociology or other social sciences, which aspire to describe and predict the behavior and functioning of *groups*. The term *individual* signals that we widely accept that two persons are not two copies of the same, but that they differ from one another in many regards. Understanding individuals, predicting their individual behavior, and finding the matching treatment for the right person at the right time is currently a defining trend in many disciplines, including the fields of personalized learning (e.g., Dumont, 2019; Bulger, 2016), personalized medicine (e.g., Senn, 2016; 2018), personalized advertisement (Zhu & Chang, 2016; Bang & Wojdynski, 2016), and many more. These goals of understanding, predicting, and treating individuals all require statistical methods that make it possible to describe how experiences develop, and how they interact within individuals (i.e., within-person methods). However, much previous research in psychology, education, medicine, and many social sciences, relies largely on methods that *do not* examine such within-person patterns (e.g., Molenaar, 2004). Consequently, it is often unknown what happens inside of an individuals' mind; the goals of a tailored understanding, prediction and treatment

for specific individuals may be difficult to reach, and many psychological studies attempting to study individuals may fail to actually do so.

A methodological revolution appears imminent, indicated for instance by several "manifestos" in which eminent experts have demanded "bringing the person back into psychological science" by using within-person methods (Molenaar, 2004, p. 201; Barbot et al., 2020; Renner et al., 2020). Various within-person methods have been proposed as solutions for better descriptions of individuals (e.g., Beltz et al., 2016; Völkle et al., 2014; von Eye, 2003), but neither do they solve all the known methodological problems of describing individuals, nor have they been sufficiently applied in studies on psychological theories, in personalized learning, or other disciplines aiming for personalization, such as personalized medicine.

This article gives an overview of some crucial limitations in the commonly used between-person methods in order to explain why and in what regards there may be an imminent risk for the emergence of a new credibility crisis in psychology (called a validity crisis by Lundh, 2019), due to systematic discrepancies between the conclusions about groups that our methods allow us to draw, and the conclusions about individuals that we tend to draw based on these methods. The article then proceeds to give an overview of some available within-person methods that solve some of the limitations of the between-person methods. After that, the article discusses why the limitations of between- person methods, the potential of within-person methods, and the demand for personalization in many applied fields together are likely to lead to paradigmatic changes in psychological research, including methods, theories, and applications. The article closes with the conclusion that a pro-active reckoning about the theory-method gap in psychology is needed to avert a new credibility crisis (or validity crisis; see Lundh, 2019), and proposes a research program for this proactive self-improvement of psychological science.

1. A Theory-Method Gap in Psychology: Limitations of Common Between-Person Methods

Although psychology aims to describe and predict how individuals feel, think, and behave, many of the analytical and diagnostical methods used to study such questions about individuals focus mostly on group-based statistics, such as mean-score differences between groups, or group-based correlation or regression coefficients. This section summarizes reasons why and in what cases such group-based between-person statistics may fail to describe some, or even all, of the individuals in the studied sample. The leading theme in this section is the concern about a systematic theory-method gap in many areas of psychology, in the sense that we often study theories and hypotheses about within-person patterns or processes with between-person methods that are unsuited to tell us anything about the within-person pattern or processes of interest.

1.1. Between-Person Methods Do Not Sufficiently Describe Change Within Individuals Due to Simpson's Paradox

One reason why group-based statistics may fail to describe many individuals in the studied sample is the possible presence of Simpson's (1951) paradox. Understanding processes of change in a person requires multiple assessments of the same person over time, in combination with methods to analyze the within-person trajectories over time (see Reitzle & Dietrich, 2019; Curran et al., 2014). Such methods can help teasing apart the aspects that change and fluctuate (states) from those that distinguish one person from the other and remain stable in a person across multiple measurements (traits).

For instance, teachers want to know what they have to do in a teaching situation to change a learner's interest or knowledge; doctors want to know how to treat a patient in a way that best changes their health for the better; advertisers want to know which malleable attitudes and behaviors they can affect with which advertisement stimuli; and all these protagonists may want to know how stable person characteristics should inform decisions about situation-specific and person-specific assessments and treatments. Different individuals may change in different ways, and to understand such between-person differences in within-person change processes it is helpful to first examine for each person how their experiences change over time, and then examine whether groups of individuals with similar change patterns can be identified (e.g., Beck & Jackson, 2020; Beltz et al., 2016).

If between-person methods are applied without the previous step of identifying the trajectories within individuals first, then misinterpretations can arise. For instance, imagine data consisting of multiple individuals with multiple measurement time points per person that are used to study whether a construct changed over time. A typical between-person approach would either calculate a group-based average of the variable for each time point (averaging the variable across individuals within each time point) and then examine changes in these group averages from one time point to the next (called mean-level stability), or examine between-person correlations of the variable from one time point to the next (called rank-order stability; for studies using both techniques, see e.g., Mõttus et al., 2012; Specht et al., 2011). A problem with both approaches is that the individual trajectories of the persons in that sample can look very different from, and therefore *cannot be deduced from such group-based statistics* (e.g., Reitzle & Dietrich, 2019). For instance, a construct can appear to increase over time according to between-person methods, while in fact it tends to decrease within each person, or vice versa, which is the longitudinal version of the problem known as Simpson's (1951) paradox or as a lack of ergodicity (e.g., Molenaar, 2004; for its role in longitudinal studies, see Kievit et al.,

2011; 2013; Yarnold, 2013). Thus, neither mean-level stability nor rank-order stability estimated with between-person methods tell us whether, how much, and in which directions, a construct changed within a person, or which trajectories were observed in which – and how many – individuals.

1.2. Simpson's Paradox May Also Obfuscate the Structure of One Construct, or the Relationships Among Multiple Constructs

The problem that within-person patterns occasionally differ from group-level trends (also described as Simpson's paradox[1] or lack of ergodicity[2]) does not only mean that within-person trajectories over time in one variable cannot be deduced from between-person analyses of mean-level or rank-order stability. *The same problem also applies to analyses of the structure of a construct, or the relations among more than two psychological constructs*. It implies that interrelations (e.g., correlations, regression coefficients) among multiple construct indicators or constructs often look vastly different if examined with within-person methods than with between-person methods (e.g., Molenaar, 2004). This means, for instance, that the factor structure of one construct, the results of a factor analysis, or the relations among more than two variables in structural equation models can look different in between-person analyses than in within-person analyses. It means that a between-person analysis can propose a structure of interrelations among multiple variables that fails to describe how the very same variables relate to each other within individuals. Some between-person findings may properly describe the experiences of at least a subgroup of individuals, others may fail to describe any individual's experiences at all.

Detecting such within- versus between-person differences in the structure or interrelations of constructs typically requires multiple measurement time points per person (i.e., intensive longitudinal data), so that within-person correlations or regression coefficients can be examined across these multiple time points (e.g., Beck & Jackson, 2021; Cattell, 1946). Therefore, many of the studies addressing either this problem or the solutions to it can be found in research using such intensive longitudinal data, many of which conclude that interrelations among psychological constructs are best described with combinations of intra- and between-person methods (e.g., Brose et al., 2020; Völkle, et al., 2014; Murayama et al., 2017).

The lack of ergodicity or Simpson's paradox is currently the most frequently mentioned argument for within-person methods and a leading argument in the various manifestos calling for within-person methods (e.g., Barbot et al., 2020; Molenaar, 2004; Renner et al., 2020).

1.3. Between-Person Approaches Report One-Size-Fits-All Effects, But Heterogeneity Can Hide Behind Such Overall Trends

Between-person approaches typically report one result for the entire sample (e.g., one correlation, or one regression coefficient), which follows a one-size-fits-all logic. However, it is known that *heterogeneity can hide behind such overall trends*, and that, for instance, individuals with patterns opposing the overall trend may be overlooked (see e.g., Anscombe, 1973; Matejka & Fitzmaurice, 2017). This problem has been long known and is yet often ignored (e.g., Asendorpf, 1993; Asendorpf, 2000; Kuhl, 1977; Lewin, 1930; Wottawa, 1981). It implies that *without examining scatter plots and bivariate distributions more in detail, we do not know how many individuals in the sample behave in the way a sample's inter-individual correlation or regression coefficient suggests*. A sample's correlation or regression coefficient may fail to describe the variable patterns experienced by some or even all individuals in the sample: As Anscombe's (1973) fourth quadrant (see Figure 1 below) shows, it is even possible to find a strong correlation or regression coefficient in a sample where there is no systematic relation between the X and the Y variable at all.

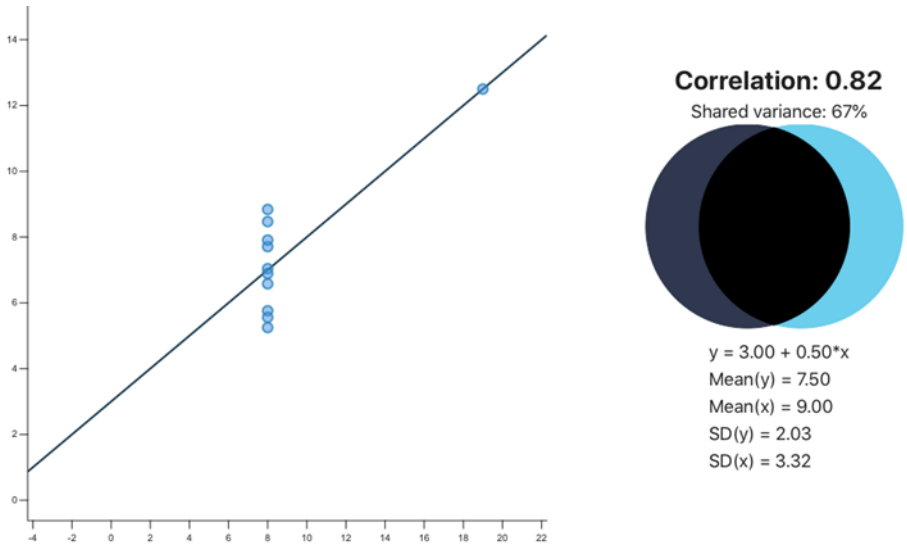
Vice versa, as Matejka and Fitzmaurice (2017) show, it is possible to find a zero correlation in a sample where about half of the individuals show patterns in line with a perfectly negative correlation between X and Y, while the other half of the individuals show patterns in line with a perfectly positive correlation between X and Y (see the dots in the scatter plot in Matejka and Fitzmaurice's [2017] Figure 2, where roughly half of the dots/individuals lie along a diagonal from top left to bottom right, representing a negative correlation, whereas the other half of the observations lie along a diagonal from the bottom left to the top right, representing a positive correlation, while the overall correlation for that sample is around zero). For these reasons, inter-individual correlations or regression coefficients cannot automatically be assumed to describe the patterns of variables within individuals. Asendorpf (2000) calls this fallacy of misinterpreting an inter-individual finding as if it applied to individuals the "idiographization of a nomothetic finding" and

[1] Simpson's paradox: An effect (e.g., a positive correlation) which appears within groups of observations (e.g., groups of multiple measurements per person) but disappears when these separate groups are combined and the effect is recalculated without taking the former grouping into account. For instance, a correlation coefficient being negative within individuals (across repeated measures) but becoming zero or positive once the nestedness of repeated measures in individuals is ignored.

[2] Definition of ergodicity: Ergodicity in a dynamical system exists if the average behavior of the system (e.g., a sample of individuals) can be deduced from the behavior of individual measurement points (e.g., one individual). For instance, ergodicity would be given if the developmental trajectory of one given person over time is a good indicator for the average development of the system over time. Lack of ergodicity implies a heterogeneity between the individual measurement points (in this example: individuals) in the sense that one given individual's trajectory is not necessarily a good estimate for how we can expect the system to develop.

Figure 1.

Anscombe's fourth quadrant, data source: Anscombe (1973). The figure was created using the interactive correlation simulation provided by Magnusson (2021).



proposes to report the percentage of individuals who show patterns in line with the sample-level nomothetic finding. So far however, misinterpretations of correlations or regression coefficients as within-person patterns are and have been persistent among even the highest educated psychologists (Valsiner, 1986).

In particular, any research attempting to understand heterogeneity and the aspects in which individuals may deviate from general trends requires methods that are able to discover such heterogeneity hiding behind between-person overall correlations or regression coefficients. This is particularly relevant for any research attempting to personalize diagnostics and treatment, such as personalized learning, inclusive education, personalized medicine, or personalized advertisement.

1.4. Covariance Is Not Co-Endorsement, But Is Often Mistaken for Such

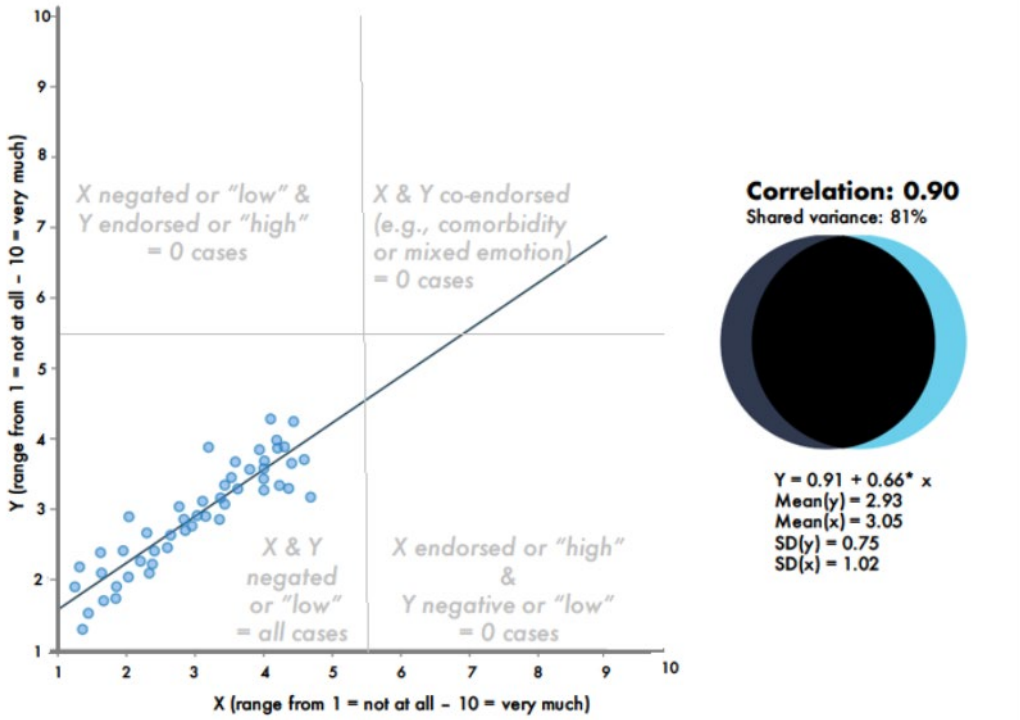
Another example of frequently misinterpreted between-person coefficients is the interpretation of correlation or regression coefficients as if they revealed whether one variable Y is high if the other variable X is high (positive correlation) or low (negative correlation). Many researchers conclude that a negative correlation implies that individuals with "high" levels in one variable show "low" levels in another variable; and that a positive correlation implies that individuals tend to show similar levels in both variables (i.e., high levels in both variables, or moderate levels in both, or low levels in both). However, this frequent interpretation ignores that *between-person correlations or regression coefficients say nothing about whether an individual endorses two items, or denies two items, or has a high score in reference to an absolute response scale* (see e.g., Asendorpf, 2000; Moeller, 2018a). Correlations and regression coefficients are independent of whether any item is *endorsed* alone or together with another item by anyone in the sample. Covariance is not the same as co-endorsement (see Figure 2)

Thus, any research trying to find out whether two items are endorsed together, such as research on comorbidity (asking which pathological symptoms are experienced together) or mixed emotions (asking which emotions are experienced together), needs methods capable of describing *within-person co-endorsements* of sets of items (see e.g., Moeller et al., 2018a). For a summary of the discussions about reasons why correlation coefficients are not well suited to study mixed emotions, please see Larsen and McGraw (2014).

Statements about co-variance do not carry information about the *absolute scores* in variables, only information about the *relative rank* of individuals in reference to the distribution of all individuals. For instance, two variables can be positively correlated, seemingly suggesting that if X is high, Y is also high, and yet the inter-individual average of one of these variables can be much higher (e.g., variable X can have a mean of seven on a scale from one to ten) than the other (e.g. variable Y can have a mean of three on a scale from one to ten), and this relation of absolute scores of X > Y can be true for all individuals in that sample. In this example, it could be argued that variable Y is never "high" in terms of an item endorsement, despite of the positive correlation between both variables. For example, the desirable (harmonious) form of passion and the undesirable (obsessive) form of passion described in Appendix B are positively correlated with each other, and nevertheless obsessive passion is typically denied by most individuals and in most cases lower than the same individual's harmonious passion (Moeller et al., 2015). Thus, it is important to keep in mind that covariance refers to individuals' *relative* ranks in the distribution of individuals, and that this can be a vastly different from what many people understand when they hear that a variable

Figure 2.

A case of a strong correlation (r = .90), despite lacking endorsement in both variables (assuming for the sake of the argument that the midpoint of either response scale – here: score 5 – represents the distinction between item endorsement and item rejection). The figure was created using the interactive correlation simulation provided by Magnusson (2021), with the dotted lines, grey comments and axis labels added by this article's author.



is "high" or "low", by which many researchers mean the absolute position on a response scale (e.g., score 3 of 10 versus score 7 of 10).

The problem that we tend to mix up a person's rank in relation to others with a person's score on a bound response scale extends beyond the issues of interpreting correlation or regression coefficients. The same principle is responsible for the problem that we often misinterpret above-average *z*-standardized scores as if they represented item endorsements, when in fact they may represent an item rejection. For instance, due to the relatively low mean score (or high item difficulty) of obsessive passion, an above-average *z*-score of $z_{obsessivePassion} > 0$ often represents individuals who rejected the items of the obsessive passion scale, but due to mix-ups of relative rank and absolute positions on response scales, such individuals have been interpreted as obsessively passionate individuals (for a critique and summary, see Moeller et al., 2015; for further discussions of this problem, see also Moeller, 2015; 2020). Thus, while inter-individual *z*-scores only hold information about a relative rank of a person *in relation to other individuals* (between-person comparison), they are often interpreted as if they reflected information about item affirmation of individuals. If ranks in inter-individual comparisons are misinterpreted as information about item endorsement or as information about "high" or "low" scores in terms of a response scale, the between-person reference is confused with information that could be interpreted intra-individually. Consequently, misinterpreting relative ranks as information about positions of responses on an absolute response scale can lead to several fallacies:

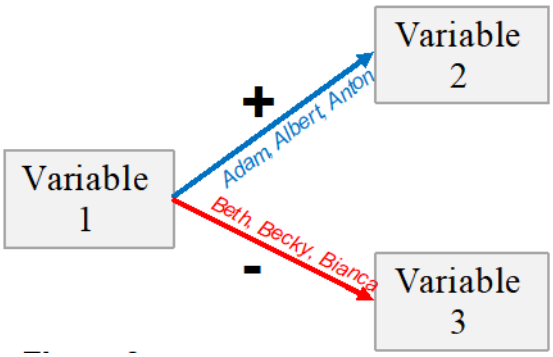
Fallacy 1: The first fallacy is to confuse co-variance with co-endorsement by mistaking a positive correlation as evidence for variable Y being "high" if variable X is "high" or by interpreting a negative correlation as evidence for variable Y being "low" if variable X is "high". A more precise interpretation would be to interpret a positive correlation in the way that individuals with *higher* scores in variable X (*compared to other individuals in the same sample*) tend to have *higher* scores in Y (*compared to other individuals in the same sample*), but please also consider the heterogeneity and ergodicity problems noted above.

Fallacy 2: The second fallacy would be to interpret relations between two or more ranks in inter-individual comparisons as if they described within-person relations between responses on absolute response scales. If two variables are *z*-standardized using the inter-individual mean score and standard deviation, then an individual's relation between the *z*-scores of variables X and Y does not translate into this person's raw scores on the same unstandardized variables. For a given person with $z_{VariableX} > z_{VariableY}$, the raw scores of X and Y may come in any one of the constellations

$raw_{VariableX} < raw_{VariableY}$, $raw_{VariableX} > raw_{VariableY}$, or $raw_{VariableX} = raw_{VariableY}$. As an example of research committing this fallacy, many studies interpret above-average *z*-scores as if they represented item endorsements when in fact they may represent item rejections. See for instance the research on flow where above-average *z*-scores of situational challenge are interpreted as "high challenge" despite the fact that many above-average ratings of situational challenge represent item rejections due to the rather low mean score (e.g., Schneider et al., 2016). Another example is the research on passion where above-average *z*-scores on obsessive passion are interpreted as "high obsessive passion" despite the fact that many above-average ratings of obsessive passion represent item rejections due to the low mean score (for an overview, see Moeller et al., 2015).

Figure 3.

Different people can account for the covariance represented by different paths in between-person path analyses and between-person structural equation models



As a further example of the same Fallacy 2, when examining Figure 3 we may be tempted to believe that the relations between the regression coefficients of the blue paths to the red paths could be interpreted as relations in the *scores* of variables 2 to variable 3 when controlling for variable 1. That is, we might be tempted to believe that when controlling for variable 1 in Figure 3, variable 2 should be higher than variable 3, because variable 1 is positively correlated with variable 2 but negatively with variable 3. However, again, the ratios of the two between-person rankings, represented by the correlation or regression coefficients, *do not translate* into within-person ratios of the scores of any of the variables in terms of absolute response scales. Even though variable 1 is positively correlated with variable 2 and negatively with variable 3, variable 3 can still be higher than variable 2 in every individual in the sample. It is therefore important to keep in mind that between-person ranks do not translate into within-person positions of scores on response scales, nor can ratios among between-person ranks be interpreted as if they reflected within-person patterns in terms of response scale positions or item endorsements.

1.5. Between-Person Structural Equation Models Are Often Interpreted as If They Reveal Within-Person Relationships Among Variables, Which They Do Not

Many studies examine between-person relations (including causal relations) among sets of *more than two* psychological variables by examining the between-person covariance among these variables with path analyses or other structural equation models (SEM; e.g., Ullman & Bentler, 2003; see Figure 3).

Typically, many researchers assume that the various paths in the model describe experiences of the same person. Therefore, we can expect to see a model such as the one depicted in Figure 3 being interpreted as if it indicated that people with relatively high scores in variable 1 tend to have relatively high scores in variable 2, but low scores in variable 3. However, it has been pointed out that such a commonly seen interpretation is based on a misunderstanding. *A path analysis like the one depicted in Figure 3 does not imply that any individual "walks" all the paths* (Reitzle, 2013).

Instead, it is possible for one group of individuals (e.g., Adam, Albert, and Anton) to account for the positive covariance depicted in one path (such as the blue path in Figure 3, or the positive covariance between math ability and math self-concept depicted in Figure 5), while an entirely different group (e.g., Beth, Becky, and Bianca) may account for the covariance depicted in another path (such as the red path in Figure 3, or the negative covariance between the math performance and the English ability self-concept depicted in Figure 5). Thus, path models do not necessarily describe the patterns among the studied variables *within* a person, which has been demonstrated both theoretically and empirically (Reitzle, 2013).

This has tremendous implications for the debate about the use of cross-lagged panel models and mediation analyses in the study of causality (Hamaker et al., 2015; Rogosa, 1980). That different sets of individuals can drive the covariance behind different paths means that, for mediation models the path from a predictor to a mediator may be driven by different individuals than the paths from the mediator to the outcome, or the direct path from the predictor to the outcome. For cross-lagged panel models, it means that *we cannot interpret paths as causal relations occurring within individuals*, which comes in addition to all the other limitations of a causal interpretation of cross-lagged panel models (e.g., Hamaker et al., 2015; Rogosa, 1980).

Causality in terms of psychological processes often refers to intra-individual relations among variables (e.g., eating breakfast causing a person to feel less hungry in the next two hours; having a conflict with a parent causing a teenager to be grumpy afterwards, see Lichtwarck-Aschoff et al., 2009). Inter-individual covariance-based analyses (such as CFA/SEM, path analyses, and cross-lagged panel models), however, *do not examine such intra-individual causal relations*

among constructs.

It should be noted that some of these above-mentioned limitations of between-person methods have been known and discussed by some researchers for a long time (for an overview, see e.g., Asendorpf, 2000). Some are more frequently discussed (such as points 1.1; 1.2, and 1.3 above), whereas others are less widely considered (points 1.4 and 1.5 above). In the light of these limitations that between-person methods face in describing individuals, various within-person methods have been lauded as possible solutions. The next section therefore discusses the within-person analyses that are available, which problems they solve, and what limitations they have.

2. Within-Person Methods Help Fill Some of the Gaps Left by Between-Person Methods

Within-person analytical methods examine how variables change within a person over time, or how various variables relate to each other within a person at one time point. This section particularly addresses and explains the six variants of within-person analysis displayed in Figure 4.

2.1. Analysis of Within-Person Profiles (e.g., Cluster Analysis, Latent Profile Analysis, Latent Class Analysis; Figure 4, Example 1).

Cluster analyses or the latent variants thereof (latent profile analyses) can reveal within-person profiles of two or more variables and indicate how many individuals experience which intra-individual profile pattern. For example, latent profile analyses revealed that despite a negative correlation between burnout and engagement (a form of motivation), about 20-30% of all students and employees in various samples experienced high levels of both engagement and burnout (Moeller et al., 2018b; Salmela-Aro et al., 2016; Tuominen-Soini & Salmela- Aro, 2014). Thus, this method can be used to report the percentage of individuals who do, or do not, show patterns in line with sample-level correlation or regression coefficients or path models, as proposed for instance by Asendorpf (2000). The above-mentioned group of engaged but burned-out individuals would be overlooked if only the significantly negative between-person correlation of these variables were examined.

Profile analyses can also be used to identify in-the-moment profiles For example, they can show one profile (e.g., high scores on motivation, stress, and anxiety) occurring in situation 1; another profile (e.g., a high score on motivation co-occurring with low scores on anxiety and stress) occurring in situation 2; and a third profile (e.g., high motivation combined with high stress but low anxiety) occurring in situation 3. In this case, one person can experience multiple profiles, one at a time, which may change from one moment to the next (e.g., Bergman et al., 2012; Dietrich et al., 2019).

2.2. Within-Person Correlations Between Two Variables That Are Measured Repeatedly in Each Person in Multiple Situations (Figure 4, Example 2)

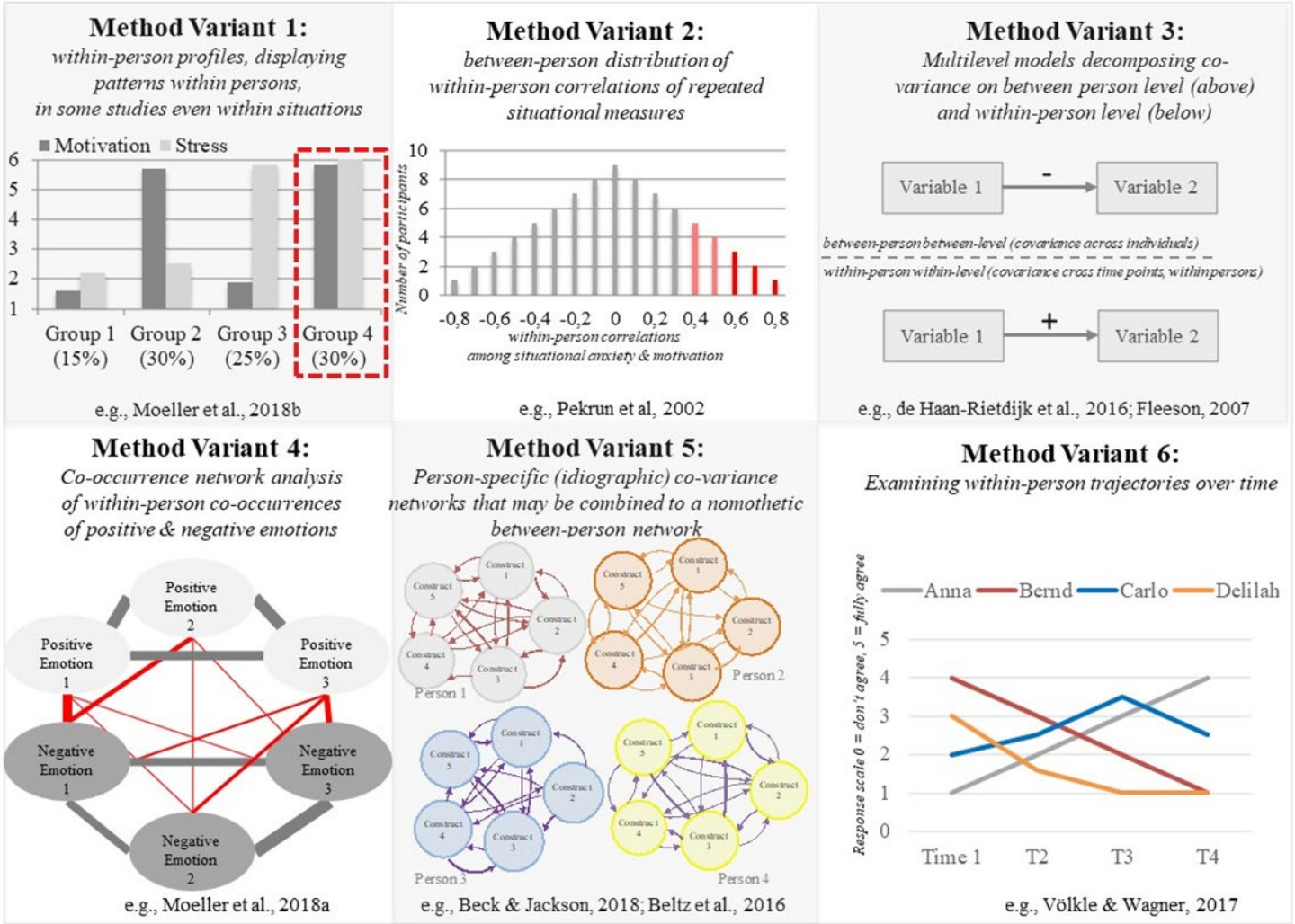
To find out how two variables are related to each other within individuals, it can be useful to examine the within-person correlation or regression coefficient among two (or more) repeatedly measured variables. This method is typically used if a large number of longitudinal measurement time points (approximately $N > 30$) are available for each person, which is often the case with intensive longitudinal data (e.g., Moeller et al., 2015; Pekrun et al., 2002). One variant of this approach has been used by Pekrun et al., (2002) and Moeller et al., (2015), who plotted the inter-individual distributions of such within-person correlations among repeatedly measured variables (in these studies: situational anxiety and situational measures of motivation and various positive emotions; see Figure 4, Example 2).

In a next step, the inter-individual distribution of the intra-individual correlations can be examined to address heterogeneity (e.g., the within-person correlation between anxiety and motivation being positive for some but negative for other individuals, see Pekrun et al., 2002) or to address possible moderators (e.g., the within-person correlation between situational anxiety and negative emotions being stronger for female than for male students, see Moeller, Salmela-Aro, et al., 2015).

2.3. Multilevel Models Decomposing Within- and Between-Person Variance (Figure 4, Example 3)

As an extension of or as a basis for the previously described within-person correlations (see Figure 4, Example 2 above), multilevel modeling can be used to decompose the variance of a repeatedly measured variable into the variance between multiple time points within each person ("within-level") and the variance between individuals in regard to each person's average across time ("between-level"). For examples, see Brose et al. (2020); Völkle et al. (2014); Moeller et al. (2017; 2020c), and Figure 4, Example 3. The advantage of this method is that it can reveal discrepancies between the covariance patterns of the within-person versus the between-person level (i.e., the compositional effect, see Raudenbush & Bryk, 2002), which is why it is currently being lauded as a solution to the problems of Simpson's paradox or lack of ergodicity in longitudinal data (described in section 1.1 above; see e.g., Brose et al., 2020; Völkle et

Figure 4.
Variants of within-person methods



al., 2014).[3] The multilevel variance decomposition can be extended to include structural equation models, including path analyses or confirmatory factor analyses, to model covariance patterns among larger sets of variables that may differ between the within-person and the between-person levels.

2.4. Co-Occurrence or Co-Endorsement Network Analysis (Figure 4, Example 4)

To account for the problem that some research questions require answers about within-person co-endorsement patterns that analyses of covariance cannot provide (see section 1.4 above), the co-variance network analysis examines how often two variables are affirmed (i.e., endorsed) together, or mentioned together, by the same person (for a description of the method, see Moeller et al., 2018a). The co-endorsement analysis is particularly relevant for all studies that focus on joint experiences, such as research on co-morbidity (asking which clinical symptoms are experienced together) and research on mixed emotions (asking which emotions of mixed valence are experienced together). In the co-endorsement network, the psychological variables (e.g., emotions) are represented by bubbles (called *nodes*), and every time a person affirms a pair of these variables together, the line (called *edge*) between these nodes becomes thicker. For example, the thin red paths in Example 4 in Figure 4 indicate that positive and negative emotions are affirmed together occasionally, but not as often as emotions of the same valence are affirmed together (thicker grey lines).

2.5. Network Models Examining Estimates of Co-Variance (Figure 4, Example 5)

As an extension of the analysis of within-person covariance (correlations or regression coefficients; see method examples 2 and 3 in Figure 4), covariance-based network analyses can be used to examine within-person correlations or regression coefficients among multiple variables. There are various unique advantages and insights to such covariance-based within-person networks. For instance, some variants of co-variance-based networks distinguish between person-specific networks of individual persons (called idiographic) and general networks representing between-person trends (called nomothetic). Employing the variance decomposition described above in point 2.3 (multilevel models), this approach acknowledges that each individual may show unique within-person patterns of how the studied variables co-vary over time.

In addition to these person-specific networks, of which there are as many as there are individuals in the sample, one separate between-person (nomothetic) network can be estimated to display those aspects of these person-specific networks that were empirically found to generalize reliably across individuals. A statistical approach of integrating person-specific (idiographic) and general between-person (nomothetic) approaches is called GIMME (group iterative multiple model estimation; Beck & Jackson, 2020; Beltz & Gates, 2017; Gates & Molenaar, 2012; Beltz et al., 2016; Wright et al., 2019). The GIMME method can also identify subgroups of individuals sharing similar sets of associations among the studied variables to account for heterogeneity between individuals in regard to their within-person correlational associations (see e.g., Gates et al., 2017).

2.6. Analysis Of Within-Person Trajectories Over Time (Figure 4, Example 6)

Understanding the changes that people go through requires an understanding of how things change within individuals. To study such within-person trajectories, we need methods that estimate person-specific intercepts and slopes, such as the analysis of within-person trajectories in within-person growth curve models (e.g., Pasyugina et al., 2015). This principle of estimating person-specific trajectories is often combined with the multilevel variance decomposition into within-person and the between-person levels (see Figure 4, Example 3 above) in so-called random coefficient regression models (e.g., for a description of the method, see Cohen et al., 2003). To find inter-individual clusters of similar within-person trajectories, growth curves can be combined with clustering approaches, by clustering similar individual trajectories into homogeneous groups and counting the distinct groups' frequencies (e.g., Muthén & Muthén, 2000), the latter of which is a way to report the percentage of individuals who show patterns in line with sample-level results, as proposed by Asendorpf (2000).

2.7. Which Gaps Left by Between-Person Methods Are Filled by Which Within-Person Methods?

Within-person methods are expected to be game-changing when it comes to describing and predicting individuals. The reasons for this expectation are mainly that within-person methods provide unique insights that go beyond those provided by the more common between-person approaches, whereas previous research in psychology and many other social sciences has largely relied on the limited between-person methods so far. Table 1 gives an overview of the above-described within-person methods and which of these are expected to solve which of the described problem in describing individuals.

[3] Please note: Such multilevel models on repeated measures nested in individuals constrain the model to be invariant across cases (individuals), and typically report only one within-person coefficient for the entire sample, in line with this assumption of homogeneity of such coefficients across individuals. This may not be adequate if ergodicity is lacking (if for instance the within-person correlation coefficient differs between individuals or if the in-the-moment profiles of the studied variables differ between situations and their prevalence differs between individuals).

Table 1.

Problems of between-person methods (rows) and within-person methods that may help to solve them (columns). Numbers 1.1 – 2.6 refer to the matching numbered sections in this article.

Within-person methods described above	2.1. Within-person profiles and scatter plots[1]	2.2. Between-person distributions of within-person correlations[2]	2.3. Multilevel correlation or regression; situations nested in individuals[3]	2.4. Network analysis showing within-person co-endorsements[4]	2.5. Combinations of person-specific/within-person and between-person co-variance-based networks[5]	2.6. Analyses of within-person trajectories (e.g., within-person slopes & intercepts)[6]
Limitations of between-person methods described above						
1.1. Understanding change requires analyzing within-person trajectories. Between-person methods may misinterpret trajectories[7]	*If scatter plots show repeated measures separately for distinct individuals.*				*If networks examine moment-to-moment autoregressive paths within persons*	**Problem 1.1 can be solved by solution 2.6**
1.2. Processes and structures of psychological constructs often differ in within-person versus between-person analyses (lack of ergodicity)[8]	*If scatter plots show repeated measures separately for distinct individuals.*	**Problem 1.2 can be solved by solution 2.2**	**Problem 1.2 can be solved by solution 2.3**	*If only single edges are interpreted or if idiographic and nomothetic networks are distinguished*	**Problem 1.2 can be solved by solution 2.5**	*If between-person differences in regard to within-person trajectories are addressed.*
1.3. Heterogeneity & unexpected patterns hiding behind a between-person coefficient[9]	**Problem 1.3 can be solved by solution 2.1**	*Does reveal differences between individuals but does not solve the problems described by Anscombe (1973)*	*Does reveal differences between individuals but does not solve the problems described by Anscombe (1973)*	*Partially, can be used to distinguish covariance from co-endorsement, but does not represent entire bivariate distribution*	*Does reveal differences between individuals but does not solve the problems described by Anscombe (1973)*	*Does reveal differences between individuals but does not solve the problems described by Anscombe (1973)*
1.4. Co-variance mixed up with co-endorsement[10]	**Problem 1.4 can be solved by solution 2.1**			**Problem 1.4 can be solved by solution 2.4**		
1.5. Different people 'walking' different paths in path models[11]	*Only profile analysis, not scatter plots*			*Adaptation needed to distinguish idiographic and nomothetic networks*	*Solved only in idiographic networks*	*If all trajectories in the model were examined within same person.*

Note. [a] = Moeller et al., 2018b; [2] = Pekrun et al., 2002; Moeller et al., 2015; [3] = Brose et al., 2020; Dietrich et al., 2017; Völkle et al., 2014; [4] = Moeller et al., 2018a; [5] = e.g., Beck & Jackson, 2020; Gates & Molenaar, 2012; Beltz et al., 2016; Wright et al., 2019; [6] = Moeller et al., in press; [7] = e.g., Reitzle & Dietrich, 2019; [8] = e.g., Molenaar, 2004; Yarnold, 2013; Kievit et al., 2013; Kievit et al., 2011; Vansteenlandt et al., 2015; Völkle et al., 2014; [9] = discussed by Anscombe, 1973; Matejka, & Fitzmaurice, 2017; Asendorpf, 1993; 2000; [10] = discussed in Moeller et al., 2018a; [11] = discussed in Reitzle, 2013. In the cells, complete solutions to problems are marked bold, partial solutions are marked in italics.

Examining intra-individual scatter plots of bivariate distributions is a powerful and much under-used instrument. It can help to detect Simpson's paradox (see for example the group-wise scatter plots by Kievit et al., 2013), which helps solving the problems 1.1 and 1.2, both of which refer to lack of ergodicity in longitudinal data. Scatter plots also help detecting unexpected bivariate distributions that deviate from the pattern that a between-person regression or correlation coefficient suggests, which solves the problem described by Anscombe (1973) and Matejka and Fitzmaurice (2017). Scatter plots also reveal differences between covariance and co-occurrence discussed in section 1.4. As a limitation, bivariate scatter plots do not reveal whether the same individuals account for multiple paths in path analyses or other structural equation models (the problem described in section 1.5 and by Reitzle, 2013).

Analyses of within-person profiles (or clusters of individuals with distinct profiles) can be used in similar ways. They can be used to examine each person's intra-individual trajectory (see Figure 4, Method Variant 6 and the problem described in section 1.1) or each person's intra-individual profile of multiple variables (Figure 4, Method Variant 1 and the problem described in section 1.2). Also, the analyses of within-person profiles may be somewhat helpful in discovering some of the heterogeneity that can hide behind overall trends (see the problem described in section 1.3 and e.g., Moeller et al., 2018b) and they can be useful to address the distinction between covariance and co-occurrence (see Moeller et al., 2018b). In contrast to bivariate scatter plots, the analyses of within-person profiles can show a person's score in more than two variables, which can be a helpful tool in finding out "who walks which path" in terms of the problem addressed in section 1.5 and by Reitzle (2013).

Plotting the between-person distribution of within-person correlations described by Pekrun et al., (2002) and Moeller et al. (2015) can be used to examine between-person differences and heterogeneity in regard to within-person covariance (see section 1.2). However, the method does not fully solve the problem of unexpected bivariate patterns hiding behind correlation or regression coefficients, described by Anscombe (1973) and Matejka and Fitzmaurice (2017) and in section 1.3.

Multilevel correlation or regression analyses decomposing variance into Level 1 (variance within individuals and across situations) and Level 2 (variance between individuals) have been used as solutions to the problem of lacking ergodicity (different coefficients within versus between individuals; see section 1.2 and Brose et al., 2020; Völkle et al, 2014).[4] Such decomposition of (co-)variance into within- and between-person sources can help to reveal heterogeneity between individuals in regard to within-person methods, but it does not solve the problem that unexpected bivariate patterns can hide behind correlation or regression coefficients (as discussed in Anscombe, 1973).

Co-endorsement network analyses (for an introduction, see Moeller et al., 2018a) could help to address aspects of possible discrepancies in within- versus between-person structure of variables (the problem described in section 1.2). One edge (line) in this network by definition shows the sample's frequency of a given within-person co-endorsement of two variables and thus combines a within-person statistic (co-endorsement: yes or no) with a sample-level aggregate (relative frequency of that co-endorsement). However, the overall network pattern cannot be interpreted as a within-person pattern, because it is not guaranteed that the same individuals account for multiple edges (lines). To change that and to more precisely address the possible discrepancy between within-person versus between-person structures of variables, the co-occurrence network analysis needs to be adapted to distinguish between person-specific (idiographic) networks and sample-level networks, as it has been proposed for covariance-based networks (e.g., Beck & Jackson, 2020; Beltz & Gates, 2017; Gates & Molenaar, 2012; Beltz et al., 2016; Wright et al., 2019).

Co-endorsement network analysis do reveal the possible existence of co-endorsement patterns that defy the interpretations suggested by a correlation or regression coefficient (see Moeller et al., 2018a), which is arguably a partial solution to the problem of possible unexpected bivariate distributions hiding behind such covariance coefficients (the problem described in section 1.3). However, the co-endorsement network analysis does not examine the entire range of the bivariate distribution, and instead focuses on the upper right quadrant[5] of a scatter plot, as Figure 2 illustrates. The method could be adapted to also examine the other three quadrants pattern of (1) joint item rejection, (2) endorsed X-variable with rejected Y-variable and (3) rejected X-variable with endorsed Y-variable.

The main purpose of the co-endorsement network analysis is to reveal possible discrepancies between the interpretations following from covariance versus co-endorsement analyses (the problem discussed in section 1.4). As with the previously discussed problem, we need to keep in mind that covariance examines the entire bivariate distribution, whereas the co-endorsement (network) analysis focuses on the upper right corner of a scatter plot (see Figure 2). This is practical for some research areas (e.g., the research on mixed emotions and comorbidity), but it may be impractical for others. This limitation can be removed by adapting

[4] Please note: Such multilevel models on repeated measures nested in individuals constrain the model to be invariant across cases (individuals), and typically report only one within-person coefficient for the entire sample, in line with this assumption of homogeneity of such coefficients across individuals. This may not be adequate if ergodicity is lacking (if e.g., the within-person correlation coefficient differs between individuals or if the in-the-moment profiles of the studied variables differ between situations and their prevalence differs between individuals).

[5] That is, assuming that the cut-off that distinguishes between item endorsement and item rejection is located at the middle of the response scales for both variables, otherwise the area representing co-endorsement is not necessarily equivalent to a quadrant in the scatter plot.

the item dichotomization for the method so that all other constellations of item endorsements and rejections can be examined (see above). Some researchers will state the often-heard problem that dichotomizing metric variables leads to a loss of information, but in this case, it also leads to an increase in information – about item endorsement – that is lost if the metric variables are correlated.

The problem that multiple paths are not necessarily driven by the same individuals (described in section 1.5 and Reitzle, 2013) cannot be solved with the co-endorsement network analysis, unless it is adapted to distinguish between person-specific (idiographic) networks and sample-level (nomothetic) networks, as it has been proposed for covariance-based networks (e.g., Beck & Jackson, 2020; Beltz & Gates, 2017; Beltz et al., 2016; Wright et al., 2019).

The combination of person-specific, idiographic networks examining covariance among a set of variables and between-person, nomothetic versions thereof, as described for instance by Beck & Jackson (2020) and Beltz et al. (2016) can be used to describe within-person trajectories and thus provide a solution to Simpson's paradox in longitudinal studies (the problem described in section 1.1), but only under the condition that the network examines such within-person trajectories from one measurement time point to the next, such as moment-to-moment autoregressive paths in idiographic networks (for examples, see Beck & Jackson, 2020; Chaku et al., 2021). The method does provide a full solution to the possible discrepancy between the within- versus between-person structure of constructs (the problem described in section 1.2), as this is the main purpose for which this method was developed. The method does reveal differences between individuals with regard to their within-person covariance patterns, but it does not solve the problem of possibly unexpected bivariate distributions that can hide behind coefficients based on covariance (the problem discussed by Anscombe, 1973 and in section 1.4). The idiographic networks included in this method also make sure that the multiple paths in the network can be interpreted as describing the same person (solving the problem described in section 1.5 and by Reitzle, 2013).

Finally, the analyses of within-person trajectories (e.g., within-person slopes & intercepts) does solve the problem of Simpson's paradox in longitudinal data (described in section 1.1). The method can also be used to address possible between-person differences with regard to within-person trajectories (the problem discussed in section 1.2), if the between-person range and distribution of the individual within-person intercepts, slopes, and curves are addressed (which is the case in some studies and applications of this method). Thus, the method can be used to address heterogeneity between individuals with regard to within-person trajectories, but it cannot solve the problem that unexpected bivariate distributions can hide behind coefficients based on covariance (the problem discussed by Anscombe, 1973 and in section 1.4). Examining within-person trajectories is one way of describing the path that a person "walks", which is arguably one solution to the problem discussed in section 1.5 and by Reitzle (2013).

Importantly, the within-person methods described in this article do not solve all of the problems described in the context of between-person methods. A detailed discussion of further limitations of the within-person methods addressed in this article can be found in Appendix A.

3. Personalized Descriptions and Predictions Require Person-Oriented Methods

Due to the fact that previous research in psychology and many other social sciences has so far relied largely on the limited between-person methods, it remains unknown how many theories and conclusions will change once within-person methods are applied. Appendix B shows how two influential theories in the field of motivation seem likely to change substantially once within-person methods are used to examine their within-person statements and hypotheses. Since many psychological theories make assumptions about within-person patterns and developments, but have so far relied largely on between-person methods for the study of these assumptions, we currently find ourselves facing a widespread theory-method gap. This is a risk to the credibility of research findings. Although the debate about this problem is not new (e.g., Asendorpf, 1993; 2000; Kuhl, 1977; Lewin, 1930; Molenaar, 2004; Valsiner, 1986; Wottawa, 1981), the current time may be a crucial moment for new attempts to bring more appropriate methods to the research on theories about within-person patterns.

The threat to research's trustworthiness that is brought about by the theory-method gap is met with a currently vibrant research on novel and re-discovered within-person methods (e.g., Reitzle & Dietrich, 2019; Beck & Jackson, 2021). The need for personalization in many research areas and applied areas puts further pressure on scientists and methodologists to provide appropriate solutions for the *description of individuals.* The debate about the limitations of between-person methods and about the unique contributions of within-person methods appears to see a renaissance since Molenaar's (2004) manifesto on "Bringing the person back into scientific psychology" (see also Barbot et al., 2020, Renner et al., 2020). With these developments converging, this may finally be the moment for a proactive revision of psychological theories with the goal of reaching truly personalized descriptions, and in the long run, personalized treatments.

One justification that has often been brought up in defense of nomothetic, between-person methods is the goal of trying to find laws and rules of behavior that apply universally to everyone (e.g., Eysenck, 1954; Whitely, 1983). However, this article has summarized the reasons why between-person statistics may fail to describe even a single person in the studied sample, which implies that nomothet-

ic methods do not necessarily describe universal rules or laws applying to everyone, or anyone at all. In addition, there is mounting evidence that many phenomena are more heterogeneous, less universal, than nomothetic approaches assume (e.g., Hoemann et al., 2017; see also Halvor Teigen, 2002).

Moreover, ever since the primary focus on nomothetic research topics was established in psychology, both research and real-life applications have moved forward. Although one-size-fits-all questions have resided in many fields, personalized approaches and studies of between-person heterogeneity have gained terrain. In education, personalized learning aspires to assess person-specific learning needs to provide each student with the tailored support that they currently need (e.g., Dumont, 2019). In personalized medicine – including personalized psychiatry – the patient's individual needs are assessed for instance by sequencing their genome to identify individually matching treatments that would not necessarily work for other patients displaying similar symptoms (e.g., Jain, 2002; Senn, 2016; 2018). In personalized advertisement, people's individual preferences are mapped to target them with advertisement fitting their individual personality and preferences (e.g., Zhu & Chang, 2016; Bang & Wojdynski, 2016). In real-life settings, large online sellers, such as Amazon or Netflix, employ individual predictions to target individuals with the offers of products that best fit their individual person characteristics. Personalization seems to be the theme of the hour both in business and retail as in many research areas.

In psychology, this represents a paradigm change away from nomothetic approaches towards more person-specific descriptions, predictions, and treatments. The demand for personalized solutions in applied fields, such as education or psychiatry, meets the supply of suitable methods for personalized predictions in many research fields, such as the pragmatic data science (e.g., Cho et al., 2002; Zhang, et al., 2013; Sarker et al., 2019), informatics (e.g., Jiang et al., 2013; Yin et al., 2010) and medicine (e.g., Senn, 2018; Zeevi et al., 2015). Considering that a clear demand is met with such a supply of personalized methods in other research fields, psychologists may want to start asking themselves if they want to leave such a core psychological topic – the description and prediction of human behavior, attitudes, preferences, and motivations – up to the applied fields while clinging onto a nomothetic paradigm limiting their scope to universal laws of behavior that appears increasingly unrealistic and of little use in many areas of life and research (see also Halvor Teigen, 2002; Salvatore & Valsiner, 2009).

This is not to say that nomothetic approaches should not be used. On the contrary, using within-person and idiographic methods does not rule out a nomothetic approach. However, based on the logical and empirical arguments summarized above, it seems a better strategy to treat generalizability and invariance of within-person findings across individuals as an empirical question itself that should be empirically tested and supported by evidence. This would mean that instead of assuming that nomothetic and between-person findings represent universal laws, we should test whether, how often, and under which circumstances within-person findings generalize across individuals before we report between-person trends expected to translate to individuals (for an example of such a bottom-up building of nomothetic insights, see the GIMME method; e.g., Beltz et al., 2016).

4. Do These Limitations of Common Between-Person Methods Imply That We Are at the Dawn of a New Credibility Crisis in Psychology?

The following main reasons suggest that a new credibility crisis might dawn upon psychological science: (1) the limitations and frequent misinterpretations of between-person methods that have been described here; (2) the widespread and often predominant use of these between-person methods in psychology; (3) the fact that these limitations have been criticized for many years by various authors in combination with the fact that these method critiques are often ignored in current psychological research (e.g., Molenaar, 2004; Simpson, 1951; Reitzle, 2013; Rogosa, 1980; Hamaker et al., 2015); (4) the fact that many people, including practitioners interested in personalized solutions, turn to psychology with questions about individuals that cannot be answered by the between-person methods that are applied to study these questions; (5) the fact that diverse within-person methods have been available for a long time, solve some of the limitations of within-person-methods, but are not yet embraced fully in many psychological research fields; and (6) the fact that within-person methods and personalized descriptions and predictions are not only needed but frequently used in applied fields that take a pragmatic data science approach and are interested in making trustworthy decisions about how to treat individuals, including banking (e.g., Galal et al., 2016; Hernández-Nieves et al., 2020;), advertisement (Zhu & Chang, 2016; Bang & Wojdynski, 2016), medicine (Senn, 2016; 2018), law enforcement (e.g., Tayebi et al., 2016), personalized content recommendation tailored to customer's preferences in for example Amazon or Netflix (e.g., Gomez-Uribe & Hunt, 2015; Smith & Linden, 2017), and many more.

Together, these considerations imply, first, that we psychologists draw possibly false conclusions about our discipline's main research object – persons; second, that we (could) have known about this fact for years; third, that we have chosen to not use the more appropriate and available methods, which would prevent us from drawing such false conclusions; and fourth, that other disciplines do this job better than us. Many psychological studies and entire research fields have chosen to turn a blind eye to the question

whether and under which circumstances the predominant between-person methods properly describe the individuals in their samples and whether the nomothetic one-size-fits-all effects identified on the group level translate to the experiences of actual people. Many psychologists learn in their introductory statistics course that "Thou shall not draw conclusions about individuals based on group trends", and then we spend the rest of our careers doing essentially just that.

Years and years of theoretical reasoning and empirical evidence show that the examination of within-person profiles and scatter plots can add to the examination of linear correlation or regression coefficients (e.g., Anscombe, 1973). Still researchers who employ profile analysis hear all the time from reviewers and colleagues that "someone still has to convince me that cluster analysis reveals anything that regressions do not" (personal experience of the author, many times repeated), and this despite of the fact that cluster and profile analysis is constantly used to great value as a standard data science tool in many disciplines and applied fields (e.g., de Oña, J., de Oña, & López, 2016; Perrotta & Williamson, 2018). It looks as though psychological science lags behind the insights that the more pragmatic use of data science offers to real-life decision making, possibly because psychologists tend to be committed to the a priori epistemological idea that a nomothetic approach should always take priority over the idiographic idea that different individuals may function in different ways. This epistemological belief has been challenged by numerous empirical findings, which show that much heterogeneity exists and how insightful it is to describe such heterogeneity (e.g., Kievit et al, 2013), as well as by many theoretical and epistemological arguments (e.g., Jacomy, 2020; Molenaar, 2004; Salvatore & Valsiner, 2009; 2010), novel integrations of idiographic with nomothetic approaches (e.g., Beltz et al., 2016), and by the fact that real-life practitioners who are interested in making efficient, trustworthy decisions and predictions about individuals seem to fare pretty well with their pragmatic use of both within- and between-person methods (e.g., DeMatteo et al., 2010).

Together, these considerations suggest that psychology may stand at the dawn of a new credibility crisis similar to the one we faced during the replicability crisis (e.g., Anvari & Lakens, 2018; Pashler & Wagenmakers, 2012) – a credibility crisis due to the fact that between-person methods and nomothetic theories fall short of the prediction of individual human behavior that real-life practitioners achieve by using pragmatic data science. Whereas the replicability debate addressed the problem that research findings may not be as invariant across samples and contexts as expected of them, the imminent next credibility crisis may reckon that *we do not even understand properly what happens in the original sample at hand*. In conclusion, the method critiques summarized here imply that psychological studies attempt to describe laws of behavior that are assumed to apply to everyone in the sample (nomothetic approach) and that actually may fail to describe many, or even all, persons in the sample. By doing that, psychology misses out on the understanding of what it should be most interested in: individuals (Barbot et al., 2020; Carlson, 1971; Molenaar, 2004). If we want to find out how groups function, we can ask sociologists. When we ask psychologists a question, we expect that their answer will help us understand what goes on *inside of people's minds and lives*, and the current use and predominance of between-person methods lets us miss exactly these points.

5. Pathways Towards a Pro-Active Reckoning in Psychological Science

The theoretical and methodological arguments that have been presented here may give reasons to worry about the question whether we know what we think we know about what goes on in people's minds and lives. Systematic empirical research is needed to find out how many studies in various fields of psychology may be affected by the possible misinterpretations of results achieved with between-person methods being interpreted as if they described within-person patterns.

In order to avert a loss of credibility similar to the one that followed from the previous replicability crisis (Anvari & Lakens, 2018; Pashler & Wagenmakers, 2012), psychologists may want to strive for a proactive reckoning by exploring the prevalence of the various possible misinterpretations of between-person methods in their field, by exploring the possibility of complementing their usual methods with a larger variety of within-person methods, and by adapting their theories. Further theory development could strive to adapt theories to the fact that most of them have so far been examined mainly with between-person methods and should therefore limit their statements to the conclusions that can be drawn based on such methods. In addition, or alternatively, further theory development could strive towards extending nomothetic theories to include reflections on possible heterogeneity in within-person patterns, by making sure that within-person assumptions are systematically tested with within-person methods.

The following research questions could serve as directions for a pro-active reckoning about the current limitations of psychological methods, for the purpose of first understanding and then closing the current theory-method gap:

(RQ1) How many studies in different fields of psychology show misinterpretations of between-person methods in the form of results obtained with between-person methods being interpreted as if they described within-person patterns[6] without providing the required evidence that such an

[6] Within-person patterns being defined as a) the covariance among two or more variables, b) the profile (or co-affirmation / co-negation) of more than one variable, c) the change in one (or more) variable(s) over multiple time points

intra-individual interpretation is appropriate?

(RQ2) How many studies in different fields of psychology consider the known limitations of between-person methods that have been described in this article, such as heterogeneity hiding behind overall trends (as described in Simpson, 1951; Anscombe 1973; Matejka & Fitzmaurice, 2017), or different individuals driving different paths in structural equation models (Reitzle, 2013) when they use methods for which these considerations are relevant?

(RQ3) Which psychological theories/conclusions will have to change substantially due to contributions by within-person methods, and in what ways do the conclusions have to change? What exactly are the unique insights provided by within-person methods that go beyond those provided by between-person methods in psychological theory development? For some first considerations on this question, see Appendix B.

(RQ4) What are the current limitations of available within-person methods, and what are the solutions to these limitations? For some first considerations on this question, see Appendix A.

(RQ5) What unique contributions can between-person and within-person methods, respectively, make to applied fields aiming at personalized diagnostics and treatments, as for example personalized medicine?

The empirical reckoning of psychology with the goal to overcome possible theory-method gaps could apply for instance the following methods to examine the research questions suggested above:

Reviews could re-examine the published research to find out how frequently between-person methods are used to examine within-person assumptions, or how often known limitations of between-person methods are (not) reflected in the analysis and interpretation of between-person methods.

Surveys could ask psychological researchers across various research topics and subdisciplines about (1) their awareness about limitations of between-person methods, available within-person methods, and possible theory-method gaps in their field, and (2) the frequency of between- versus within-person methods being applied in their psychological studies.

Revisiting established theories: Theories that have been addressing within-person assumptions but have so far been supported mainly by between-person methods could be revisited by adopting appropriate within-person methods suited to test their within-person assumptions (for suggestions regarding specific theories, see Appendix B). This could be done either by conducting new studies collecting new data, or by using existing datasets associated with well-established theories that have previously been examined mainly with between-person methods and re-analyzing them in secondary data analyses with within-person methods.

If taking this proactive approach of a systematic reckoning to avert a possible new credibility crisis, we can build upon the lessons learned from the previous replicability crisis. The replicability debate can teach us that it may be wiser to acknowledge systematic problems in our use of research methods rather than sweeping them under the carpet by ignoring or downplaying the warning voices. We can also keep using the power of crowdsourcing for the systematic empirical revisiting of established theories, for instance by adopting a many-analysts approach (e.g., Aczel et al., 2021; Bastiaansen et al., 2020; Silberzahn et al., 2018) on open data to answer the question whether and in what cases a re-analysis of data used in previous studies supports the published conclusions or contributes new insights. Panel discussions could be organized to explore the scope of the problem, its implications, and available solutions. Open data could speed up the systematic re-analysis of data on established, well-studied theories. The experience of having faced the replicability crisis together as a field and having grown from it, can teach us to keep in mind that research always changes and hopefully increases in its wisdom and that such changes, even when affecting one's own much-loved research field, do not have to be perceived as a threat, nor as a failure, but can be welcomed as a form of insight and growth. In sum, the flourishing research on within-person methods promises to enrich the psychological methods portfolio in likely game-changing and highly relevant and timely ways. Embracing and discussing the novel insights it contributes, along with their methodological, theoretical, and epistemological implications (and their own limitations), promises to be a valuable and important task for psychological science.

6. Getting Ready for the Future by Learning from the Past

It should be noted that this article is far from being the first to point out a mismatch between the within-person theories being studied with between-person methods. That this mismatch has led psychological research into a validity crisis has been pointed out before by Lundh (2019). The same author also already pointed out that the currently trending attempts to personalize medicine are particularly in need of person-oriented (within-person) methods, a thought echoed by Lundh & Falkenström (2019) and also mentioned by Wiedermann et al. (2016). The calls for more within-person methods and notes of concern about the theory-method mismatch addressed in this article go back many decades to early works by, for instance, Allport (1961; 1962), Carlson (1971), Stern (1911), Magnusson (1988; 1999), Cattell et al. (1947) and Windelband (1894/1998). The various historical roots and decades-old but much-needed solutions of idiographic and within-person methods have been summarized in the early issues of the Journal for Person-Oriented Research (see e.g., Lundh, 2015; Valsiner, 2015; Bergman & Lundh, 2015; Valsiner, 2016).

This article aspired to build upon but reach beyond these previous considerations by pointing out the extent to which

ignoring the long-known theory-method mismatch has brought psychological science to the brink of a new credibility crisis and by linking the addressed problems to an overview of available solutions and a research program aimed at helping psychological science meet the demands of the currently trending personalized diagnostics and intervention.

7. Further Theory and Method Development Needed to Better Describe and Predict Individuals

It should also be noted that this article's list of limitations of between-person methods with regard to the aspects about individuals that these methods fail to describe is far from complete. For instance, while this article has emphasized the theory-method gap with regard to dangers of using standard group-oriented *analyses* to draw conclusions about laws applying to individuals, other authors have pointed out that already the previous step of *measurement* has to be improved in order to avoid a further loss in the credibility of psychology and the social sciences. Please see for instance Flake and Eid (2021) for a general discussion of prevalent measurement issues in psychology; and Bergman (2017) for a discussion of changes to measurement practices that would be needed to obtain more trustworthy measures for the assessment of individuals.

Likewise, although the list of within-person methods that has been presented here may help solving some of the listed limitations of between-person methods, it is far from complete. Particularly promising avenues are current developments building on dynamical systems theories, such as nonlinear dynamic systems and nonstationary dynamical systems approaches (e.g., Molenaar et al., 2016). While the main aim of this article was to point out that there are many widely studied within-person methods available to overcome the listed limitations in between-person methods, there is a rapid development of novel and groundbreaking within-person methods reaching beyond the solutions presented in this article.

Finally, there will be more to do, even if future studies set out to avoid the here-described theory-method gaps by making sure that theories about within-person patterns are examined with appropriate within-person methods. Many psychological theories only make statements about between-person differences, possibly in part due to the fact that the available between-person methods may have shaped and limited the ways we dare to think about psychological processes. On the other hand, many of the studies that use within-person methods (e.g., cluster analyses) or idiographic methods present their approaches as exploratory and not theory- but data-driven. One challenge for further research on individuals will be to close this gap by developing theories and theory-driven hypotheses that explicitly address within-person patterns and processes and by linking within-person methods more strongly to such theories and theory-driven hypotheses (see also the recent debate about the theory crisis and possible solutions, e.g., Eronen & Bringmann, 2021; Fiedler, 2017; Fried, 2020; Guest & Martin, 2021; Haslbeck et al. (in press); Muthukrishna & Henrich, 2019; Meehl, 1967; 1990; Oberauer & Lewandowsky, 2019; Smaldino, 2019; Vaidyanathan et al., 2015). It also seems that the available confirmatory hypotheses-testing features of within-person analyses (e.g., confirmatory latent profile analysis or confirmatory network analysis) are underused and widely unknown. And finally, there is a long way between describing or predicting individuals and finding the right treatment for the right person at the right time. We have only started this long journey. Let's take a first step by applying within- and between-person methods mindfully so as to match theories and hypotheses.

Declaration of interests

The author declares that there are no conflicts of interests.

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Appendix A: Limitations in the currently available within-person methods

Not all of the above-mentioned limitations of between-person methods with regard to their capabilities of describing individuals are solved by the available within-person methods. *Essentially, within-person methods just bring the problems concerning between-person analyses mentioned above to the next more specific level concerning differences between multiple measurement time points*.

Example 1: Where between-person methods fail to describe heterogeneity between individuals (see above section 1.3), within-person methods may fail to describe heterogeneity between situations, within individuals. If the heterogeneity between situations comes in the form of non-stationary effects (e.g., a cross-sectional or lagged correlation coefficient or a lagged auto-regression changing over time), then certain time series models capable of revealing non-stationarity can be a solution (see e.g., Molenaar et al., 2016). However, if the heterogeneity between situations comes in the form of distinct state profiles that may differ between the various situations that an individual experiences (e.g., profile 1: interesting, easy but emotionally taxing situations; profile 2: uninteresting, easy and emotionally taxing situations; profile 3: interesting, difficult and emotionally taxing situations, profile 4: interesting easy but not emotionally taxing situations, etc.), then correlation-based methods may not reveal the full scope of such within-person heterogeneity because the heterogeneous profiles may hide behind the person-specific and time-specific correlation coefficients.

Example 2: Where the analysis of intra-individual profiles or co-occurrences solves the problem that unexpected groups of individuals with distinct within-person patterns can hide behind overall trends, these within-person profile and co-occurrence network analyses may overlook that within each person, there may be groups of situations with profiles or co-occurrence patterns defying the cross-situational trend seen in the same individuals. Dynamic models for non-stationary processes can accommodate for covariance patterns differing between time points or time periods within a person, but they cannot solve the problem that distinct state profiles may hide behind such correlation trends.

Example 3: Where multilevel analyses decomposing the within- and between-person (co-)variance help to detect Simpson's paradox by showing differences between the covariance patterns of the within-person level and the between-person level (please see sections 1.1; 2.1; and 1.2; 2.2 above, as well as Brose et al., 2020; Völkle et al., 2014), they may overlook Simpson's paradox with regard to differences in covariance patterns between other groups of situations (please see below).

A.1 Limitations with regard to understanding heterogeneity within individuals

While the within-person methods described in this article may help to describe *heterogeneity between individuals* in regard to within-person patterns, they are limited in regard to their capacity to describe *heterogeneity within individuals between situations*. Heterogeneity does not only exist between individuals in regard to their within-person trajectories, covariance patterns, profiles or co-occurrences. *Heterogeneity can also exist within a person, in the way that situation-specific profiles of variables may differ from one moment to the next.*

For example, within a person, different situations may differ with regard to (1) the in-the-moment profiles (see e.g., Dietrich et al, 2019), or (2) the co-occurrence pattern among repeatedly measured variables. The first of these problems can be addressed by analyzing in-the-moment profiles, for instance with multilevel latent profile analyses, see Dietrich et al., 2019, or the so-called ISOA approach[7] (Bergman et al., 2012), whereas the second problem can be addressed by analyzing in-the-moment co-occurrence patterns (Moeller et al., 2018a).

Third, there may be unexpected bi-variate distributions hiding behind within-person correlations or regression coefficients of repeatedly measured variables (as described by Anscombe, 1973 and Matejka, & Fitzmaurice, 2017), which implies that a within-person correlation or regression does not imply that the studied variables show the relation that we expect based on their within-person correlation or regression coefficient. Scatterplots, which show the bivariate distributions of the repeated measures for each person, may help to rule out such unexpected bivariate distributions hiding behind within-person trends.

Fourth, the problem of Simpson's paradox or lack of ergodicity is not entirely and automatically solved by the variance decomposition into within- and between-person variation that is often hailed as its solution, because of the following reasons: It is possible that within a person there are groups of situations in which a regression or correlation coefficient looks different than the overall coefficient found in the same person. For example, in an intensive longitudinal study with repeated surveys per person, the relation between two emotions could be negative within each school lesson, but might look positive for the same person if we only examine the within-person versus between-person covariance and instead of the here more appropriate decomposition into (1) the within-lesson – within-person variance, (2) the within-person – across lesson variance, and (3) the between-person variance. Some readers may argue that this can be easily solved by taking additional layers or clusters of observation into account by adding a third level to the multilevel analyses that are often used to address lacking ergodicity (e.g., Brose et al., 2020; Völkle et al., 2014), but there is a crucial limitation to that option: *We cannot make sure to know all the relevant boundary conditions that may influence a within-person regression or correlation coefficient*, and in intensive longitudinal data, there are typically many possible – often unknown – contextual sources of variation, combined with the expectation that the studied phenomena are highly time- and context-sensitive (hence the need to study them with intensive longitudinal methods). If

[7] ISOA approach: In intensive longitudinal samples of including many repeated measurement time points per person, latent profile analyses (for continuous response scales) or latent class analyses (for nominal response scales) are used to identify state (in-the-moment) profiles of variables (nestedness of measures in persons is typically ignored).

fluctuation across contexts and time is expected in intensive longitudinal studies, and we keep in mind that many studies and findings in research with intensive longitudinal studies are rather new and often the first of their kind, we have strong reasons to believe that there is much that we do not know about the context- and time-related sources of variation in intensive longitudinal studies. Consequently, we should assume that within-person correlation or regression coefficients may depend on certain characteristic of the contexts and times in which the data were collected. This in turn implies that Simpson's paradox or a lack of ergodicity can exist without us being able to detect it, for instance a within-person regression or correlation coefficient is positive within the units of an unknown/unexplored boundary condition (e.g., within school classes, within days, within certain contexts such as school, home, leisure, or within each interaction with a person's significant other), while the same within-person regression or correlation coefficient can be negative for the same person across these units of an unknown/unexplored boundary condition. While there are some tutorials helping to detect and address Simpson's paradox in longitudinal data (e.g., Kievit et al., 2013), we can expect the problem of unknown boundary conditions to limit the trustworthiness of coefficients concerning incompletely understood phenomena that are expected to fluctuate strongly between contexts and time points.

A.2 The question of who walks which paths (Reitzle, 2013) remains unanswered in within-person co-occurrence analyses

A limitation of the co-occurrence network is that it currently does not distinguish between person-specific (idiographic) networks and between-person (nomothetic) networks, as some of the co-variance-based network approaches do (e.g., Beltz et al., 2016; Beck & Jackson, 2020). While each bivariate edge (line between two variables) can be interpreted as the number of times these variables were co-endorsed or co-occurred within the same person[8], we do not know whether the same individuals[9] account for the co-occurrences represented by the other edges (lines) in the same network. Thus, we currently do not know "who walks which path" (see Reitzle, 2013 and section 1.5 above), nor do we know the percentage of individuals showing intra-individual patterns in line with the *entire* inter-individual co-occurrence network, as proposed by Asendorpf (2000). To solve this, we need to examine person-specific co-endorsement networks.

Appendix B: What Psychological theories can we expect to change when we start using more within-person methods?

It may seem surprising how many established, much-cited psychological theories still predominantly use between-person methods to study within-person patterns, considering the many years of debates about the limitations of between-person methods (e.g., Asendorpf, 2000) and the number of available within-person approaches (e.g., Barbot et al., 2020; Fleeson, 2004; Molenaar, 2004; Renner et al., 2020; Reitzle & Dietrich, 2019; Beck & Jackson, 2021). Some research areas have widely adopted some (but usually not all) within-person methods described above. For instance, a number of studies in the area of personality psychology, clinical psychology, and emotion research examine within-person variance with multi-level models and within-person network models (e.g., Beck & Jackson, 2019; Haslbeck & Ryan, 2020; Hoffart & Johnson, 2020). In contrast, in some fields of motivational psychology, research on education, and in research related to problems of personalized learning, there may still be a need for within-person approaches to complement the currently widely prevailing between-person approaches. In the following, I introduce research areas that currently use mostly between-person methods to study within-person patterns. All subsequently addressed theories are relevant for the improvement of personalized learning that this article has addressed. Please note that these theories are mentioned as exchangeable examples or placeholders for many other theories relying on mostly between-person methods, in order to make the point that future studies may want to make more use of the available within-person methods. The following discussion is not meant as a critique but rather as a proposal for new directions in further studies on these important, plausible, and well-studied theories, because of the importance, plausibility, and popularity of these theories, which can serve as role-models for the further development and paradigmatic changes in other, not subsequently mentioned, theories.

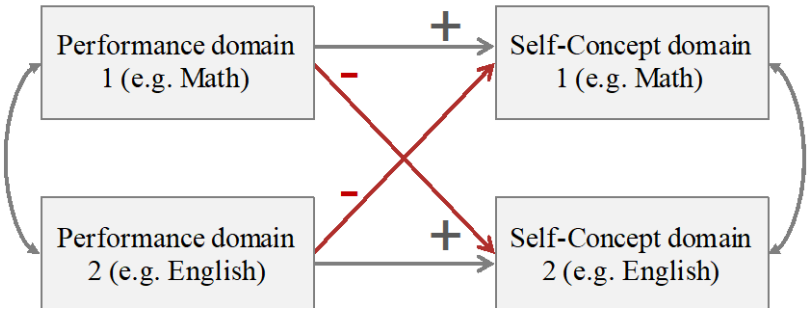
Dimensional Comparison Theory (DCT)

Dimensional comparison theory (DCT) is a much-cited theory in education (e.g., Möller & Marsh, 2013; Möller et al., 2016). It states that students compare within themselves how they perform in one school subject (e.g., Math) with how they perform in another (e.g., English). A central conclusion in the literature on the DCT is that "Students performing better in the math than in the verbal domain tend to have lower self-perceptions of their verbal ability than do students with identical verbal ability but lower math ability (and vice versa)" (Möller & Marsh, 2013, p. 544).

[8] and within the same situation, in the case of this method being applied to intensive longitudinal data

[9] or the same situations, in the case of this method being applied to intensive longitudinal data

Figure 5.
Assumed relationships between performances and self-concepts in different school subjects in dimensional comparison theory.



Although DCT at its core draws conclusions about within-person comparisons of performances and self-concepts within and across school subjects, the large amount of studies on DCT almost exclusively rely on between-person analyses of covariance among these variables (but see also the less frequent experimental and longitudinal studies on DCT, e.g., Helm et al., 2016; Möller & Husemann 2006; Niepel et al., 2021; Strickhouser & Zell, 2015). The prevalence of between-person methods in DCT research implies that there is a theory-method gap in many previous studies on DCT, and that research on DCT risks being affected by the limitations of between-person methods in describing individuals that were described in the sections 1.1 to 1.5 of this article). In particular, research on the DCT may or may not be affected by the problems of heterogeneity hiding behind overall regression coefficients (see section 1.3 and Anscombe, 1973), the problem that separate groups of individuals may drive the covariance in different paths in the model (section 1.5 and Reitzle, 2013), and the problem that the examined covariance does not carry any information about the relations in raw scores (section 1.4 and Moeller et al., 2018a). Thus, based on the prevalence of between-person methods that have been used to study the DCT, we do not know whether and for whom the self-concept in one subject is "higher" than the self-concept in another, we do not know how much heterogeneity may hide behind any one path in the model, and we do not know how many individuals are described properly by one or by multiple paths of the model. Therefore, it is unknown whether the conclusions of the much-cited DCT will be confirmed by within-person analyses. There are first studies assessing within-person variation in DCT-related measures with experimental, diary and experience sampling method approaches (Helm et al., 2016; Möller & Husemann 2006; Niepel et al., 2021; Strickhouser & Zell, 2015). These and further within-person studies on the hypothesized within-person relations among performances and competence self-concepts in different school subjects are needed to find out whether the hypothesized within-person comparisons can be found within persons.

These questions matter, because DCT is a very influential theory about students' motivation. Its findings and their implications are crucial for the understanding of how students can be motivated, which in turn is crucial for the understanding of how personalized learning can address individual students' motivation. That makes it necessary to test the statements of the DCT with the appropriate methods. DCT is so popular and known that it is a beacon with signaling effects. Demonstrating that within-person methods contribute unique insights to such a widely-studied subject is likely to inspire researchers working on other theories to revisit theirs.

The Dual Model of Passion

The Dual Model of Passion is another extremely influential theory about motivation. It states that passion is a form of motivation that consists of a person liking an activity, finding that activity important, investing time and energy with that activity, identifying with the activity, and calling the activity their passion (e.g., Vallerand et al., 2003). The Dual Model of Passion distinguishes between a positive, desirable form of passion called harmonious passion (HP) and a harmful, undesirable form of passion, called obsessive passion (OP). Both forms of passion are measured with different scales (Vallerand et al, 2003). While HP is correlated with and predicts all sorts of positive outcomes, such as positive emotions, indicators of wellbeing, and achievement, OP is correlated with and predicts all sorts of negative outcomes, including negative emotions, the risk of getting injured, burnout-symptoms, and many other undesirable experiences (for overviews, see e.g., Vallerand, 2010; Vallerand & Houlfort, 2019). The vast majority of studies on the Dual Model of Passion employ between-person methods, mostly the analyses of between-person correlations or regression coefficients, often in the context of structural equation models. Nevertheless, one frequently finds that these studies are interpreted as if they described within-person patterns of the passion forms HP, OP, and their respective outcomes, for instance in sentences such as "people with a harmonious passion should be able to fully focus on the task at hand and experience positive outcomes" (Vallerand, 2012, p. 4) and "people with an obsessive passion can thus find themselves in the position of experiencing an uncontrollable urge" (Vallerand, 2012, p. 3). Even some studies that explicitly set out to examine within-person combinations of harmonious and obsessive passion use between-person methods (mostly analyzing between-person covariance; Schellenberg et al., 2019). This demonstrates that there are many different layers and aspects to

within-person variation that are often hard to address in one single study, leaving room for future studies on further aspects of within-person patterns of HP and OP (e.g., co-occurrence analyses or profile/cluster analyses).

A frequent statement posits that passionate individuals experience either a predominant HP or a predominant OP, and could thus be classified into “mainly harmonious individuals” and the “mainly obsessive individuals” (Philippe et al., 2009). However, the few within-person studies on within-person profiles of HP and OP revealed that there were almost no individuals with higher obsessive than harmonious passion, and that most individuals experienced HP and OP together in either both high, or both moderate, or both low levels (e.g., Moeller et al., 2015; Wang et al., 2008). This suggests that within-person methods can really change our interpretations of the experiences that individuals make and that further within-person studies on the Dual Model of Passion promise to make interesting contributions to our understanding of the ambivalent motivation experienced by individuals.

The insights gained by discussing these two examples of the DCT and the Dual Model of Passion are likely to transfer to many other psychological theories. The goal of this discussion was not to call out specific theories, but to describe these insights as an inspiration to demonstrate how much more insight and what tremendous paradigmatic changes may be gained from integrating the available within-person and between-person methods more systematically.

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Causal inference on human behaviour

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Making causal inferences regarding human behaviour is difficult given the complex interplay between countless contributors to behaviour, including factors in the external world and our internal states. We provide a non-technical conceptual overview of challenges and opportunities for causal inference on human behaviour. The challenges include our ambiguous causal language and thinking, statistical under- or over-control, effect heterogeneity, interference, timescales of effects and complex treatments. We explain how methods optimized for addressing one of these challenges frequently exacerbate other problems. We thus argue that clearly specified research questions are key to improving causal inference from data. We suggest a triangulation approach that compares causal estimates from (quasi-)experimental research with causal estimates generated from observational data and theoretical assumptions. This approach allows a systematic investigation of theoretical and methodological factors that might lead estimates to converge or diverge across studies.

Many human behaviours and experiences are difficult or impossible to manipulate in controlled settings, and yet their underlying causal mechanisms are at the heart of research questions in several fields, such as economics[1], epidemiology[2], political science[3,4], psychology and neuroscience[5–8], and sociology[9,10]. Advances in the study of causal inference have increased the potential for causally informative analyses from observational data of human behaviour[1,2,11,12]. Combining these with technological advances that have made longitudinal data more available[13] holds great potential for behavioural research.

In this Review, we bring together perspectives from a variety of fields. We describe issues of causal interpretation that are common to these disciplines, as well as available approaches to address them, with the aim of providing a non-technical, integrative conceptual overview (a glossary of key terms is provided in Box 1). Although many issues we discuss (such as lack of specificity in language) apply to causal inference in general, we focus on longitudinal data because scholars have made many exciting methodological advances in their analysis that may provide partial solutions. Even with longitudinal data, inferring causality is difficult—perhaps even more so than researchers imagine. In this Review, we hope to provide a potential future road map for researchers seeking to answer causal questions pertaining to human behaviour. We identify key challenges to causal inference on human behaviour

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BOX 1

Glossary of key terms

Treatment or intervention (synonyms): Any deliberate change made to one or more variables in a model (often with the goal of uncovering causal effects).

(Randomized) experiment: A controlled study in which the treatment is allocated to the participants at random.

Quasi-experiment: A study in which naturally occurring events determine who receives a treatment (for example, country of birth or school of participants' choice).

Observational study: A non-experimental study (that is, one in which no random perturbations are given) that records naturally occurring phenomena.

Causal effect: The hypothetical difference in a variable *Y* given different conditions of an intervention *X* for a population (average causal effect), a part of a population (conditional causal effect) or a person (individual causal effect).

Estimand: The quantity (for example, a parameter) representing the (causal) effect of interest in a statistical analysis.

Estimate: The statistically estimated or calculated value derived from data to approximate the true value of an estimand.

Directed acyclic graph (DAG): A visual representation of the causal assumptions of a theory or model. Nodes in a DAG correspond to variables and are either connected by directed edges (indicating a causal influence) or not (indicating the absence of a causal influence). DAGs help to clarify whether possible covariates serve as mediators, confounders or colliders.

Mediator: If a treatment *X* causally influences *M* and *M* causally influences the outcome *Y*, then *M* is the mediator of the treatment. Whereas conditioning on *M* biases the estimate of the total effect, it might be necessary to condition on mediators when one is interested in estimating which path or mechanism is the reason for a causal effect.

Confounder: If a variable *C* causally impacts both the treatment *X* and the outcome *Y*, then *C* is a confounder for the treatment effect. Not conditioning the treatment effect on *C* biases the estimate.

Collider: If both the treatment *X* and the outcome *Y* causally influence a variable *C*, then *C* is a collider. Conditioning the treatment effect on *C* biases the estimate.

BOX 2

Asking appropriately specific research questions

Unspecific research questions impede selecting the optimal design, models and techniques to answer them.

Tentative solutions

- Clarify the research question: do you attempt to estimate causal effects, and, if so, which kinds of effects?
- Can the question be answered by the data at hand?
- Can the estimate be bounded within a range of plausible values under realistic assumptions?

Literature suggestions

- Gelman and Imbens[15]
- Hamaker et al.[64]
- Hernán[18]
- Hernán and Robins[2]
- Lundberg et al.[65]

and provide tentative solutions for single challenges. Boxes 2–8 summarize each identified challenge, point to challenge-specific solutions and suggest literature for further reading. We further suggest how to deal with imperfect causal effect estimates in a more general manner.

Major challenges

Failing to ask appropriately specific research questions

It may sound obvious that causal inference can succeed only if we specify the causal effect of interest. However, an underappreciated reason why a large number of research questions have not yet been addressed using recent advances in the tools and language of causal inference is that researchers are often unclear (even to themselves) about what question their analysis is attempting to answer—specifically, whether the researchers are interested in estimating a causal effect at all. Although in some fields the default answer may be a clear 'yes' (for example, in economics[14]), researchers in other fields (for example, psychology) may show less consensus on this question.

Let us start with the kinds of questions we hear every day in scientific settings and in our daily lives:

(1) 'Why are some people happier than others?'
(2) 'Does happiness predict a person's later health status?'
(3) 'What are the economic returns to completing an additional year of schooling?'

The ease with which these questions can be mapped onto well-defined causal effects varies. Question 1 is a natural and perhaps useful starting place. However, although it is unambiguously causal, it specifies no cause at all, resulting in a 'reverse' causal question[15] that cannot be directly addressed with methods intended to address forward causal inferences (what is the effect of *X* on *Y*?). Such a 'why' question is also sometimes called a 'backward causal question', because it starts with the effect and asks for a cause. Question 2 is framed as a statistical, not a causal, problem (does *X* predict *Y*?). The answer could be used for actuarial purposes when trying to forecast future health status or to select people for training or intervention. Only question 3 is both specific and (unambiguously) causal: the cause and effect of interest are reasonably clear.

Which question we ask has implications for data collection and analysis. A researcher interested in question 1 might be best served by conducting a literature review or, if there is little prior work, pursuing an exploratory line of research. A researcher interested in question 2 would not need to worry about confounding because the question is concerned only with prediction without further qualification; additional variables would be of interest only to improve the predictive accuracy of the model. But researchers often still attempt to control for third variables as confounders, which highlights the fact that such analyses are often associated with an underlying causal question, even if it is not explicated—a source of substantial confusion[16–18].

Under-control

Even clearly specified questions like question 3 do not guarantee clear causal answers. A researcher attempting to estimate the effect of happiness on health by statistically controlling for potential confounders (that influence both earlier happiness and later health) will have a difficult time convincing a reasonable sceptic. A positive estimate, a sceptic might argue, is plausibly still confounded by unmeasured common causes (for example, existing health conditions[19]). Omitted confounders are a problem for causal inference approaches that

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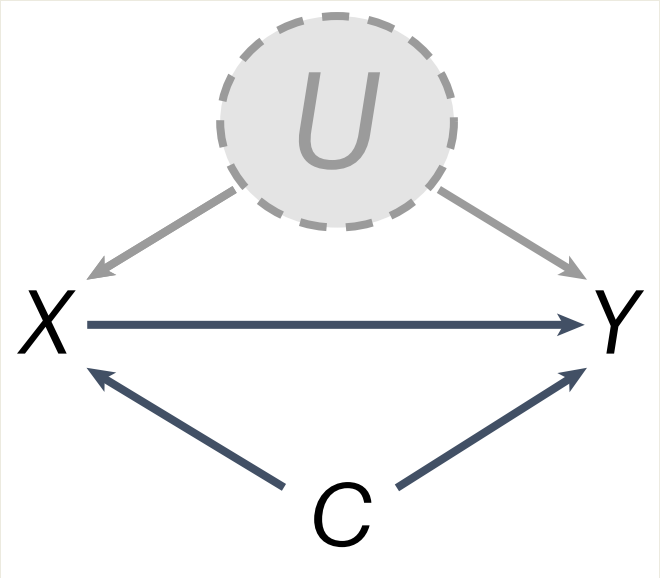
BOX 3

Under-control



Confounders bias the estimate of causal effects if not adequately controlled for. If a variable *C* (confounder) is known to cause both *X* and *Y*, then the estimate of the causal effect of *X* on *Y* will be biased if *C* is not adequately controlled for. Unknown or unobserved confounders (depicted as *U*) are a main threat to valid causal inference.

Example: The estimated effect of years of schooling (*X*) on earnings (*Y*) may be upwardly biased if childhood human capital (*C*) is not ruled out by the design or analysis.

Tentative solutions

- Clarify the causal model before data collection to assess potential confounders.
- Measure as many confounders as possible and adequately adjust for them when estimating the causal effect.
- Actively discuss assumptions under which your estimate approximates a causal effect.
- Consider a design-based solution or methods that control for unobserved confounders.

Literature suggestions

- Angrist and Pischke[48]
- Cinelli and Hazlett[22]
- Cinelli et al.[30]
- Pearl et al.[100]
- Rohrer[7]
- Usami et al.[84]

BOX 4

Over-control

Controlling for colliders biases the estimate of the causal effect. For example, if *X* and *Y* both cause another variable *C* (collider), conditioning on *C* biases the estimate of the causal effect of *X* on *Y* (even if *X* and *Y* are unconditionally independent).

If the causal estimand is a total effect, controlling for mediators biases the estimate. For example, if the effect of *X* on *Y* is mediated by *M*, conditioning on *M* biases the estimate of the total effect of *X* on *Y*.

Example: The estimated effect of years of schooling (*X*) on earnings (*Y*) may be downwardly biased if adult neighbourhood (*C*) or post-schooling occupation (*M*) is statistically controlled.

Tentative solutions

- Clarify the causal model to decide which variables (not) to control for.
- Do not merely control for all covariates in your dataset.
- Actively discuss assumptions under which your estimate approximates a causal effect.

Literature suggestions

- Achen[101]
- Cinelli et al.[30]
- Elwert and Winship[32]
- Greenland et al.[27]
- Hoyle et al.[29]
- Wysocki et al.[31]

rely on third-variable adjustment (as opposed to well-implemented design-based approaches[1]). In the study of human behaviour, individuals may anticipate what they expect to happen, and many behaviours or emotional reactions[20] precede the anticipated events. For longitudinal research on causal mechanisms affecting human behaviour, anticipated events may confound prospective associations between psychological variables such as happiness and subsequent outcomes.

One may argue that, as long as researchers control for 'major' sources of confounding, they can obtain relatively accurate causal estimates. In principle, for some types of data and some kinds of research questions, major confounders may be limited in number, but we suspect that this is rarely the case for observational studies of human behaviour, as it is often complex and influenced by numerous factors. At the least, there is no way to prove that all sources of confounding have been controlled for. There have been attempts to identify most relevant confounders using a robustness check approach[21]; however, measured confounders are often limited by design. As can be seen in various formulas that quantify the bias due to multiple omitted confounders[22], the cumulative impact of those omitted confounders (including their higher-order interactive effects) could indeed rarely be ignorable. This may explain some of the widespread contradictions between findings from experimental and observational studies across fields[5,23,24].

In addition, even when confounders are known, the quality of their measurement can limit causal inferences. For example, when health recommendations change, the associations between behaviours labelled as potentially harmful or helpful (for example, vitamin consumption) and other health-related behaviours (for example, smoking) and outcomes (for example, heart health) sometimes change as well, which suggests that more health-conscious individuals are more likely to respond to health recommendations[25]. But statistically controlling for measures of demographics, socio-economic status and other health-related behaviours is not sufficient to eliminate the time-varying association between labelled behaviours and health-related outcomes. This suggests that 'health-consciousness' cannot be sufficiently captured by these measures—and thus, we cannot fully adjust for them.

Finally, even if researchers include sufficient measures of the right confounders, causal estimates can still be biased if the model is misspecified—for example, if it fails to account for measurement error or inadequately captures the function through which changes in the confounder affect the causal variable of interest and the outcome[26].

BOX 5

Effect heterogeneity

The effect of a treatment *X* on the outcome *Y* may differ between individuals, time points, contexts and other conditions. Average treatment effects provide estimates for central trends that may fail to describe many of the individuals in a heterogeneous population. Systematic sampling and systematic analysis of heterogeneity can enrich our understanding of causal effects in heterogeneous populations.

Tentative solutions

- Systematically sample heterogeneous populations.
- Use statistical methods that detect heterogeneity.
- Whenever possible, test assumptions of homogeneity before applying them.
- Actively discuss assumptions under which your estimate approximates an average causal effect.

Literature suggestions

- Athey and Imbens[102]
- Bryan et al.[34]
- Geng et al.[103]
- Gische et al.[80]
- Moeller[104]
- Montoya et al.[59]
- Pearl and Bareinboim[105]
- Wager and Athey[106]

Over-control

A different sceptic might argue that a reported smaller-than-expected estimate of the effect of earlier happiness on later health has resulted from statistical over-control. Over-control happens when researchers (wrongly) control for the variables that are the causal consequence of the causal variable of interest, resulting in post-treatment bias[27–29]. Post-treatment bias can arise from conditioning on mediators (variables caused by the causal variable of interest that affect the outcome of interest) or colliders (variables influenced by the causal variable of interest and the outcome).

Using the previous example of the causal effect of happiness on health, imagine that a researcher controlled for sleep quality (assessed at the same time as happiness). Statistical control for sleep quality may reduce the estimated effect of happiness on health. But if happiness truly causally influences sleep quality, which in turn influences health (rather than sleep quality influencing both happiness and health), the researcher has controlled for a mediator—a pathway through which happiness might affect health. Thus, they underestimate the true effect of happiness on health. The issue is further complicated because sleep quality may also be a confounder that affects both happiness and health; thus, if we do not adjust for it, we may overestimate the true effect of happiness on health. It is thus unclear whether it is a 'good' or 'bad' control[5,30,31] because of uncertainty about the causal order of variables: plausibly, sleep quality is both a confounder and a mediator and thus cannot be used to recover the effect of happiness on health without additional information.

Furthermore, if the researcher controls for a factor plausibly influenced by both health and happiness, this may result in collider bias[32]. Such 'control' can even occur without explicit statistical control if the sample is selective. If both happiness and health make it more likely that an individual participates in the study, study participation becomes a collider. In such a scenario, even if happiness causally affects health, one may fail to detect such an effect or underestimate its magnitude.

A well-known real-world example of collider bias comes from research using administrative data from police stops to estimate the magnitude of racial discrimination in police officers' use of force. These studies are based on the following logic. If officers are more likely to use force in otherwise similar situations (with regard to suspect behaviour) on members of one group than on members of another, this would be a sign of discrimination. Discriminatory use of force implies a causal effect of the suspect's perceived race (*X*) on the use of force (*Y*) that is not mediated via suspect behaviour (*M*). However, in reality, perceived race directly influences officers' decisions to stop suspects in the first place. Being stopped is thus a collider, influenced by both perceived race (*X*) and criminal activity (*M*). Members of the racial group discriminated against may thus be on average less likely to be engaged in criminal activity when they are stopped. Paradoxically, in the presence of discriminatory stopping but not discriminatory use of force, we might expect a lower rate of use of force against the group stopped more, conditional on being stopped. Thus, relying on data from stops alone to attempt to adjust for suspect behaviour may lead us to underestimate racial discrimination in police officers' use of force[33].

Effect heterogeneity

Many popular causal inference techniques from observational data are useful for estimating the average causal effect. A focus on average causal effects is convenient because they provide a simple summary of causal effects, averaged across time and individuals within a sample; and under certain circumstances, which need to be tested, they allow for generalizations to the population. Furthermore, averaging can be useful to estimate causal effects, because counterfactuals for a specific individual cannot be directly observed. However, people are unique and change over time[34,35], and such heterogeneity is easily overlooked if only a single average treatment effect is examined. A failure to appropriately model the heterogeneity of causal effects across covariates, across individuals or across time can even lead to biased estimates of an average causal effect under some reasonable conditions[36–38], particularly in the absence of a randomized experiment.

A practical implication of effect heterogeneity is that it makes the interpretation of a new finding that conflicts with the previous literature challenging. In the absence of a strong causal design, researchers cannot tell whether the unexpected finding originates from bias (a lack of internal validity), heterogeneity of the causal effect of interest (a lack of external validity) or random variation. Consider a hypothetical finding where, among individuals with a severe medical diagnosis, optimism is more strongly predictive of subsequent mortality than in other populations. A possible explanation for this finding is that the causal effect of optimism is moderated by one's condition, such that it is particularly important to stay positive when one's health is compromised. Another is that there is no such differential effect of optimism on health in this population, but rather unmeasured prognostic details in the patient population affect both current levels of optimism and later mortality. The theoretical underdetermination of why conflicting findings arise complicates the process of attempting to make cumulative progress in the study of human behaviour.

Interference

Many causal inference techniques assume the absence of interference (also referred to as spillover effect or contamination), meaning that the potential outcome of one individual is not affected by the causal variable of interest for other individuals[39]. However, when observational data are collected in a sample of people who could potentially interact with each other (for example, data from school or other social settings), there is a good possibility that the assumption is violated, in that the treatment of interest affects not only treated individuals but also untreated individuals used to approximate the counterfactual.

For example, if we attempt to estimate the effect of years of schooling on later earnings by comparing children with more years of schooling to their siblings with fewer years of schooling (whose outcomes

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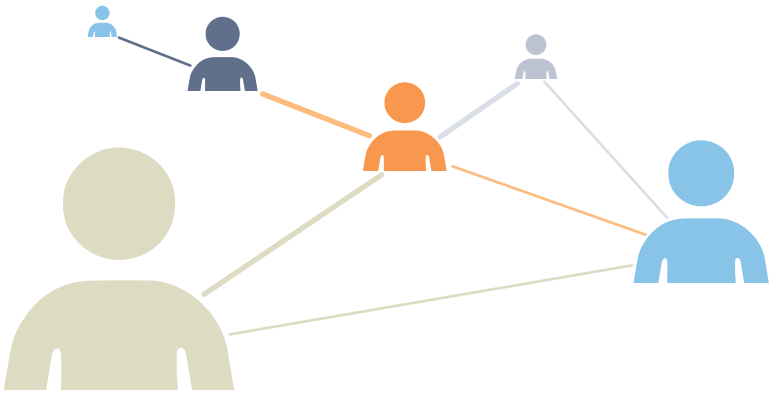
BOX 6

Interference/spillover



A standard assumption of most approaches to causal inference is that there is no interference or spillover. That means that the potential outcome of an individual is not affected by the treatment status of other individuals. However, in some practical applications, individuals participating in a study do interact or communicate with each other, which increases the risk of potential spillovers.

Tentative solutions

- Consider analysis- or design-based solutions.
- Conduct robustness checks with subgroups or settings under which spillovers are unlikely.
- Consider alternative estimands that explicitly take interference into account.

Literature suggestions

- Benjamin-Chung et al.[107]
- Hudgens and Halloran[108]
- Imai et al.[109]
- Tchetgen and VanderWeele[110]
- Zhang et al.[111]

may approximate the counterfactual had their siblings not received additional schooling), an important assumption is that one's earnings are not affected by one's sibling's schooling. However, research on the effect of people's education on their siblings' subsequent educational choices or performance suggests that spillovers are common—siblings' educational experiences appear to affect each other[40–42]. If individuals influence each other, an individual influenced by the causal variable of interest, X, subsequently influences others with different values of X, complicating the accurate estimation of causal effects on Y; associations between X and Y are likely to be biased even when they are adjusted for confounders.

Timescales of effects

We have already argued that effects can vary across individuals, and average and individual effects can also vary substantially across timescales. An estimate of an intervention effect may be specific to the time at which the intervention occurs and when the outcome is measured[35]. Having an alcoholic drink is likely to increase one's self-confidence for a short period; not finding an effect on self-confidence measured a week (or even years) after having a drink does not indicate that alcohol does not influence self-confidence. The effects of a different intervention (for example, the effects of parenting on subsequent illegal behaviour), however, may only show up after long periods; conversely, daily diary or ecological momentary assessments would not be useful to assess such effects directly, although they may offer insights into some of the intervening processes. How to connect the various timescales, and how to make sense of this, poses a major challenge[43], particularly when behaviours affect each other bidirectionally[44].

Fat-handed and other complex interventions

Returning to the question of the effects of happiness on health, let us assume that a researcher was able to address these issues and obtained an unbiased causal estimate of the effect of a year-long sustained increase in happiness on various health outcomes years later. Using this estimate to gauge the benefits of various interventions to increase health still would not be straightforward. The problem is that there are different ways to change happiness (for example, a self-help workshop, exercise or a magic happiness pill), and the hypothesized causal consequences of happiness on health could be different depending on the intervention implemented. One school of thinking about causality refers to this as a lack of consistency; in this line of reasoning, asking for the effects of happiness on health would not be a well-defined question to begin with, as happiness is not (directly) manipulable[45,46]. Another school of thinking about causality maintains that such effects can be well defined[47], but that still leaves it open that the intervention used to induce happiness interacts with the effects of happiness on health. In any case, moving from causal effects of variables that cannot be directly manipulated to interventions would not be straightforward.

Moving the other way around—from interventions to conclusions about the effects of variables that cannot be manipulated directly—is equally challenging. For example, to estimate the effect of earlier happiness on later health, perhaps a researcher conducts a randomized

BOX 7

Fat-handed and other complex interventions

Sometimes interventions alter not only the treatment *X* but also other variables. If we intervene on *X* to see whether it is a cause of *Y*, an intervention that not only changes *X* but also changes some other causes of *Y* (that are not on the path from *X* to *Y*) is called a fat-handed intervention. For example, an intervention on a person's daily diet will change the daily calorie intake (treatment) but might also change the daily intake of vitamins (other variable). Both calorie intake and intake of vitamins affect the body mass index (outcome). The intervention on daily diet is thus fat handed.

Tentative solutions

- Consider triangulating across various interventions with different likely side effects.
- Measure and report potential side effects of your interventions.
- When possible, intervene on variables that are well defined and isolatable from other variables.

Literature suggestions

- Eberhardt and Scheines[112]
- Eronen[49]
- Mooij et al.[113]
- Peters et al.[114]
- Scheines[50]
- VanderWeele[26]

BOX 8

Timescales of effects

The latency of an intervention on an effect may be unclear. If the effect of *X* on *Y* has a latency of several years, it will not show when the time interval between measurements is just a few months. Moreover, effects may be non-stationary, meaning they can change over time, whereas most estimates of time effects assume stationarity and overlook the fact that over time, an effect can emerge, speed up, slow down or cease to exist.

Tentative solutions

- Consider the potential timescale of effects before data collection to measure causes and effects at appropriate intervals.
- Consider continuous time modelling.

Literature suggestions

- Aalen et al.[115]
- Driver and Voelkle[116]
- Røysland[117]
- Ryan and Hamaker[118]
- Voelkle et al.[56]

controlled trial of an intervention that targets participants' happiness directly, but not their health. In this case, the impact of the intervention on later health could provide a strong test of the causal effects of happiness on health (in such a design, assignment to the intervention would be an 'instrumental variable' for estimating the effect of happiness on health[48]). However, this would depend critically on the assumption that the intervention did not also directly influence other factors (for example, health-related motivation) that increase later health (in the instrumental variables framework, this is called the exclusion restriction assumption). Interventions that violate this assumption and thus change variables beyond the causal variable of interest are labelled 'fat-handed' interventions[49,50]. Interventions that target psychological constructs such as happiness are extremely likely to be fat-handed: how could one intervene on just happiness without also changing related psychological constructs, such as positive affect or elevated mood? One source of this problem is that constructs in the behavioural sciences are often unclearly or ambiguously defined, or conceptually overlap with other constructs[51], making them challenging for causal inference[52]. This makes it difficult to test the causal effects of such variables even when randomized controlled trials are feasible (for example, did the intervention affect health via happiness or via frequently elevated mood?). Such effects of constructs (rather than well-defined interventions) also have important practical implications: if happiness is hypothesized to mediate the effects of an intervention on health over a long period, it is far more efficient to optimize iterations of the intervention to generate larger impacts on happiness than on health. However, if the wrong mediator is identified, this strategy may not work.

Summary and implications of challenges

The major challenges above all come down to human behaviour being embedded in complex causal systems of which we can observe only a limited part. This leaves researchers in a difficult position. Sceptics with strong priors on either side of a scientific disagreement can offer plausible reasons why any study or set of similar studies presenting evidence against their position should not convince them of the opposite view. Observational studies under-control (or over-control) for various factors influencing the outcome, experiments directly influence more (or less) than the causal variable of interest and causal effects could apply to a small subgroup or no individual at all, limiting the usefulness of such estimates.

Tentative solutions

Given the challenges to drawing causal conclusions from longitudinal data outlined above, what can be done? Recent years have brought many potential (if partial) solutions. For under-control, there are methods that attempt to control for unobserved time-invariant confounders with the help of longitudinal data[53–55]. We may be able to get a more informative causal estimate for different timescales using continuous time modelling[56]. Effect heterogeneity in longitudinal data could be partially tackled using methods that stratify on post-treatment variables[57], through the estimation of nomothetic effects in bottom-up procedures by empirically identifying those patterns that generalize across person-specific effects (for example, group iterative multiple model estimation[58]) or through dynamic treatment regimes[59]. Methods for estimating precise causal effects for various data structures such as panel designs, intensive longitudinal data and neuroimaging have also been proposed[60–63].

However, in the absence of a strong causal design, it is often difficult to establish whether such solutions have generated unbiased estimates. The capacity of causal inference from observational data is substantially determined by how we design the research, not solely by advanced statistical techniques. We offer several potential approaches below.

Clarifying the research question

Perhaps the lowest-hanging fruit for improving inference is to improve the clarity of the research question, which may often involve narrowing its scope. Such clarity helps readers, but most importantly the researchers themselves, to determine (1) whether the purpose of the study is to estimate causal effects at all, (2) which kinds of effects the researcher is interested in[64–66] and (3) whether the question asked can be answered by the data at hand.

An important example that demonstrates the need for well-defined research questions comes from the Many Analysts Project[67]. The authors asked researchers to reanalyse a dataset to answer the question of whether soccer referees are more likely to give red cards to players with a dark skin tone. Worryingly, research teams obtained a wide range of effect sizes using the same data. However, Auspurg and Brüderl[68] observed that research teams appeared to interpret the research question differently, with some teams attempting to answer the descriptive question of whether players with a darker skin tone were more likely than lighter-skinned players to receive a red card, but other research teams inferring a more complex research question about the causal mechanisms through which such a difference might occur: "the *direct causal effect* of skin tone that remains after netting out confounders and 'productivity-relevant' mediators"[68]. When analyses were restricted to the latter category, the variability of estimates was much narrower.

In the previous case, the study would have benefited from more specificity in the research question. A clearly specified cause, effect, timescale and population of interest are necessary to provide credible estimates of causal effects. At the same time, it is important to keep in mind that the research question of interest may be broader, with individual estimands only providing partial answers. Likewise, there is still a space in science for backward causal questions (such as 'Why are some people healthier than others?') that often motivate the quest for answers to narrower forward causal questions. Such questions have an important role in human thinking and decision-making and can be useful for checking the assumptions of causal models and revising theories (Gelman and Imbens[15] discuss them and give examples). For example, noticing patterns across space and time in which people become ill may be useful for forming theories about why some people are healthier than others, which can be further tested with stronger causal designs. Furthermore, a researcher who has considered the backward question, 'Why do students in the same school frequently have such different levels of academic achievement?' might be less likely to mistakenly attribute test score differences to the causal effects of specific school or teacher characteristics in the absence of a strong set of baseline statistical controls. Considering more general questions is useful for thinking about generalizability, mechanisms, alternative hypotheses, and matches between data and model assumptions and thus should be considered in service of asking better forward causal questions and for better assumption-checking when trying to answer them.

Choosing the right design and variables

Once a causal question is set, we need to think carefully about the right design to permit appropriate causal inference[69]. This includes the decision about potential controlling variables to be measured, the time interval of measurement, the number of time points, sample characteristics, sample size and so on. Tools that help with such decision-making processes are increasingly available[7]. Researchers must pay careful attention to the measurement of constructs in the study of human behaviour (for example, depression and poverty), which pose unique challenges related to reliability and validity, along with conceptual challenges (for example, whether measurements are better understood as effects or causes of the construct of interest), all of which can affect causal inferences[26,70,71].

Actively discussing the causal informativeness of a model

We argue that observational work should be discussed in terms of its implications for causal theories, where the causal implications are judged in a continuous, not a binary manner. A current norm in many fields studying human behaviour is to deliberately avoid making causal statements if the study lacks a randomized experimental design[18,72]. Although researchers should attempt to appropriately calibrate the strength of their claims to the strength of the evidence, failing to consider the causal implications of one's model (even a tentative one) may make it easier to overlook indications that assumptions are violated while probably conveying a causal interpretation to the reader anyway[16,17]. For example, consider a hypothetical study that regresses earnings on measures of previously measured cognitive test scores, social skills and demographic controls and interprets the implications of such findings for the (presumably causal) 'importance' of cognitive and social skills. Although the analysis is at risk for bias from all the potential factors listed in the previous section, the analyst might dismiss a critique about potential bias from the exclusion of other potential confounders from the model under the grounds that the analysis uses non-experimental data and is thus not 'causal'. However, this argument conflates the goal of the study (causal estimation) with the degree of certainty the method affords in its findings, paradoxically missing an opportunity to improve the analysis. Researchers should be able to communicate the goal of the study (for example, causal estimation) and the certainty with which estimates are causally informative (for example, low in an observational, cross-sectional setting; higher in a setting that rests on fewer assumptions) separately, rather than lumping them together under the single term 'causal' (causal estimation with high certainty). Presenting the study as a rough attempt at bounding causal estimates of the returns to cognitive and social skills would allow researchers to convey a great deal of uncertainty while still directly addressing the causal questions at hand. An approach to rule out different threats to causal inference of estimates could be to use a range of estimation strategies. Deming[73], Pion and Lipsey[74], and Ritchie and Tucker-Drob[75], for example, compare results across multiple identification strategies.

Triangulating on causal estimates

Clarifying research questions is a necessary but far-from-sufficient solution to identifying causal effects of constructs, particularly when only fat-handed interventions or observational data are available. How might a researcher interested in the causal effects of such a construct proceed, knowing that any finding can be dismissed for statistically over- or under-controlling or including an overly broad or overly narrow experimental manipulation, depending on the observed effect? Like others before[76–78], we argue that successful examples from the study of human behaviour triangulate a causal effect (or set of causal effects) of interest by considering multiple findings using various identification strategies and reasoning through them in the context of tentative theories and assumptions (Fig. 1). Triangulation may happen on different levels, across studies in the field as a whole, within a particular research programme (and thus potentially across studies) and within individual studies.

Researchers may consider two types of sources of evidence: (1) strong causal estimates of well-defined (but probably fat-handed) interventions (broadly defined, including both applied interventions and exogenous manipulations of behaviour designed solely for the purpose of improving causal understanding) and (2) non-experimental estimates from observational data. Estimates should be compared with each other whenever possible and interpreted considering theory and assumptions. Combinations of theories and assumptions that accommodate existing estimates are re-examined, with a focus on research questions and designs that can plausibly distinguish among them[78,79]. We describe an example below.

A researcher interested in the effects of a complex or fat-handed intervention on a set of outcomes might approach the problem from several different perspectives. For example, a researcher interested in the effects of cognitive skills on later educational and occupational success might be interested in this problem so that they can (1) better understand the causal dynamics between cognitive skills and educational inputs and/or (2) design interventions to better prepare students for educational or occupational success.

A researcher following the first approach might consider a large longitudinal dataset (Fig. 1, left box) and use recent advances in longitudinal data analysis techniques to address confounding via matching

and latent variable modelling[56,80–82], justifying the use of statistical controls from a causal inference perspective. They might test whether their conclusions are robust over variations in model specification[83,84] and assumptions about how cognitive skills produce variation in test scores[70,85]. They might attend to whether estimates are similar across a range of subgroups and time periods as well.

In many cases, estimates will vary, sometimes predictably across specifications (Fig. 1, bottom arrow). Even if estimates fall within a narrow range of values, they might differ across individuals, subgroups, measures and outcomes. The researcher might attempt coherent pattern matching[69], where in the absence of an intentional intervention, the researcher tests a list of theory- and assumption-driven predictions and calibrates the strength of the match between theory and results. The researcher might use robustness checks or falsification tests to attempt to quantify the amount of remaining bias in the estimate of interest. For example, concerned about bias in the estimated effects of corporal punishment on children's later antisocial behaviour, Larzelere and colleagues[86] compared estimated effects of corporal punishment to the estimated effects of other interventions not hypothesized to increase antisocial behaviour (for example, grounding or psychotherapy). Finally, the researcher might include some bounding exercise that allows for a lower bound of plausible estimates[87], such as a sensitivity analysis testing the magnitude of unobserved confounders necessary to produce the estimated effect[22].

In contrast, a researcher who is interested in developing an effective intervention on later educational or occupational success might approach the problem by reviewing the literature on field experiments or strong quasi-experimental evaluations of exogenous factors found to improve students' cognitive skills (Fig. 1, top box). Interventions are likely to vary in their fat-handedness (the extent to which they influence education via pathways other than students' cognitive skills alone). For example, a brief cognitive training intervention might plausibly influence later outcomes primarily via changes to cognitive skills. However, a broader intervention, such as an additional year of schooling, includes exposure to different mentors, peers and other social contexts over an extended period and is much more fat-handed (Fig. 1, right arrow). Thus, the researcher might consider whether within and across categories of interventions, subgroups and outcomes, improvements to cognitive skills are reliably related to improvement in educational or occupational success[88].

Finally, both researchers might attempt to obtain estimates of the effects of cognitive skills on educational or occupational success on the same scale (for example, what is the estimated effect of a hypothetical minimally invasive intervention that influences cognitive skills at age 18 by half a standard deviation on wages at age 35?) so they can be compared (Fig. 1, left arrow). In the presence of an exogenous intervention, the researcher might be able to systematically compare estimates of the causal effect of interest using an experimental design and an observational analysis in the same sample. This method, sometimes called a within-study comparison, sometimes finds converging[89,90] and other times diverging results[91]. Key to attempting to reconcile these estimates is a judgement about the likely net bias of both of them: if statistical controls are poor and interventions are all substantially fat-handed, for example, perhaps the net bias of both kinds of studies will go in the same direction[19].

If estimates differ substantially across methods, several possible explanations should be considered: Are the meanings of outcome scores measured in observational studies qualitatively different from the outcome scores in stronger causal designs, despite sharing the same label and even the same measure[92]? Are the statistical controls in observational studies likely to miss plausible confounders[93]? Are effects heterogeneous across samples and settings, which differ systematically across timescales and designs? If so, are estimates different primarily because they are unbiased estimates of different effects, or because they are estimates of the same underlying effect but are differently biased? Is there a structural model that can reconcile these findings, and if so, what predictions would it make about the results of future observational or experimental studies (Fig. 1, right box)? A large number of potential explanations can account for non-overlapping estimates[76,94]. Importantly, though, triangulating between experimental and observational analyses makes it possible to probe more of them, relative to an approach that views observational and experimental research as irreconcilable approaches to studying fundamentally different kinds of questions[79].

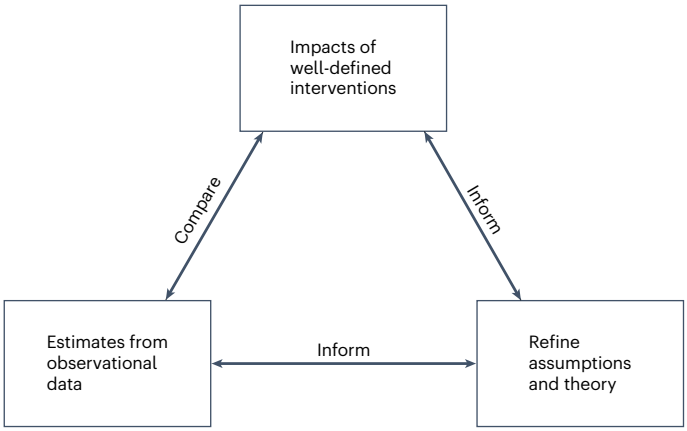


Fig. 1 | A framework for triangulating on causal effects. Designs of intervention and observational studies should be theory informed, and the results of interventional and observational studies need to be compared as well as to be interpreted in light of and to inform theory.

Several scientific practices enacted by researchers, but perhaps incentivized by systems, could enhance the efficiency of this process. In some cases, it may be possible a priori to design a research programme where a research question is well defined and constructs are operationalized sufficiently similarly across studies that estimates are comparable and reconcilable after they are conducted. For example, systematically exploring whether heterogeneity can account for apparently different findings would require larger, more systematic samples across both experimental and non-experimental designs (many such corresponding research programmes have been undertaken in recent years[34,95–97]). Meta-analytic models can incorporate estimates obtained from multiple identification strategies, samples and outcomes to attempt to identify sources of heterogeneity of estimates.

Many of the methods above can be integrated with a structural model—a formal model of behaviour that makes predictions about the results of future experiments[98]. For example, Todd and Wolpin[99] proposed and estimated a formal model of how children choose to attend school each year through high school, using data from the control group of a randomized experiment in which children and families in the treatment group received nutritional supplements and a subsidy conditional on children attending school. They showed that the subsidy increased school attendance by the amount predicted by their model, demonstrating its potential usefulness for both improving our understanding of why children (do not) attend school and forecasting the impacts of potential policies on school attendance in the future. We view such formalizations of causal models that make accurate predictions about the effects of future interventions as a worthy goal for the social sciences to aspire to. However, we note that a useful early step for many social scientists will be to understand that estimates yielded by different analyses can be mutually informative, either because they approximate the same estimand or because they approximate estimands that have a theoretically important relation to each other that can be used to triangulate on the estimand of interest. For example, a researcher studying the effects of different implementations of the same psychotherapy programme might catalogue the features of the

evaluation design, participant and programme features (for example, the baseline symptomology of different samples or the dosage of different implementations), and outcomes (for example, whether they are self-reported or tracked by administrative data) associated with larger or smaller estimated effects. Understanding what set of plausible explanations best account for variation in estimates may be useful for both understanding what features to include in a structural model and guiding future experimental efforts.

Triangulation will not necessarily provide a simple satisfactory answer to the broad question, 'What are the effects of cognitive skills on later educational and occupational success?' or the more specific question, 'What would be the effect of a hypothetical minimally invasive intervention that influences cognitive skills at age 18 by half a population standard deviation on wages at age 35?' But it might yield potentially useful answers to more specific causal questions, along with a broader model for predicting the impacts of future similar interventions. These predictions might be useful for improving the effectiveness of future interventions[80] and individualizing treatments[34,59,100] and for making testable predictions about underlying data-generating processes. Under such an approach, evidence from a wide variety of sources is admissible, and no single study is definitive. Estimates—even for a reasonably well-defined research question—can vary within and across studies for a large range of reasons reviewed above, and more estimates allow for a more comprehensive model of the underlying causal process(es) of interest.

Conclusion

Causal inference is hard, complicated by many logical problems and methodological challenges. Appreciating them is crucial, for we can overcome only the obstacles we are aware of. Yet, causal inference is at the heart of the study of human behaviour. Giving up on the goal of ascertaining causal mechanisms is therefore not an option. Recently, there have been substantial methodological advancements and debates about methods for causal inference. Studies attempting causal inference can now integrate the available tools from different disciplines and triangulate across the theories, methods and findings they have generated; this interdisciplinary dialogue seems vital to us.

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The Limits of One-Size-Fits-All Patterns:
A 5-Step Approach to Testing Homogeneity of Within-Person Correlations

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Abstract

Psychological research increasingly uses situated measures, requiring intensive longitudinal and within-person methods. Such data are typically examined with multilevel models clustering multiple measurement time points within individuals. The intercorrelation patterns obtained with such multilevel models rely on various homogeneity assumptions that should be, but rarely are, tested before using multilevel modeling and interpreting the results.

This study proposes a five-step approach to systematically testing homogeneity assumptions in order to obtain more trustworthy estimates of associations among measures when examining clustered multilevel data.

To provide an educational illustration of this five-step approach, 1,028 experience sampling method surveys from 37 students were collected over one semester in one university lecture. Twenty-one items measured motivational components and academic emotions. Multilevel analyses estimated an overall within-person and an overall between-person network of zero-order correlations among motivation and emotion facets. Idiographic within-person networks of the 37 participants revealed substantial heterogeneity in their within-person intercorrelation patterns. Heterogeneity was further indicated by one third of the within-person correlations being bi- or multimodally distributed across individuals. We conclude that heterogeneity in within-person patterns must be examined, otherwise, methods relying on homogeneity assumptions, including multilevel analyses, may fail to describe many if not all individuals' idiographic correlation patterns. This is crucial for the trustworthiness of measurement and structure models, the generalizability of ESM research, and any research aiming for personalized diagnostics and interventions.

Keywords: Experience sampling method, academic emotions, situated expectancy-value theory, control-value theory, heterogeneity, intra-individual analyses, bottom-up generalizability, multimodality

The Limits of One-Size-Fits-All Patterns:

A 5-Step Approach to Testing Homogeneity of Within-Person Correlations

Introduction

Intensive longitudinal data are increasingly often used in research on emotions and motivation to study their context- and time-specific aspects (Dietrich et al., 2022; Hamaker & Wichers, 2017; Moeller et al., 2023). Research on situated learning-related motivation and emotion mostly relies on the analysis of within-person correlations among emotions, motivational components, and their predictors, correlates, and outcomes. Unnoticed by many, one recent study has raised crucial doubts about the trustworthiness of the common methods used to estimate such within-person correlations: Haslbeck et al. (2023) showed that measures of emotions were often bi- or multimodally distributed in several intensive longitudinal datasets. This discovery has crucial implications for research using intensive longitudinal data: For variables that are bi- or multimodally distributed, their bivariate associations may be ill-described with covariance-based statistics. Similarly, their univariate distributions may be ill-described with mean scores and standard deviations, and be better described with mixture distributions, or at least statistics that are robust against violations of unimodality assumptions.

Understanding multimodality and other forms of heterogeneity is particularly relevant for the work with intensive longitudinal data, because of their manifold complex sources of variation and boundary conditions (e.g., Moeller et al., 2025). Further reasons why appropriate methods capable of detecting such heterogeneity should be considered have been introduced during the debates about the heterogeneity revolution (Bryan et al., 2021) and about the generalizability crisis (Yarkoni, 2022a; 2022b). A crucial take-home message of these debates has been: Many psychological phenomena may be heterogeneous for many reasons, and our typical statistical methods rely on homogeneity assumptions that might not be realistic. The implication is that many statistical coefficients in research on emotions and

motivation may be untrustworthy. Fortunately, there is a simple solution to this threat: To test the homogeneity assumptions required by a statistical procedure before choosing this procedure. Unfortunately, many studies currently ignore such homogeneity assumptions, neither testing nor reporting on them, while habitually choosing statistical methods relying on these untested homogeneity assumptions. The present study aspires to help changing that. We propose a five-step approach (see Table 1) to testing the homogeneity assumptions that need to be tested before Level 1 correlations in a multilevel model can be considered trustworthy. These homogeneity assumptions need to be tested on various levels of analysis, namely within and between persons and with regard to univariate, bivariate, and multivariate patterns. While we developed this 5-step approach for intensive longitudinal data with multiple measurement time points nested in individuals, the approach can be adapted to other data with a 2-level structure, such as students nested in classes at school, for instance.

The study we report applies this 5-step approach to research on academic emotions and achievement motivation. This research area increasingly focuses on situational and contextual fluctuations and determinants (e.g., Dietrich et al., 2022; Eccles & Wigfield, 2020; 2024; Pekrun, 2006, 2024; Pekrun & Marsh, 2022). These are often studied with the experience sampling method (ESM; e.g., Dimotakis et al., 2013; Myin-Germeys & Kuppens, 2022; Van Berkel et al., 2017). Building upon recent work by Moeller, Viljaranta, et al. (2022), we examine associations among ESM measures of the situated expectancy-value theory (SEVT; Eccles & Wigfield, 2020) together with ESM measures representing control-value aspects of academic emotions (CVT; e.g., Pekrun, 2006, 2024a; 2024b).

Table 1

Testing Homogeneity Assumptions on all Levels: Univariate, Bivariate, Multivariate, Within- and Between Person

Step	*What to ask?*	*How to test?*	*What to conclude?*	*Relevant for what?*
Step 1	Is each item distributed unimodally within each person, across measurement time points? **univariate, within-person**	**Unimodality tests** – e.g., visual inspection, – Hartigan's dip test, – Silverman's test, – the bimodality coefficient	– If bi- or multimodal: use robust statistics to estimate this person's measure(s) of the central tendency (e.g., median), or describe the heterogeneity of that distribution. – If unimodal, (and if other assumptions are met), use the average (or ML estimate).	Needed for trustworthy calculations of each person's measure of central tendency (e.g., average, median) of an item over time, and any statistics building on that, including within-person correlations, which require unimodal items or robust statistics.
Step 2	Is the within-person aggregate of each item (aggregated across time points) distributed unimodally across individuals? **univariate, between-person**	**Unimodality tests** *see above*	– If bi- or multimodal: use robust statistics to estimate this person's measure(s) of the central tendency (e.g., median), or describe the heterogeneity of that between-person distribution. – If unimodal, (and if other assumptions are met), use the average (or ML estimate).	Needed for trustworthy calculations of each person's measure of central tendency (e.g., average, median) of an item across individuals, and any statistics building on that, including between-person correlations, which require unimodal items or robust statistics.
Step 3	Is each pairwise within-person correlation distributed unimodally across individuals? **bivariate, between-person**	**Unimodality tests** *see above*	– If bi- or multimodal: use robust statistics to estimate this sample's measure(s) of the central tendency (e.g., median), or describe the heterogeneity of that between-person distribution. – If unimodal, (and if other assumptions are met), use the average (or ML estimate).	Needed for the generalizability & meaningfulness of bi-variate inter-correlation patterns, including within-person correlations and all bivariate coefficients built on them (incl. regression coeff., paths in structural equation models, edges in correlation-based networks, etc.).
Step 4	Are the networks of within-person correlations similar across individuals? **multivariate, between-person**	e.g., **Multi-sample test** for high-dimensional covariance matrices by Zhang et al. (2018)	– If the idiographic networks turn out to not have originated from the same population, then methods for describing heterogeneity in multivariate correlation patterns are needed. – The aggregated within-person network resulting from typical multilevel analyses may not generalize to all, or even to any, of the individuals in the sample.	Generalizability of multi-variate inter-correlation patterns, including multivariate interpretations of networks and structural equation models.
Step 5	Do the within-person correlations and the between-person correlations lead to the same conclusions? **Bi- or multivariate, within versus between-person**	Disentangling within-person and between-person covariance using **multilevel modeling**	– If the within-person network differs from the between-person network, then the phenomenon is not ergodic; – In that case, it is crucial to test within-person hypotheses with within-person methods and to not generalize between-person findings to within-person theories or vice versa.	Generalizability of bi- and multi-variate correlation patterns from one level of analyses to another (within-person to between-person or vice versa), including interpretations of networks and structural equation models.

To Determine Whether Nomothetic Methods of Generalization Can be Used, Their Homogeneity Assumptions Must be Tested

Except for single case studies, intensive longitudinal studies typically use statistical methods that originate from a nomothetic tradition and therefore rely on many homogeneity assumptions. For instance, ESM studies typically use data with many measurement time points nested in individuals and employ multilevel analyses to estimate one overall (nomothetic) within-person relation (Level 1) and one overall (nomothetic) between-person relation (Level 2). In our study of correlations among motivation and emotion measures, this means that one overall within-person correlation is estimated on the within-person Level 1. This overall estimate of a sample-level within-person fixed effect represents a between-person aggregate across all individual's within-person correlations (random effects). Such aggregates representing one-size-fits-all overall estimates in multilevel and other models typically assume that the individual effects are more or less normally distributed across individuals, with one mode (one hill in the distribution). Such unimodality is a necessary requirement for an average or a similar one-size-fits-all coefficient to represent the trend (i.e., in statistical terms: "the central tendency") of that distribution. In contrast, in a heterogeneous mixture distribution with more than one hill (i.e., a bi- or multimodal distribution), a one-size-fits-all coefficient not only fails to describe that heterogeneity, but may fail the majority or even everyone in that sample. In intensive longitudinal data, heterogeneity is likely, since many sources of variation can generate heterogeneity, including differences between time points, differences between contexts, differences between individuals, and differences between larger groups of individuals and their cultural background, nationalities, etc. (see Moeller et al., 2023; Moeller et al., 2025).

Since intensive longitudinal data typically include these multilayered and complex sources of heterogeneity, they are therefore particularly likely to violate homogeneity assumptions, compared to many other types of data that keep many sources of heterogeneity constant (e.g.,

lab experiments) or fail to include them in the data collection (e.g., cross-sectional data). Thus, the homogeneity assumptions that underlie the analyses typically applied to multilevel models can be unrealistic.

In order to correctly interpret the multivariate covariance patterns in intensive longitudinal data, homogeneity assumptions, such as assumed unimodality, should be tested in univariate, bivariate, and multivariate distributions both with regard to within-person and to between-person distributions. An overview of these required homogeneity tests and their relevance for the choice of analytic methods and interpretations of results are depicted in Table 1. For a two-level model, we suggest that five consecutive homogeneity tests are needed for any trustworthy interpretation of covariance patterns.

Steps 1 and 2 examine homogeneity assumptions underlying the interpretation of descriptive item statistics such as mean scores, separate for within-person distributions (Step 1) and between-person distributions (Step 2). If the within-person distributions of a person's item scores followed a unimodal form (Step 1), then one could assume homogeneity across the different situations experienced by this individual, with some variation around an arithmetic mean. If, however, in an extreme case that distribution had peaks at the lower end and at the upper end of the scale, the mean calculated from this distribution would not be very informative as a central tendency (Derrible & Ahmad, 2015). The same logic applies to the between-person level (Step 2), where a unimodal distribution of within-person coefficients is a requirement for between-person aggregates. For example, the estimation of each item's intercept in multilevel modeling (i.e., each item's mean score on the between-person Level 2) assumes that the individual item scores (i.e., individuals' trends across time points) be distributed unimodally across individuals. If the item was distributed multimodally across individuals instead, then the between-person average would not be a suitable measure of this item's central tendency, and the distribution will then be better described using different statistics (e.g., median, modi, skewness, statistics for mixture distributions, etc.).

Step 3 then addresses the estimation and interpretation of bivariate within-person associations. As one approach to determine the best method to obtain generalizable results about within-person associations, we suggest to empirically test the assumption that these within-person correlations are unimodally distributed across individuals, which is the condition that must be met for an average or maximum likelihood estimate to represent the overall sample-level trend (central tendency) of the between-person distribution of these individual within-person correlations.

Step 4: While Step 3 considers each bivariate association separately, it leaves unanswered whether the entire system of multivariate associations is homogenous across individuals (see e.g., Moeller, 2021; Molenaar, 2004; Reitzle, 2013). In other words, do the individuals in the sample function in a similar way? In the literature on intensive longitudinal data, systems of associations are often examined in network analyses (e.g., Borsboom et al., 2021; Jähne et al., 2024). Consequently, Step 4 tests if the idiographic within-person correlation networks originated from the same homogeneous population. If that homogeneity assumption was supported, then the common method of reporting one multivariate network of within-person correlations for the entire sample would represent an appropriate method of generalization. If, in contrast, the idiographic multivariate correlation networks did not originate from the same homogeneous population, then there would be a need for different methods for summarizing the generalizable and the non-generalizable aspects of the networks across individuals.

Step 5 tests the equivalence of within-person and between-person bivariate and multivariate correlation patterns, because it frequently happens that effects, including bi- and multivariate inter-correlation patterns, differ between the within-person level and the between-person level; a problem known as Simpson's Paradox (Simpson, 1951) or lacking ergodicity (e.g., Molenaar, 2004). Lack of equivalence of the within- versus between-person

patterns would limit the generalizability of any pattern found between individuals to the within-person level and vice versa (see e.g., Moeller, 2021; Molenaar, 2004).

Whether Homogeneity is Confirmed or Not has Implications for Choosing the Correct Bottom-up Versus Top-down Procedure of Generalization

There are many methods to derive generalizable conclusions. Some of these methods assume that the variables under study be relatively normally distributed, with one peak (mode) of that distribution. Multilevel models can be considered top-down methods of generalization: At the within-level the same model is imposed for each individual (Hamaker et al., 2021; Reitzle & Dietrich, 2019). Between-person differences (random effects) in the strength of associations are allowed to vary around an average (the fixed effect).

Other methods acknowledge that some variables are not normally and unimodally distributed, and instead follow heterogeneous mixture distributions (for supporting arguments, see Bryan et al., 2021; Gelman, 2015; Haslbeck et al., 2023). In the latter cases, bottom-up methods of generalization, that determine which pattern (i.e., form of distribution) was observed how often in the sample, may be more useful (see e.g., Asendorpf, 1993). Bottom-up generalization asks which individual (idiographic) patterns can be found, how often these patterns can be found, and whether they can be summarized with a one-size-fits-all coefficient or with alternative statistics describing their heterogeneity. This can lead to conclusions about homogeneity, if for instance a normal distribution with one peak is found, or to conclusions about heterogeneity, if multiple groups with different patterns are found (see also Kuper et al., 2024 and Lane, Gates, et al., 2019).

Applying the Five Steps to Academic Emotions and Achievement Motivation

Achievement motivation and academic emotions are fluctuating experiences that are influenced by characteristics of learning contexts and time (e.g., Dietrich et al., 2022; Eccles & Wigfield, 2020; Pekrun, 2024a; Pekrun & Marsh, 2022). This situated nature of emotional and motivational states has been studied with regard to the control-value theory of academic

emotions for a long time (e.g., Goetz & Hall, 2012; Pekrun, 2024; Pekrun et al., 2002; 2007; 2023), whereas the focus on situation-specificity is rather new in research on achievement motivation (situated expectancy-value theory of achievement motivation; Eccles & Wigfield, 2020). The control-value theory of academic emotions (CVT; Pekrun, 2024) posits that students' emotions in academic settings or situations can be influenced by their perceived control over an educational task and learning activity and the subjective value they assign to these tasks or activities. The CVT highlights the role of these appraisals in creating and shaping emotional experiences such as enjoyment, anxiety, and boredom, which in turn affect motivation, learning strategies, and academic performance. The situated expectancy-value theory (SEVT; Eccles & Wigfield, 2020) of achievement motivation posits that students' motivation to engage in educational activities and to perform educational tasks is influenced by their expectations of success and the subjective value they place on these tasks and activities. Both emotional and motivational states are considered crucial determinants of learning (Eccles & Wigfield, 2020; Pekrun, 2024).

Because CVT and SEVT share several similar theoretical foundations (e.g., for overviews, Beckmann & Heckhausen, 2018; Brunstein & Heckhausen, 2018; Pekrun, 2024), expectancy-value appraisals and academic emotions are considered together in the present study (see also Berweger et al., 2022; Lauermann et al., 2017; Moeller, Viljaranta, et al., 2022). In line with the DYNAMICS framework (Moeller, Viljaranta, et al., 2022), which conceptualizes academic emotions and SEVT components of achievement motivation as parts of the same state system, the present study examines their within-person associations. As a tool to display the associations among academic emotions and expectancy-value experiences, network analyses have been proposed (e.g., Moeller, Viljaranta, et al., 2022; Jähne et al., 2024; Kulakow et al., 2024; Lee et al., 2023).

Some authors have argued that such within-person correlation networks might differ between individuals (e.g., Moeller, Viljaranta, et al., 2022, more generally, see Beck &

Jackson, 2020; Lane et al., 2019). Theoretical arguments can be made both in favor of a homogeneity assumption, as in favor of a heterogeneity assumption: The two theories (CVT and SEVT) do not yet provide specific-enough information to derive hypotheses about eventual between-person heterogeneity in within-person patterns. For example, the CVT relies on universality assumptions by proposing basic mechanisms of emotions that are expected to be invariant across persons and contexts (e.g., Pekrun & Goetz, 2024). On the other hand, there are interventions aiming to change the way how people regulate, recognize, label, manage and use their emotions or appraise what they experience, suggesting that the associations between emotions, accompanying motivational experiences, and outcomes, may differ by emotion competences, or by appraisal styles (e.g., Izard et al., 2008; Quoidbach et al., 2015; Troy et al., 2018). For instance, a negative experience such as anxiety may generally be an aversive state accompanied by avoidance motivation, negative valence and high arousal for most people at most times, but that does not mean that people cannot learn to cope with their anxiety, even to benefit and thrive from it, implying that the same feeling of anxiety can feel very different and lead to different outcomes, depending on a person's coping and appraisal mechanisms (e.g., Jones et al., 2009; Meijen et al., 2020; see also Moeller et al., 2015 and Pekrun et al., 2002). Moreover, constructionist theories about emotions suggest that individuals may differ in idiosyncratic ways in how they experience, label, express, and react to emotions (e.g., Gendron & Feldman Barrett, 2009; Gündem et al., 2022). Thus, there are theoretical arguments for both universality assumptions and for heterogeneity assumptions with regard to many aspects of emotional experiences.

Thus far, explicit tests of homogeneity in within-person multivariate networks of affective-motivational experiences are largely lacking. However, some previous research is in line with the homogeneity assumption concerning specific variable intercorrelations (our Step 3). Pekrun et al. (2002) found a normal between-person distribution of within-person correlations between situated anxiety and learning motivation, ranging from -0.5 to 0.5 with

an average of roughly zero. Likewise, roughly normal, relatively unimodal between-person distributions of within-person correlations among situated motivation and emotion measures were reported by Yoshida & Murayama (2013), Malmberg et al. (2013), as well as Moeller et al. (2015). The present study investigated whether such unimodal between-person distributions of within-person correlations can be expected for a larger set of situated emotion and motivation measures.

Research Questions and Hypotheses

This study presents an approach of systematically testing homogeneity assumptions in order to estimate within-person associations in a trustworthy way. The present study focuses on within-person correlations among situated ESM measures of achievement-related emotions and motivation facets. Before such correlations can be estimated and interpreted, homogeneity assumptions need to be tested in various levels for the univariate, bivariate and multivariate distributions and patterns, within and between individuals, as argued above. Testing these homogeneity assumptions step by step is crucial for choosing the appropriate statistical analyses in any subsequent step. In order to provide an educational example of how to test such assumptions, this study demonstrates in which order and depth homogeneity assumptions are relevant to obtaining trustworthy estimates for the multivariate inter-correlation patterns that we ultimately aim to describe. While most ESM studies simply report one within-person and one between-person correlation pattern, the present study demonstrates that obtaining trustworthy estimates for these associations is a more complex multi-step process (see Table 1).

Research questions 1 through 5 therefore address the homogeneity assumptions that need to be tested before within-person correlations can be interpreted in a trustworthy way. Table 2 shows how these research questions systematically address the five steps of homogeneity assumption testing previously introduced in Table 1.

Table 2

How Research Questions of this study Relate to the Proposed 5-Step Approach of Testing Homogeneity Assumptions

Research Question	Steps; for the definition and relevance of the five steps, see Table 1
RQ1	**Step 1** of testing homogeneity assumptions: In univariate within-person distributions
RQ2	**Step 2** of testing homogeneity assumptions: In univariate between-person distributions
RQ3	**Step 3** of testing homogeneity assumptions: In bivariate within-person correlations
RQ4	**Step 4** of testing homogeneity assumptions: In multivariate idiographic within-person inter-correlation networks
RQ5	**Step 5** of testing homogeneity assumptions: By comparing equivalence of nomothetic within-person and between-person correlations

RQ1: Testing Homogeneity in Within-Person Distributions Across Time Points ("Step 1")

RQ1 asks how often the ESM measures of emotional and motivational states are unimodally, bimodally, and multimodally distributed within individuals. Previous studies have recently shown frequent bi-and multimodal distributions of ESM measures of emotions (Haslbeck et al., 2023; Ram et al., 2017). RQ1 tests whether such bi- and multimodality affects our motivation and emotion measures. We then determine for each person and each item the most suitable measure of the central tendency, that is: the mean for unimodal distributions, or the median for bi- or multimodality.

RQ2: Testing Homogeneity in Between-Person Distributions of the Idiographic Within-Person Measures of the Central Tendency of all Items ("Step 2")

Comparable to the previous step, RQ2 asks how often the idiographic statistics of the central tendencies derived in Step 1 (i.e., mean scores or medians) are distributed unimodally, bimodally and multimodally between individuals.

RQ3: Testing Homogeneity in Between-Person Distributions of Within-Person Correlations ("Step 3")

RQ3 addresses homogeneity assumptions underlying the estimation and interpretation of the bivariate correlations between motivation states (SEVT facets) and emotion states (CVT facets) that shall be examined for RQ5. Specifically, we ask how the within-person correlations among the emotion and motivation states are distributed between individuals. Building on the previous work by Malmberg et al. (2013), Moeller et al. (2015), Pekrun et al. (2002), and Yoshida and Murayama (2013), we expect that some bivariate intra-individual distributions of ESM-measured emotional and motivational states are unimodally and normally distributed across individuals. However, it is not known whether or how often such normal unimodal between-person distributions of within-person correlations can be expected among all the emotion and motivation measures.

RQ4: Do the Multivariate Idiographic Correlation Networks Originate From the Same Population (Homogeneity Assumption; "Step 4")?

As a final pre-condition to determining the appropriate statistical method of deriving generalizable intercorrelation patterns, we tested in Step 4 whether the idiographic within-person zero-order correlation networks for the 37 participants originated from the same homogeneous population. For the specific question of homo- versus heterogeneity in multivariate within-person correlation patterns among situated measures of SEVT and CVT components, we lack both specific theoretical arguments and empirical prior evidence. We

estimated correlation networks separately for the associations among emotions, among motivation components, and between emotions and motivation components to reduce complexity.

RQ5: Are the sample-level (nomothetic) associations among and between emotions and motivation components equivalent across the within-person and between-person levels ("Step 5")?

As a last step to examining homogeneity assumptions relevant to a trustworthy interpretation of correlation patterns, research question 5 asks whether the sample-level, that is, nomothetic correlation network differs on the within-person level of analyses versus the between-person level of analysis. Although this is the last step in our five-step approach, it is arguably the more widely accepted and more frequently adopted approach of taking possible heterogeneity into account: Disentangling within-person from between-person (co-)variance is now a common practice amongst researchers working with intensive longitudinal data (see e.g., Voelkle et al., 2014).

Method

The dataset used for this study can be found in Appendix H-6 (open data).

Sample and Procedure

This study used data collected as part of the ManyMoments project collaboration, a large multi-lab collaboration aiming to improve the replicability and generalizability of intensive longitudinal studies. Data were collected in one semester-long lecture series on statistics for undergraduate psychology students at a German University. Students normally attend this lecture in the second semester. This lecture series teaches analysis of variance and regression models. In addition, students learn how to apply the methods in suitable statistical software. The aims of the lecture series are thus to impart skills for the selection, application

and presentation of important inferential statistical methods and the implementation of these methods in statistical computer software.

Thirty-seven students participated in the ESM data collection in return for receiving course credits. The participants provided 1,028 ESM surveys (on average 27.78 per person; minimum 14 beeps per person, maximum 39 beeps per person). The students were on average 21.13 years old (*SD* = 3.21). The lecture was held in the summer semester of 2023. Students received ESM surveys over the course of one semester during the time of one university lecture, with three surveys per weekly lecture. Only students with at least ten fully or partially completed ESM surveys were included in the analysis sample. ESM data were collected with a smartphone app called "ManyMoments" by the company 65Security.cz.

All students received written information as well as an explanation video on how to download and use the app. They also received written information about the privacy policy of the app and of the study as well as information about the study in general. At the start of the semester, they received a study code and a personal code to log in to the specific ESM survey. This guaranteed that only students of that lecture participated in the study and that all ESM surveys appeared at the right time on every participant's smartphone. The participants were informed by an auditory signal that the ESM surveys should be answered now. After the auditory signal the students had 15 minutes to answer the ESM survey but were instructed to answer as soon as possible. It took about two minutes to answer the survey.

Each lecture lasted for 90 minutes. The participants were divided into three groups, across which the ESM survey times were rotated so that group A received ESM surveys after 10 minutes, group B after 19 minutes, group C after 28 minutes, group A again after 37 minutes, and so on. This guaranteed a total of nine measurement points per lecture without the effort per person exceeding three measurement points. This rotated survey schedule was described in detail in our pilot study (see Moeller, Viljaranta, et al., 2020).

Measures

We collected ESM-measures of emotional and motivational states with 21 items. The exact wording of all ESM measures, their response scales, instruction, and their item abbreviations used in the figures can be found in Appendix A.

Emotions and feeling-related experiences were assessed with seven items adopted from CVT research (e.g., Pekrun, 2024). We assessed the feelings excited, relaxed, bored, interested[1], stressed, frustrated, anxious with single items each, in line with how they were measured in previous ESM studies (e.g., Moeller, Brackett, et al., 2020; Moeller et al., 2018; Schneider et al., 2016). The instruction preceding the items was: "Right now, I feel …". The response scale was: 0 = *not at all*, 1 = *barely*, 2 = *somewhat*, 3 = *quite a lot*, 4 = *very strongly*. Furthermore, one item asked how challenging / demanding participants found their current activity (response scale: 0 = *strongly underchallenged*, 1 = *somewhat underchallenged*, 2 = *challenged just right*, 3 = *somewhat over-challenged*, 4 = *strongly over-challenged*).

Motivation was assessed with 14 items developed based on the situated expectancy-value theory (Eccles & Wigfield, 2020). Most of these items had been validated in previous studies (Dietrich et al., 2017; 2019; Moeller, Viljaranta, et al., 2020; 2022). Intrinsic value was assessed with one item (e.g., "I liked doing what was just discussed"), attainment value with three items (e.g., "What was just discussed is important for me personally"), utility value with two items (e.g., "What was just discussed, will still be useful to me later in my studies"), emotional cost with one item (e.g. "What was just discussed gave me a bad feeling"), opportunity cost with one item (e.g. "What was just discussed caused me to neglect other activities I would have preferred to engage in") and the effort cost with two items (e.g., "I put

[1] While there is some controversy whether interest is a motivation facet, an emotion, or both (see Moeller et al., 2018; Pekrun, 2019; and Silvia, 2005, 2008, for discussion), interest overlaps very strongly conceptually with intrinsic task value and stress overlaps strongly with emotional costs. It is beyond the scope of this study to specify the conceptual overlaps and differences between emotions and motivation with regard to interest and stress. The associations reported in this study can be used as empirical evidence in future studies aiming to delineate the overlaps and differences of interest and stress with academic emotions and SEVT components.

effort into what was just discussed"). Success expectancies were measured with three items regarding competence experience ("Right now, I feel … 0 = *strongly incompetent*, 1 = *somewhat incompetent*, 2 = *neither competent nor incompetent*, 3 = *somewhat competent*, 4 = *very competent"*) and expectations of success in the final exam as well as in explaining the contents of the lecture to other students (e.g., "What was just discussed, I could… 0 = not *explain at all*, 1 = *rather not explain*, 2 = *somewhat explain well*, 3 = *explain well*, 4 = *explain very well*" …to a fellow student).

The instruction preceding the items was: "On the following pages, you will find questions about your experiences in the minutes just before this survey. Please answer these questions with regard to the activity you were engaged in when this survey appeared on your smartphone."

Results

An overview table of all results and supplemental online materials is provided in Appendix B. Descriptives of all variables (*Variance within-person, Variance between-person, Intra-Class Correlation / ICC, Means, Medians, Modi, overall Kurtosis and overall Skewness*) are documented in Appendix D, Table D-1.

RQ1: Homogeneity in Within-Person Distributions Across Time Points

Analyses. We examined for each person the idiographic (within-person) distribution of each item across all time points available for that person, using the R package ggplot2 and the function ggplot (for the R code, see Appendix H-4). This analysis resulted in 777 idiographic graphs (37 participants * 21 items; for an example, see Figure 7; for the results for all items and individuals, see Appendix G-1 to G-21).

We determined for each person and each item the most suitable measure of the central tendency, that is: the median for bi- or multimodality, or the mean for unimodal distributions. We used four different procedures to determine the modality (uni-, bi-, or multimodal) of each item for each participant (all R-Codes for the modality tests are reported in Appendix H-4).

We used the Silverman Method using the R package “multimode” and the function cat. Additionally, we applied the Hartigan’s dip test using the R package “diptest” and the function dip.test. We also calculated the Bimodality Coefficient using the R package “mousetrap” and the function bimodality_coefficient. However, the comparison of all of these methods with the results of the visual inspections of multiple raters suggested that the visual inspection was the most trustworthy method (see also Haslbeck et al., 2023 for a similar conclusion). All results for the checks of the modalities are reported in the Appendices F and G.

We also visually inspected the plots to ensure the accuracy and reliability of our findings. Two independent inspectors were tasked with evaluating the visual representations of the data. To quantify the level of agreement between the inspectors, we calculated Cohen's kappa coefficient. To resolve occurring discrepancies, the inspectors discussed the differing interpretations and reached a consensus on the final evaluation for each plot. This collaborative approach not only enhanced the reliability of our visual inspection process but also ensured a more robust and validated set of conclusions.

Results. For Step 1, we tested the homogeneity assumptions of within-person distributions across measurement time points for all items in all individuals with unimodality tests. The results of the modality tests are described in Tables 1 and G-1-G-21. Overall, 92.15% of the within-person distributions were unimodal, 7.85% were bimodal, and 0.00% were multimodal (Table 3). Table 4 shows the percentages of uni-, bi- and multimodal distributions for each item. The proportion of bimodal within-person distributions ranges from 0% for the items attainment value 2, intrinsic value 1, and feeling challenged, to 18.92% for the items emotional costs and opportunity costs. Given the bimodal distributions in this step, we used the median as a measure of central tendency for all items and individuals to enable comparability of all within-person distributions across individuals and items.

Table 3:

Overall frequencies of unimodal, bi-, and multimodal distributions, summarizing all analyses belonging to RQ1 to RQ3

Research Question		Uni-modal	Bi-modal	Multi-modal
RQ1 Step 1	Within-person distributions across time points of all items in all individuals[1]	92.15%	7.85%	0.00%
RQ2 Step 2	Between-person distributions of all items in the sample[2]	100.00%	0.00%	0.00%
RQ3 Step 3	Between-person distributions of within-person correlations in the sample[3]	68.57%	29.05%	2.38%

Note. [1] = Baseline are 777 within-person distributions of all items (37 individuals * 21 items); [2] = Baseline are 21 between-person distributions of all items; [3] = Baseline are 210 within-person correlations (all pairs of 21 items).

Table 4:

Modalities of Within- and Between-Person Distributions of Single Items

	Within-person distributions (Step 1) *Percent of individuals*			Between-person distributions of the idiographic medians (Step 2)		
	Uni-modal	Bi-modal	Multi modal	Uni-modal	Bi-modal	Multi modal
Anxious	94.59	5.41	0.00	X		
Frustrated	89.19	10.81	0.00	X		
Stressed	91.89	8.11	0.00	X		
Bored	89.19	10.81	0.00	X		
Interested	94.59	5.41	0.00	X		
Excited	91.89	8.11	0.00	X		
Relaxed	94.59	5.41	0.00	X		
Challenged	100.00	0.00	0.00	X		
Competence	83.78	16.22	0.00	X		
Attainment Value 1	91.89	8.11	0.00	X		
Attainment Value 2	100.00	0.00	0.00	X		
Attainment Value 3	97.30	2.70	0.00	X		
Utility Value 1	86.49	13.51	0.00	X		
Utility Value 2	94.59	5.41	0.00	X		
Intrinsic Value 1	100.00	0.00	0.00	X		
Effort Costs 1	91.89	8.11	0.00	X		
Effort Costs 2	97.30	2.70	0.00	X		
Emotional Costs	81.08	18.92	0.00	X		
Opportunity Costs	81.08	18.92	0.00	X		
Exam (Expectancy 1)	91.89	8.11	0.00	X		
Explain (Expectancy 2)	94.59	5.41	0.00	X		

Note. The classifications of modality rely on visual inspections by two independent raters. See Appendix F for reasons why we chose visual inspection over three alternative modality tests (Hartigan's dip test, Silverman's method and the Bimodality Coefficient).

RQ2: Testing Homogeneity Assumptions for Between-Person Distributions of the Within-Person Measures of Central Tendency

Analyses. To examine the between-person distributions of the person-specific parameters of the central tendency (i.e., within-person means or medians), we plotted the between-person distributions of these parameters with the R package ggplot2 (for the R code, see Appendix H-5). If unimodality was not present for at least one individual and item in the sample, then we used the more robust median instead of the mean score to describe this item's measure of central tendency and applied that rule to each participant and each item in the sample to ensure comparability of the coefficients across individuals.

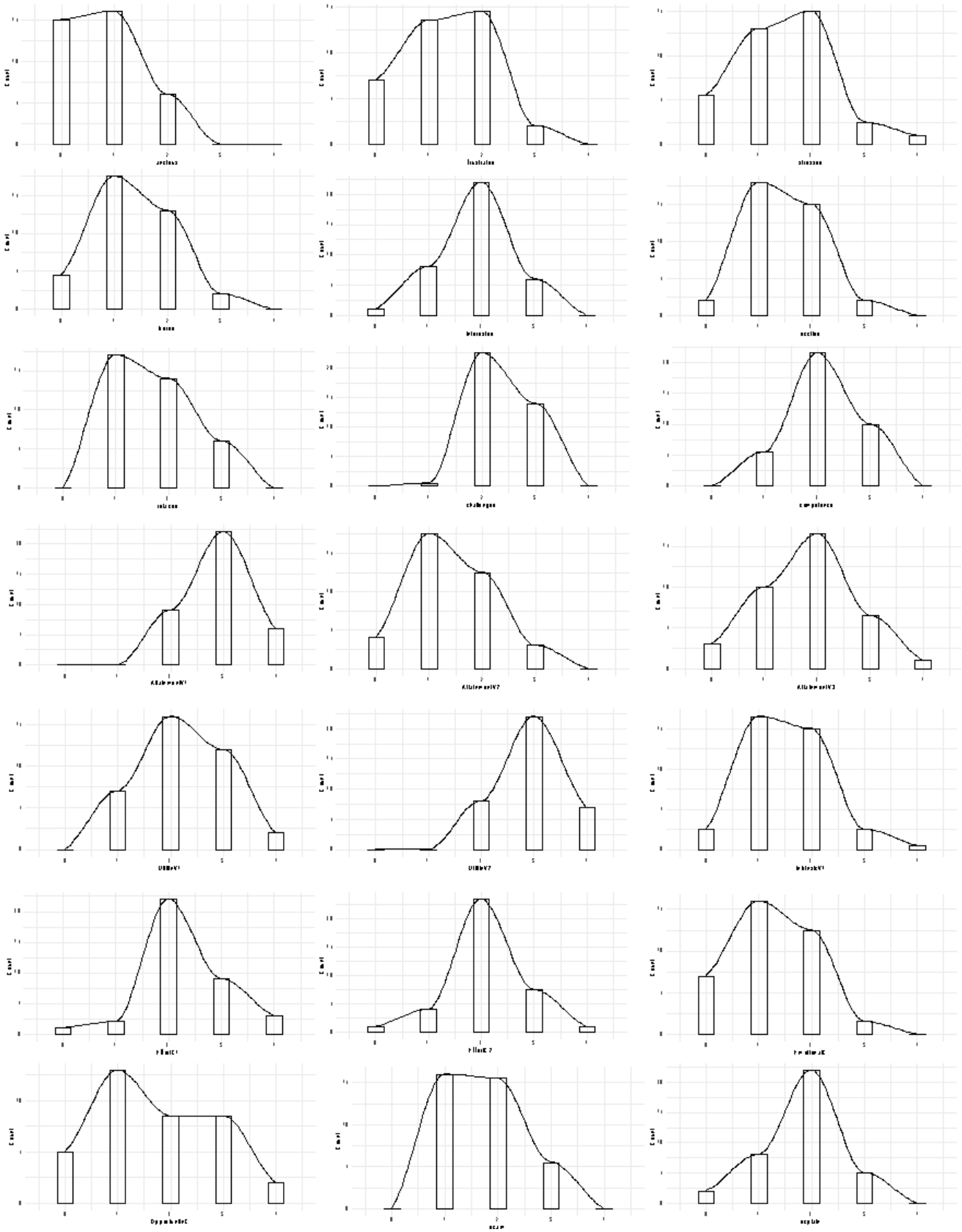
The results are displayed in Figure 2. We also estimated statistics describing the form of each distribution (skewness[2], kurtosis[2], modality, sample mode, average[2], variance within person, between-person and Intra Class Correlation) using the R package moments and the function skewness and kurtosis and the R package lavaan and the function sem (see Appendix H-5). To sum the results up across individuals, we quantify for each item how many individuals showed unimodal or bimodal/multimodal distributions (see Table 4 and see Table 3, Step 1 for the percentage of individuals showing uni-, bi- and multimodal distributions across all items). For this research question we used visual inspection to check the modality.

Results. The median rather than the mean as a measure of central tendency, because the previous analyses (for RQ1) had revealed that some items were distributed bimodally across measurement time points in some individuals. We smoothed distributions to avoid artificial multimodality (see Appendix J). Figure 1 shows the between-person distributions of the within-person medians for all 21 items. All the 21 between-person distributions of all within-person medians, 100% were unimodal, although several were skewed.

[2] insightful only for the unimodal distributions.

Figure 1:

Between-person distributions of the within-person means or medians of each item (Step 2)



RQ3: Testing Homogeneity in Between-Person Distributions of Within-Person Correlations

Analyses. In order to test homogeneity assumptions of bivariate analyses of emotional and motivational states, we examined the between-person distributions of within-person (idiographic) correlations among all emotion and motivation measures for the R code, see Appendix H-3. We first calculated for each individual the within-person correlation for each item pair. This was done using the function correlation (settings: method = "kendall") from the R package correlation; see Appendix H-3. Next, we plotted the between-person distributions of these within-person correlations, following the examples of Pekrun et al. (2002) and Moeller et al. (2015), using the R package ggplot2 and the functions "gghistogram" and "ggdensity".

We then quantified how often types of within-person correlations (insignificant vs. significant; small, medium, or large effect size) were found across all individuals, to implement Asendorpf's (1993) call for bottom-up generalization. As a further approach to bottom-up generalization, we refrained from reporting the average as the default measure of central tendency when describing each variable pair's between-person distribution of within-person correlations. Instead, we tested one of the requirements that need to be met for an average to be a useful measure of a distribution's central tendency, namely unimodality, and used the median to describe the central tendency if these requirements were not met.

Results. Calculating for each of the 37 individuals the idiographic within-person correlations for each of the 36 item pairs resulted in 37*36 = 1,332 correlations (see Tables E-2 to E-38 in Appendix E). Table E-1 in Appendix E shows for each item pair how often its within-person correlation was insignificant, significantly positive, and significantly negative, distinguishing between weak, moderate, and large effect sizes (following Cohen's 1988 guidelines). We propose this detailed analysis as a bottom-up strategy of generalizability in line with

Asendorpf's (1993) proposal to not simply report one overall effect for heterogeneous results, but to instead report how often which pattern is observed in a sample. Overall, 68.57% of these within-person correlations were unimodally distributed across individuals, 29.05% were bimodal, and 2.38% multimodal. This implies that an overall, sample-level estimate of a within-person correlation, as it is typically reported by most multilevel statistics programs and research studies, may fail to describe the central tendency and form of the distribution for one out of three within-person correlations among situational emotion and motivation measures.

Figure 2 shows examples of such unimodal, bimodal, and multimodal between-person distributions of within-person correlations. The complete set of figures showing unimodal correlation distributions can be found in Appendix E, Figure E-1. The complete set of figures showing bimodal correlation distributions can be found in Appendix E, Figure E-2. The complete set of figures showing multimodal correlation distributions can be found in Appendix E, Figure E-3.

Even for unimodal distributions, it can be insightful to examine the detailed percentages of individuals reporting different directions and effect sizes of the same within-person correlation, especially if the variance is large and/or correlations are distributed across the zero threshold. Take for example, the within-person correlation between the items competent and excited, which is unimodally distributed with an estimate of Kendall's $\tau = 0.32$ (C.I.: 0.22 – 0.41) across all individuals (see Figure E-1 in Appendix E): As Figure E-1 shows, this within-person correlation was weakly positive in 29.73% of all individuals, moderately positive in 27.03% of all individuals, strongly positive in 24.32% of all individuals, but counter-intuitively insignificant or not possible to estimate[3] in 10.81%,, weakly negative in 5.41%, and strongly negative in 2.70% of all individuals. Thus, even a unimodally distributed within-person correlation can differ much between individuals,

[3] In some individuals, there was no variance in certain variables across time points, because they had invariantly answered the same (e.g., 0 = not at all anxious) at all times. Without variance, it was impossible to calculate covariances and correlations for these individuals and these variables.

implying that the one sample-level estimate that most statistics programs provide and most ESM studies report for within-person correlations provide only very limited information about how that within-person correlation looks in how many individuals.

Figure 2: *Example plots of unimodal, bimodal, and multimodal between-person distributions of within-person correlations (Step 3)*

Figure 2.1: One example of the 68.57% unimodal between-person distributions of within-person correlations among ESM measures: Here between an item measuring attainment value and one measuring emotional costs.

For all unimodal plots of all within-person correlations, see Appendix E-1.

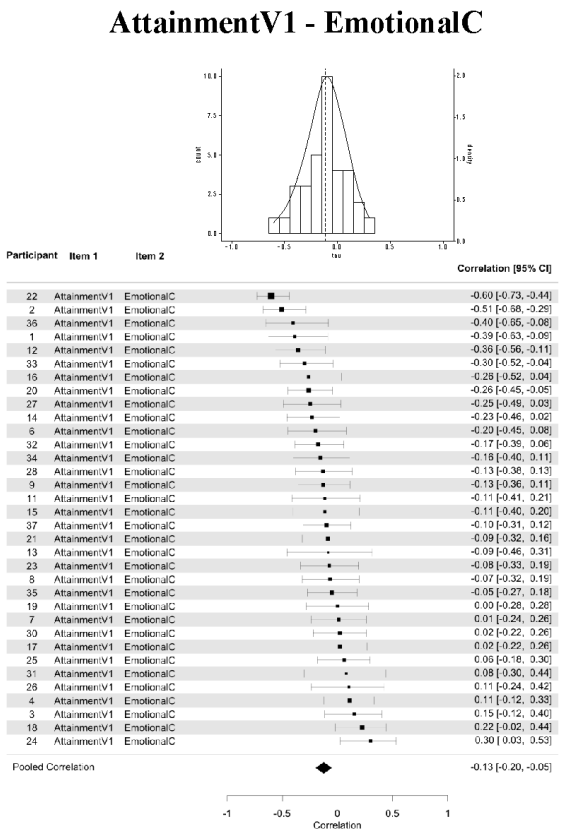


Figure 2.2: One example of the 29.05% bimodal between-person distributions of within-person correlations among ESM measures: Here between an item measuring excitement and one measuring anxiety.
For all bimodal plots of all within-person correlations, see Appendix E-2.

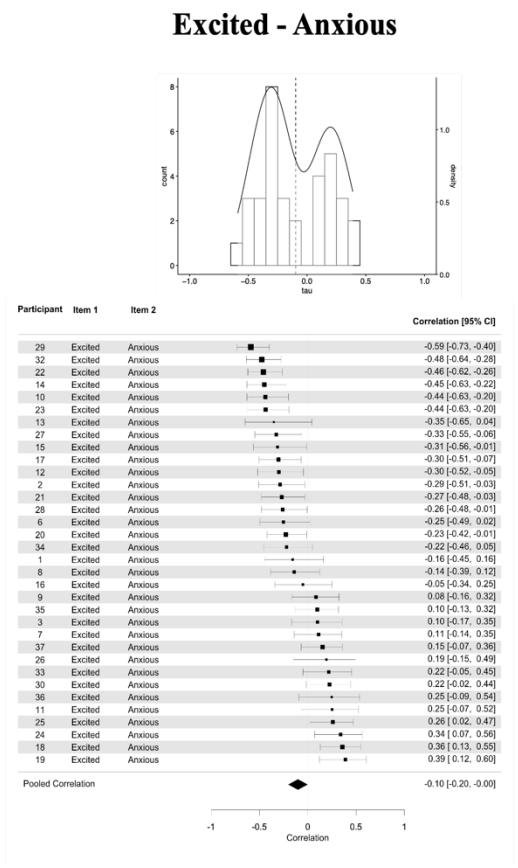
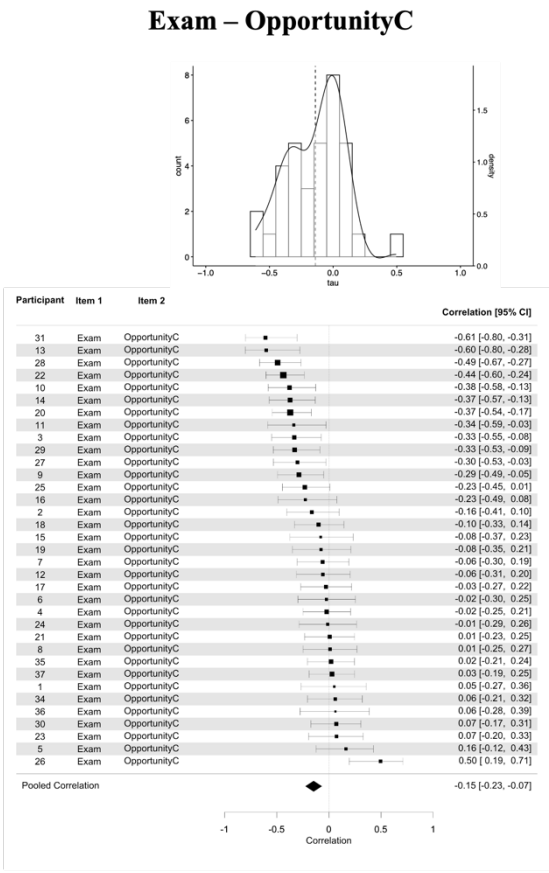


Figure 2.3: One example of the 2.38% multimodal bivariate inter-individual distributions of intra-individual correlations among ESM measures: Here between an item measuring success expectancies and one measuring opportunity costs.
For all multimodal plots of all within-person correlations, see Appendix E-3.

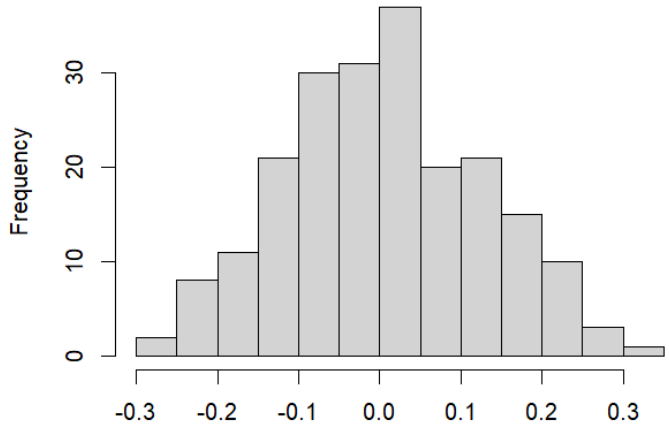


Robustness checks / sensitivity analyses. To rule out method artefacts due to missing data or response compliance, we examined whether the individual response rate to ESM surveys was a moderator of the within-person correlations. For each of the 210 pairwise within-person correlations, we estimated the between-person correlation (Kendall's Tau) with the individual number of a student's completed ESM surveys (for the results, see Table I-1). We display the distribution of all these moderator correlations in Figure 3.

The results showed that, on average across all emotion and motivation item pairs, the individual amount of completed ESM surveys was not correlated with the within-person correlations, suggesting that the size of the within-person correlations did not systematically depend on a person's ESM response rate. Of the 210 pairwise within-person correlations, 17 (8.10%) were correlated with effect sizes of 0.20 to 0.31 with the individual ESM response rate; and 13 (6.19%) were correlated with effect sizes of -0.20 to -0.30 with the individual ESM response rate. Among the correlations with effects of larger 0.20 or smaller than -0.20, bi- and multimodal distributions were not more prevalent (76.67% unimodal; 10.00% bimodal; 3.33 multimodal) than in the overall sample of all within-person correlations (68.57% unimodal; 29.05% bimodal; 2.38 multimodal). This suggests that ESM compliance rates do not explain why some within-person correlations are bi- or multimodally distributed across individuals.

Figure 3:

The distribution of effect sizes of between-person correlations of students' number of completed ESM surveys with each of the 210 within-person correlations



Note. The average of this distribution is $M = 0.01$ ($SD = 0.12$).

RQ4: Testing Homogeneity of Multivariate Within-Person Correlation Networks

Analyses. To examine inter-individual heterogeneity with regard to *multivariate* intra-individual patterns, we created for each person an idiographic network depicting the zero-order within-person correlations among the emotion and motivation patterns (for the R code, see Appendix H-2). The networks were generated with the R package ggraph and the qgraph function (for the R codes, see Appendix H-1). We examined zero-order correlation rather than partial correlation networks to avoid possible suppressor effects and other possible complications for data interpretation, following recommendations by Kulakow et al. (2024) and Jähne et al. (2024). In some of the idiographic networks, some items are not displayed, because they were either uncorrelated with all other items in that network or very weakly correlated with Tau < 0.1, or because no correlation between this item and any other could be calculated due to the lack of variance. By setting the threshold for the display of correlations to 0.1, we aimed to direct the reader's attention to more meaningful correlations. This approach helps to clarify the structure of the network.

To test homogeneity assumptions, we used the multi-sample test for high-dimensional covariance matrices described by Zhang et al. (2018) to determine whether it was plausible that the 37 idiographic covariance matrices originated from the same population. This test

compares the covariance matrices for large variable sets and provides significance tests for tests of the hypothesis that all tested covariance matrices originate from the same population. This test was performed three times, first for the set of emotion items (cf. Figure 2), then for the set of SEVT motivation items (cf. Figure 3), and finally for the relations between both item sets. The test compares the Pearson correlations, not Kendall's Tau, across the individual students.

Results. The results are displayed in Figures 4 to 6. The idiographic (person-specific) cross-construct inter-correlations between SEVT and CVT items are documented in Appendix E (Tables E-2 - E-38).

The multi-sample test for high-dimensional covariance matrices was performed separately for the set of emotion items, the set of motivation items, and the relations between the two sets. The tests rejected the hypothesis that the correlation networks were similar across all individuals. The p-values approximated zero in all three tests (i.e., for the matrices among emotions, among motivation items, and across emotion-motivation item pairs). The values of the scaled[4] test statistics were 14.5 for emotions, 11.3 for motivation items, and 13.8 for the combined item set. The R code for these analyses can be found in Appendix H-7.

[4] Scaled means here, that all values should be distributed in a standard normal distribution if the hypothesis that all individuals have the same covariance matrix was true.

Figure 4:

Idiographic networks of within-person correlations among emotions (participants 1-37, left to right)

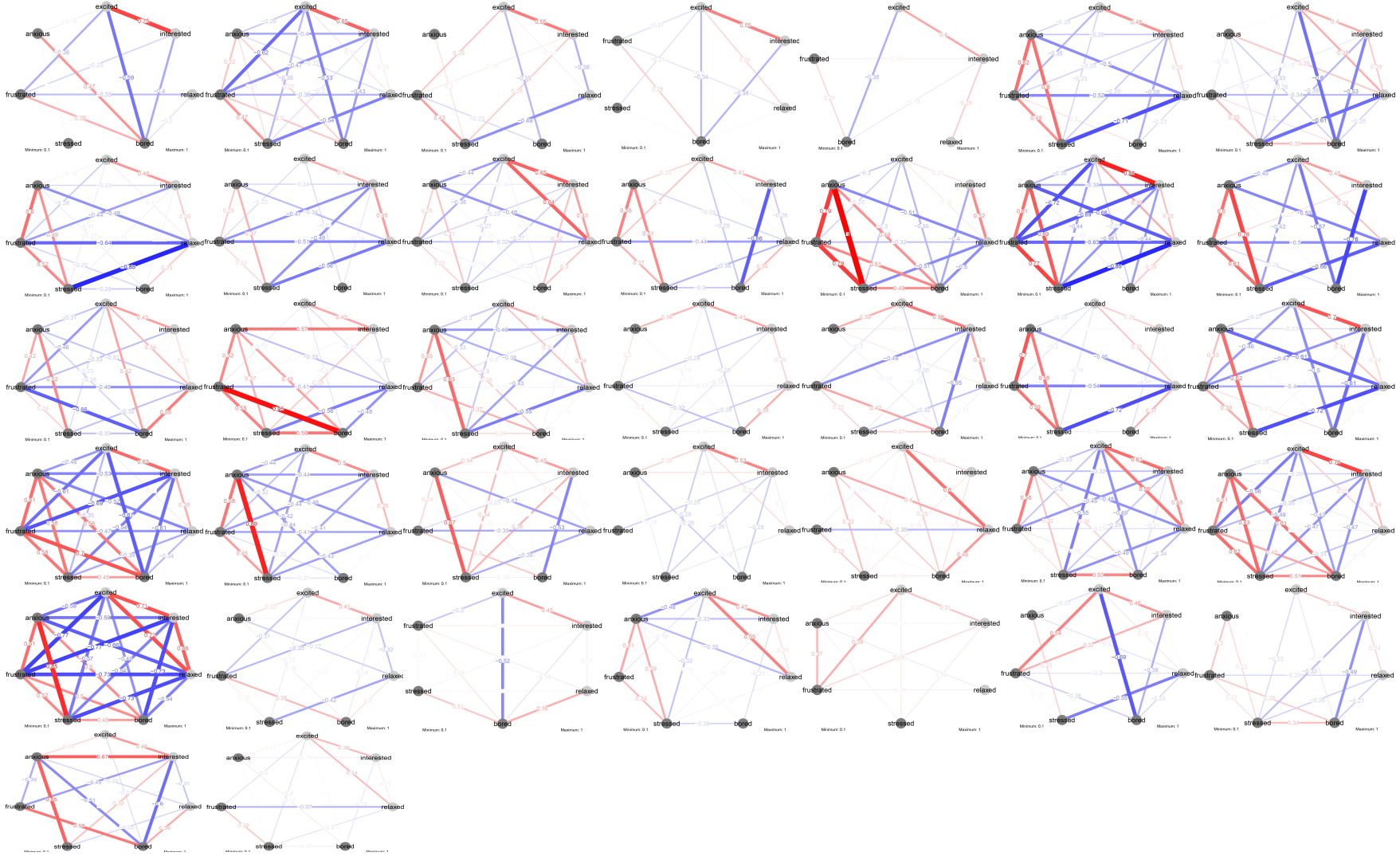


Figure 5:

Idiographic networks of within-person correlations among motivation components (participants 1-37 left to right)

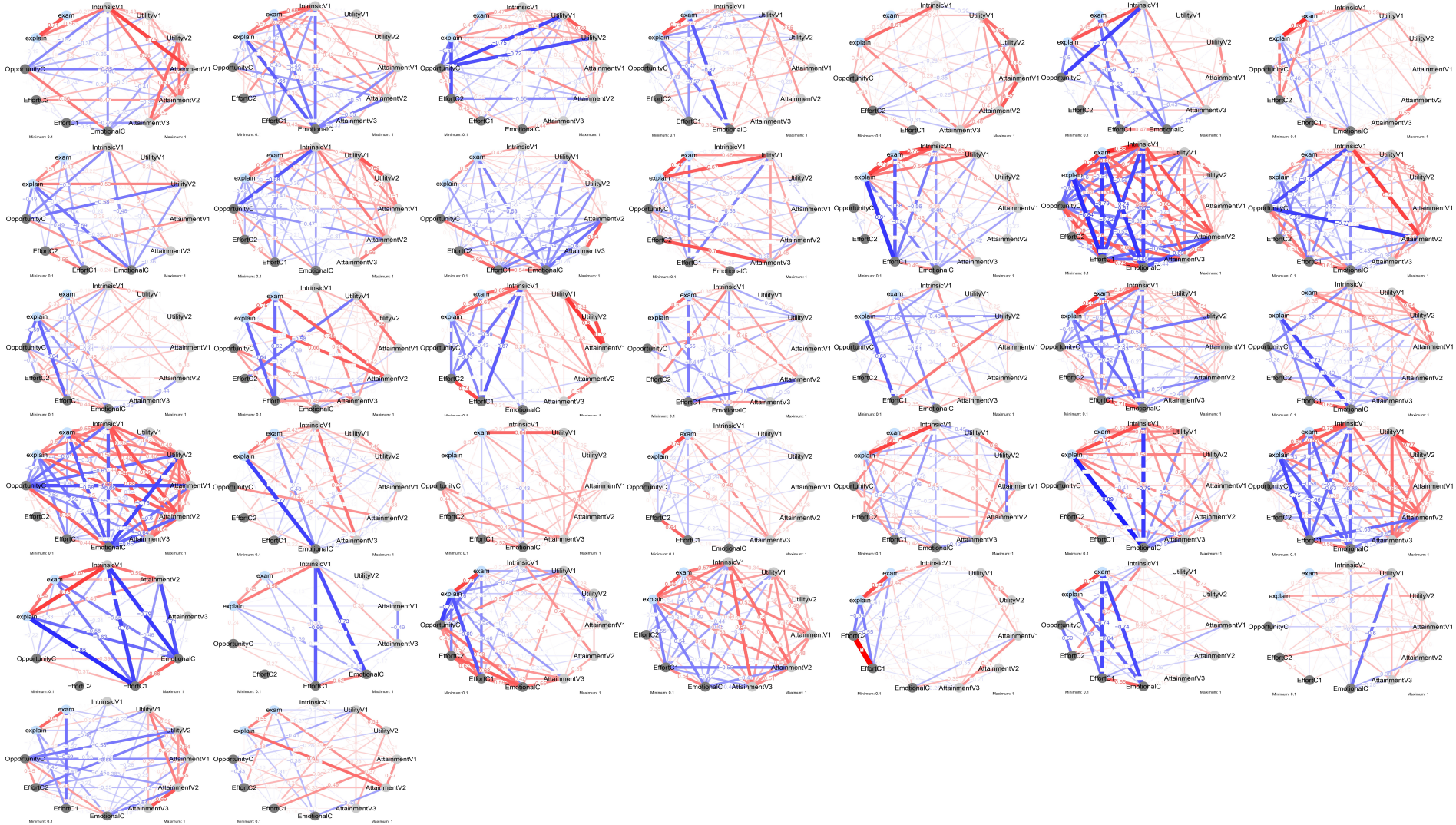
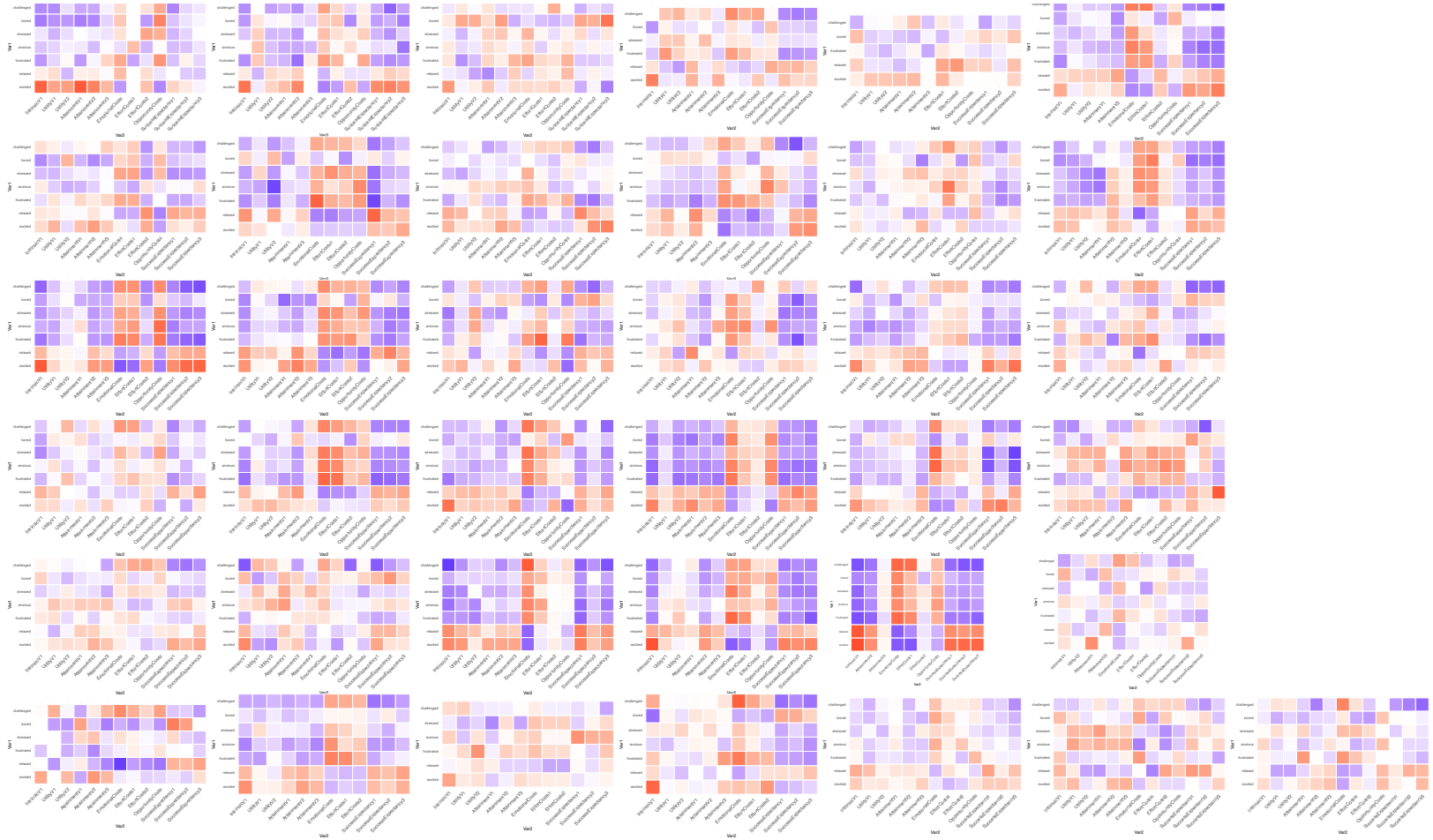


Figure 6:

Idiographic cross-construct correlations between emotions and motivation (participants 1-37, left to right)



*Note***.** Since the cross-construct correlations (Figure 7) were too many, we opted for a correlation heatmap for a more readable graph, while Figure 5 and Figure 6 display the correlations in network graphs for a better overview.

Figures 4 to 6 show substantial between-person homogeneity in the within-person patterns of correlation among emotions and motivation. Nevertheless, for each item pair there were participants who showed negative correlations for this specific item pair, individuals who showed positive correlations for the exact same item pair, and individuals showing no significant correlation at all.

Not all individuals showed the multivariate patterns that were displayed in the top-down nomothetic within-person graphs (Figures 1 to 3, left panels). For instance, the top-down nomothetic within-person network of emotions (Figure 2, panel A left) showed positive[5] correlations of $.19 \leq .23$ for three of the four negative emotions (anxious, stressed, frustrated). Only 20 of the 37 individuals (54.05%) showed a similar pattern of positive associations among these three negative emotions. The nomothetic within-person network of emotions (Figure 2, panel A left) also showed positive correlations of $.12 \leq .24$ between the three positive emotions (excited, interested, relaxed). Only 15 of the 37 (40.5%) individuals showed a similar pattern of positive associations among these three positive emotions.

The nomothetic network of zero-order correlations among motivation (SEVT) items (Figure 3, panel A) shows a pattern of positive correlations between the intrinsic value 1 item ("I liked doing what was just discussed.") and the two items representing success expectancies, namely item called exam ("If examined on what was just discussed, I would be able to do it…") and the item called explain ("I could explain what was just discussed to a fellow student…"). These three items were intercorrelated in the nomothetic within-person network with correlations ranging from $.17 \leq .32$. This pattern of positive associations among these three items was observed in 30 out of the 37 individuals (81.1%; see Figure 6). The nomothetic network of zero-order correlations among motivation (SEVT) items (Figure 3, panel A) also shows a strong negative correlation (-0.25) between the item called explain (see

[5] Positive meaning here an effect size of Tau > 0.0 and negative meaning a Tau < 0.0. We ignore p-values here because of the rather exploratory nature of these analyses.

above) and the item “Emotional C” (emotional costs; “What was just discussed gave me a bad feeling”). This negative correlation among these two items was observed in 32 of 37 individuals (86.5%).

RQ5: Equivalence of Sample-Level (i.e., Nomothetic) Within-Person and Between-Person Correlation Networks

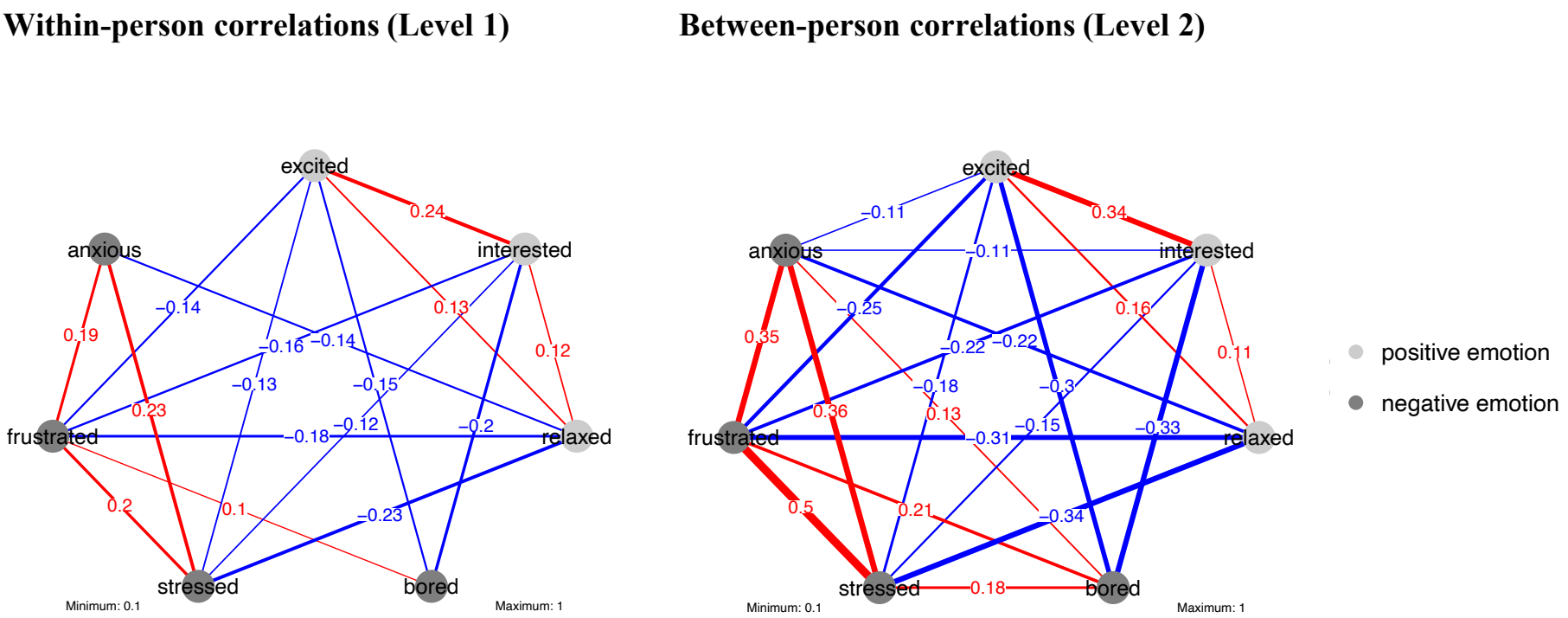
For RQ5, we calculated two nomothetic sample-level network models of zero-correlations among emotion and motivation items on the within- and between-person levels. Within- and between-person zero-order correlations were examined with R package lavaan and the function sem for the R code, see Appendix H-1.

For a better visibility, we separately display the within-person correlations among emotion items (Figure 7), the associations among motivation items (Figure 8), and the correlations between emotion and motivation items (Figure 9). Effect sizes and p-values for all sample-level within-person and between-person correlations are documented in Appendix C (Table C-1).

For the inter-correlations among emotions, the conclusions suggested by the within-person correlations resemble those suggested by the between-person correlations. In both the within- and between-person correlation networks, the three high arousal negative emotions are positively correlated, with slightly smaller effect sizes on the within level ($.19 \leq$ Kendall’s $\tau \leq .23$) than on the between-level ($.35 \leq$ Kendall’s $\tau \leq .50$). Likewise, the correlation between the two high arousal positive emotions was uniformly positive in both the within-person model (Kendall’s $\tau = .24$) and the between-person estimation (Kendall’s $\tau = .34$). The low arousal positive emotion of feeling relaxed was positively but weakly correlated with the high arousal positive emotions in both the within- and the between-person models.

Figure 7:

Nomothetic (sample-level) correlations among emotion (CVT) items



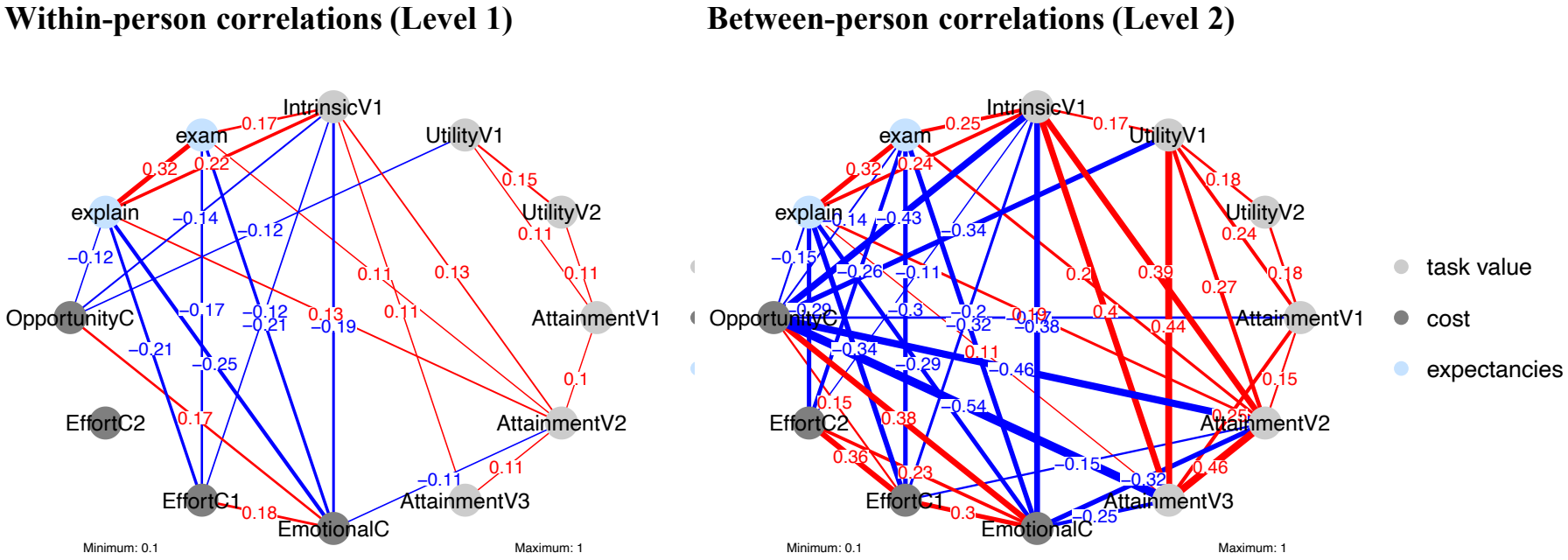
Note. For details (effect sizes and significance values) see the Table C-1 in Appendix C.

Nomothetic Within-Person Correlations Between Motivation Components

For the inter-correlations among motivation facets, the network of within-person correlations differs from the network of overall between-person correlations in various aspects: The positive associations of attainment value items to intrinsic value and utility value, as well as the negative associations of attainment value items to opportunity costs seemed less pronounced in the within-person network, compared to the between-person network. All effect sizes were smaller in the within-person than the between-person network, which is typical for such multilevel findings of ESM data.

Figure 8:

Nomothetic (sample-level) correlations among motivation (SEVT) items



Note. For details (effect sizes and significance values), see the Table C-1 in Appendix C.

Nomothetic Within-person Correlations Between Emotions and Motivation Components

As Figure 9 (left panel) shows, the positive emotions (interested, excited and relaxed) were positively correlated with intrinsic value, utility value, attainment value, and success expectancies within individuals. The effect sizes of these significant correlations were small within individuals (.03* ≤ Kendall's $\tau \leq$ .20*). An exception was the item relaxed, which was correlated positively only with the first and not with the second utility value item within individuals. The positive emotions were mostly negatively correlated with emotional costs, opportunity costs, and the first effort cost item ("Mastering the topic just discussed would require… 0 - *no effort at all*, 1- *little effort*, 2- *some effort*, 3- *quite a lot effort*, 4 - *too much effort*"). The effect sizes of these significant correlations were small within individuals (-.07* ≤ Kendall's $\tau \leq$ -.19*). The feelings relaxed and excited showed insignificant associations with the second effort cost item ("I put effort into what was just discussed."), whereas interest was weakly positively correlated with the second cost item.

The negative feelings (stressed, frustrated, anxious, bored, challenged) were mostly negatively correlated with intrinsic value, utility value, attainment value, and success expectancies within individuals. These effect sizes of these significant correlations were small within individuals (-.03* ≤ Kendall's $\tau \leq$ -.16*). Exceptions were the item anxious, which was uncorrelated with the third attainment value item ("What was just discussed is important for me personally.") and the feeling of being challenged, which was uncorrelated with the second utility value item within individuals.

The negative emotions were mostly positively correlated with emotional costs, opportunity costs, and the first effort cost item ("Mastering the topic just discussed would require… 0 - *no effort at all*, 1- *little effort*, 2- *some effort*, 3- *quite a lot effort*, 4 - *too much effort*"). The effect sizes of these significant correlations were small within individuals (.04* ≤ Kendall's $\tau \leq$ .30*). Boredom was uncorrelated with the first effort cost item ("Mastering the

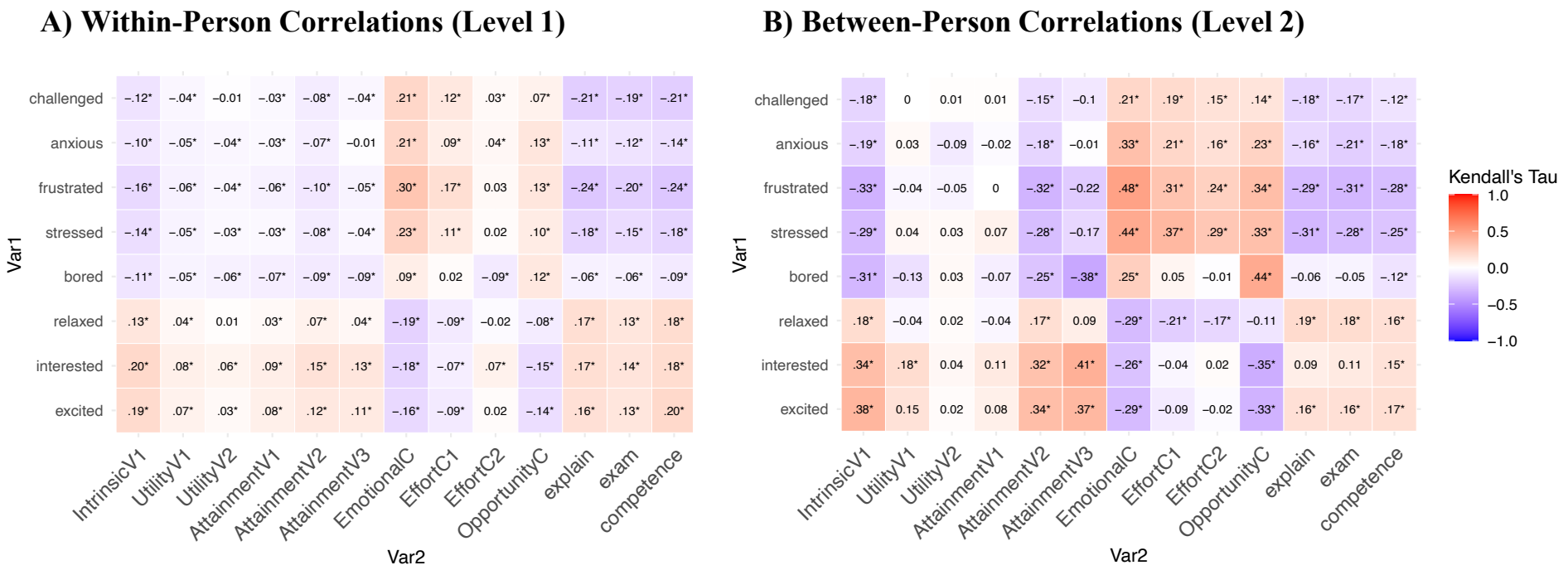
topic just discussed would require….") and weakly negatively correlated with the second effort cost item ("I put effort into what was just discussed."). The feelings frustrated and stressed were uncorrelated with the second effort cost item.

Nomothetic Between-Person Correlations Among Emotions and Motivation Components

As Figure 9 (right panel) shows, the general pattern was rather similar in the within-person and the between-person analyses, although effect sizes and significance differed in some item pairs between the two levels of analyses. Similar to the within-person analyses, the between-person analyses showed positive associations between the positive emotions and most of the positive task value facets (intrinsic, attainment and utility value) and the success expectancies, and negative associations between the positive emotions and the cost facets. Also similar to the within-person analyses, the between-person analyses showed negative associations between the negative emotions and most of the positive task value facets (intrinsic, attainment and utility value) and the success expectancies, and positive associations between the negative emotions and the cost facets.

Figure 9:

Nomothetic (sample-level) cross-construct correlations between emotions (CVT components) and motivation (SEVT components).



Note. This large number of correlations could not be displayed in a readable way in a network graph, therefore we present these heatmaps instead. For all details about significance levels and all 37 idiographic estimations of these micros-construct correlations, see Table C-1 in Appendix C. Since the cross-construct correlations (Figure 9) were too many, we opted for a correlation heatmap for a more readable graph, while Figure 7 and Figure 8 display the correlations in network graphs for a better overview.

Discussion

This study examined homogeneity assumptions underlying the methods that researchers typically use to examine within- and between-level associations among variables measured with intensive longitudinal data, in this particular case: with the experience sampling method (ESM). We first indexed the various homogeneity assumptions that underlie the methods that are typically used in the ESM literature to analyze such data and found that many of these homogeneity assumptions are simply assumed to be satisfied, but rarely empirically tested, before the methods relying on them are employed in the previous literature.

We assume that many homogeneity assumptions are not tested systematically, because the multilayered and complex structure of ESM data can make it difficult to understand the various instances and aspects of homogeneity assumptions underlying the employed statistical analyses. To help ESM researchers understand and test homogeneity assumptions more systematically, we propose a 5-step approach for testing these homogeneity assumptions. We then provide an empirical proof of concept demonstrating the relevance of such homogeneity tests by employing the 5-step approach to our dataset and demonstrating that many of the relevant homogeneity assumptions indeed turn out to be violated, once a researcher starts testing them empirically.

In the following discussion, we explain why untested and violated homogeneity assumptions pose a threat to the generalizability of research findings, and how systematic tests of homogeneity assumptions can provide valuable information needed to reach conclusions about generalizability and heterogeneity. Finally, we connect these methodological insights to ongoing debates in the current psychological literature and show how the here proposed 5-step approach contributes solutions to problems debated in discussions about the generalizability crisis (Yarkoni, 2022a; 2022b), heterogeneity revolution

(Bryan et al., 2022) and the much-needed integration of idiographic and nomothetic models (Beck & Jackson, 2020).

The Main Take-Home Message: Caution When Interpreting Aggregated Within-Person Effects

In a nutshell, the main take-home-message that this study offers is: We need to be careful when interpreting the one-size-fits-all within-person correlations that intensive longitudinal studies typically report after analyzing their data with multilevel analyses of multiple measurement occasions clustered in individuals. These within-person effects are typically labeled "within-person," but they are in fact between-person aggregates of within-person effects, and the methods used for this aggregation lead to trustworthy and generalizable findings only for variables showing unimodal distributions. Research and theory focusing on within-person effects thus needs to exert caution when interpreting the typically aggregated within-person correlations. Ideally, the assumptions required for an informative interpretation of aggregated coefficients should be tested empirically.

Testing homogeneity assumptions in four steps: univariate, bivariate, multivariate

To this end, this study aimed to serve as a model by testing the homogeneity assumptions in all of the analyses relevant to a trustworthy interpretation of multilevel correlation patterns. Specifically, we tested homogeneity assumptions in five steps: *Step 1*: the homogeneity (modality) in within-person univariate distributions of item scores across each individual's time points; *Step 2*: the homogeneity (modality) in between-person univariate distributions of item scores across all individuals; *Step 3*: the homogeneity (modality) in between-person distributions of the bivariate (pairwise) within-person correlations; *Step 4*: the between-person homogeneity regarding the entire multivariate within-person correlation networks. *Step 5*: In Step 5, we compared whether the within-person multivariate correlations were equivalent (ergodic) to the between-person bivariate correlation patterns. It could be argued that Zhang et al.'s (2024) multi-sample test used in Step 4 could be used in Step 5, too,

to determine whether the within- and the between-person covariance matrices are equivalent to each other.

In this study, resulting from the test in step 1, we calculated the more robust within-person medians, rather than the within-person mean scores, for each item, because many items showed bimodal distributions in at least some individuals (see also Haslbeck et al., 2023). Regarding step 2, in our data, all between-person distributions of the within-person medians were rather unimodal, so we can say that we ruled out bi- or multimodality of our items becoming a problem for the estimation of between-person correlations.

Moving on to step 3 we found larger indications of heterogeneity: only 70% of the examined distributions of within-person correlations were in fact unimodal with some variation around the average correlation. This shows that not only can such within-person correlations among motivation and emotion variables be larger or smaller than the sample-wide aggregate, as previous studies had shown (e.g., Malmberg et al., 2013; Moeller et al., 2015; Pekrun et al., 2002; Yoshida & Murayama, 2013; see also Murayama et al., 2017). Going further, our findings suggest that for 30% of the examined within-person correlations, the sample-wide aggregate did not represent a central tendency of the distribution, suggesting that mixture distributions were present. In such situations, multilevel models with random effects ("some variation around the average correlation") may be unsuitable to describe interindividual differences in the relations between variables, which are potentially better described by mixture models.

Finally in step 4, the multivariate tests of homogeneity rejected the assumption that the 37 idiographic correlation matrices were similar across all individuals. This suggests that different individuals in the sample accounted for different edges in the between-person networks - the correlation between variables A and B and the correlation between variables C and D may have been driven by different subgroups of individuals (see also Moeller, 2021; and Reitzle, 2013).

Our first main conclusion is that unimodality assumptions (steps 1, 2, and 3) need to be tested much more rigorously than they typically are in the current literature, because multimodality occurs frequently and limits the trustworthiness of commonly applied methods of generalization. Secondly, we conclude that in the face of such heterogeneity, we need bottom-up methods of generalization that first examine the form of a distribution of a given coefficient, before deciding on the most suited statistical method for describing this distributions' features, for instance by describing the number, height, and location of multiple peaks in a mixture distribution. Our third conclusion is that sample-level within-person correlations and associated methods (e.g., measurement models and structural models obtained with multilevel modeling) may fail to describe how constructs relate to each other within individuals (step 4).

Our discussion on homogeneity also relates to the important problem of lack of ergodicity, meaning that sometimes between-person results do not apply within persons (see, e.g., Molenaar, 2004). In much of the longitudinal literature, this problem is addressed by disentangling between-person variance from within-person variance in multilevel models, nesting time points within individuals (e.g., Voelkle et al., 2014). Our results suggest that such an approach provides only a partial solution because the overall, sample-level aggregate of within-person effects is still a between-person aggregate, which may or may not represent the relations between variables within individuals.

Our results showed a rather strong similarity of the within- and the between-level correlations in the top-down nomothetic multilevel models (step 5). For us ESM researchers who have often seen discrepancies in within- versus between-level findings, the rather strong ergodicity of our findings in this study was rather surprising. A lack of ergodicity may be due to different mechanisms affecting the relation between two variables in different individuals. Likewise, in idiographic within-person networks, different types of situations may drive different correlations. Ergodicity may be lacking when different contexts, different cognitive

ways of processing information, different demographic factors, or different time-sensitive mechanisms cause systematic differences in effects without the researcher being aware of them. Unknown boundary effects shaping relations - also called hidden moderators - might however more likely be detected if homogeneity assumptions are explicitly tested with the steps proposed in this article.

Implications for Research on Academic Emotions and Motivation

This study aspired to make not only a methodological contribution. To the research on academic emotions (in terms of e.g., Pekrun, 2024) and expectancy-value achievement motivation (in terms of e.g., Eccles & Wigfield, 2020), this study provides the arguably most systematic overview to date about within-person associations among CVT emotions and SEVT motivation components, as previously proposed in the DYNAMICS framework (Moeller, Viljaranta, et al., 2022). The information about effect sizes of within-person correlations provided in this article are suited to serve as data-driven effect size priors for subsequent studies using a Bayesian approach to hypothesis testing, which has been proposed as a promising avenue towards a better understanding of effect size heterogeneity (e.g., Gelman, 2015). We propose that future studies could use our findings as data-driven hypotheses about within-person correlations, to be tested for replicability in similar learning contexts and to be tested for generalizability across different learning contexts. To provide a stepping stone for such future studies on homogeneity and heterogeneity in within-person associations among CVT and SEVT variables, this study highlights some heterogeneity between individuals[6], in the hope to inspire future studies on how our findings generalize to other lecture formats, lecture topics, teachers, universities, countries, etc.

[6] Some of our co-authors argue that our results can be considered as support for universality assumptions, rather than as support for heterogeneity assumptions. The fact is that we found both: Patterns that generalized across many individuals, as well as patterns that differed in bimodal and other ways between individuals. It is also a fact that a bimodal distribution is still in line with assumptions of local universality, which proposes that a phenomenon generalizes to some, but not all, individuals under certain boundary conditions (see Moeller, Viljaranta, et al. 2022, for this concept). All co-authors agree that our results reveal hitherto unknown heterogeneity, as well as aspects of previously theoretically expected homogeneity. However, the heterogeneity

The DYNAMICS framework also proposed network graphs as a useful tool for visualizing and inspecting between-person heterogeneity in within-person associations among CVT and SEVT variables, which has given rise to a number of studies applying and comparing different network approaches to the SEVT and CVT literature (e.g., Jähne et al., 2024; Lee et al., 2023). The present studies' findings expand on previous network studies by demonstrating that a sample-wide or nomothetic within-person network resulting from a commonly used multilevel model may show within-person correlations that may not apply to all individuals in the sample.

The between-person heterogeneity that we found when comparing the within-person multivariate covariance matrices of different individuals suggest that it would be rather implausible to assume that one measurement model or one structure model describes the within-person associations among emotion and motivation variables well in this sample. That has potentially dramatic consequences for measurement practices using intensive longitudinal data. Most measurement approaches rely on assumptions of generalizability (homogeneity) across individuals. If we cannot assume one measurement model to apply to all individuals, then we may also lack the justification to assume that the same measurement properties apply to all individuals, across all items. Whether factor structures, item-factor loadings, aspects of concurrent, prospective or incremental validity and other psychometric properties apply to all individuals similarly is questionable if such between-person differences in within-person multivariate correlation patterns as in other study are observed. This also implies that we may lack the justification of creating general composite scales by averaging several items across single-item measures or by creating the same factor score for all individuals, because this practice also relies on the assumption that the same multivariate covariance patterns apply to all individuals. This raises complex, somewhat philosophical question for future research to

findings are particularly relevant, given that heterogeneity has been understudied and overlooked in much of the previous research (see Bryan et al., 2022; Moeller et al., 2025).

ponder about: Based on the Classical Test Theory and our nomothetic research tradition, psychological research practices have typically assumed that the silver bullet to ridding ourselves of noise and error is by averaging across individuals, assuming that individual errors and deviations cancel each other out. That logic relies on the idea that there is one true effect represented by one true hill (mode) in a normal distribution. Now that we come to understand more and more that many phenomena may be truly heterogeneous in the sense that multiple effects affect multiple subgroups in a population in manifold different ways, the idea of gaining the one effect by ignoring that heterogeneity is increasingly questionable.

Heterogeneity in measurement and structure models therefore implies that for both measurement and modeling, we need statistical and epistemological approaches that embrace and try to understand heterogeneity. In sum, neither measurement nor structure models should use one-size-fits-all statistics if their assumptions are not met.

Limitations and Future Directions

Need To Study Further Boundary Conditions Beyond Person-Specific Effects. We only examined data from one university lecture, and thus systematic comparisons of heterogeneity across university lectures are needed to gain knowledge that is generalizable across lectures. This is a mission for future research. ESM data are affected by manifold challenges to their generalizability and replicability (for overviews, see Moeller et al., in prep.; 2025), and more systematic research on boundary conditions is needed to gain knowledge about locally and universally generalizable associations among learning-related motivational and emotional states (see e.g., Moeller et al., 2025).

Sample Size. Our robustness checks (see Figure 4 and Appendix I) revealed no indications to believe that the size of the within-person correlations, nor their bi- or multimodality, depended on the individual numbers of completed ESM surveys. Apart from this sensitivity analysis though, we do not know how sample sizes of ESM surveys per person and persons in the overall sample may have influenced the results. On the between-person level, 37

individuals are hardly a sample large and representative enough to estimate robust and generalizable effect sizes and between-person distributions.

However, for the main purpose of this study, the sample size is arguably large enough, as our main goal was to demonstrate, as a proof-of-concept, that bi- and multimodality exist and that not all distributions of within-person scores and correlations are unimodal. In contrast, the exact prevalences of for instance uni- versus bi- and multimodal between-person distributions of single items and of pairwise within-person correlations should be tested for replicability and generalizability in a larger and more representative sample.

In any case, we currently do not know whether to attribute our empirical findings to true differences between individuals or to lacking robustness due to small sample sizes of ESM time points per person. Two future research efforts are needed to clarify that question: Conceptual replications of our study with ESM data containing much larger samples of time points per person, and simulation studies transferring Schönbrodt and Perugini's (2013) approach to ESM data to generally determine the sample sizes needed for obtaining robust correlation effect sizes. Guidance for the calculation of required sample sizes of time points and individuals are available for multilevel time series analyses from Schultzberg & Muthén (2018). Their estimations do not apply to this present study though because we do not report any time series analyses and therefore cannot directly translate these sample size estimations to our study.

Conclusion: Drawing Generalizable Conclusions Not Despite But Because of the Heterogeneity Revolution

Some readers may worry about the amount of differences between individuals and even feel discouraged, thinking that so much variability at all levels impedes any generalizable conclusions, other than saying "it depends". We would rather argue that explicitly testing if nomothetic conclusions apply will help researchers to make better generalizations. This is because the findings either confirm homogeneity, or, if heterogeneity

is present, they systematically account for differences between individuals rather than erroneously treating all individuals the same despite differences. One contribution of this article is that we suggested bottom-up methods to obtain generalizable findings despite such heterogeneity. In other words, systematic investigations about homogeneity and heterogeneity allow to quantify which patterns occurs how often, and thus, the methodological solution for the above-mentioned worries is to find ways to distinguish meaningful patterns from noise. Ultimately, understanding effect heterogeneity is crucial not only for developing valid theories, but also for research and applications aiming for personalized diagnostics and interventions (e.g., Gelman, 2015; Bryan et al., 2021; Nahum-Shani et al., 2015; 2018; Riley et al., 2011).

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Appendix A

Table A-1

Wording of all ESM items

Item abbreviation	Item text	Response scale	Construct
The instruction for all items was: " How did you feel when the signal for the survey appeared? On the following pages you will find descriptions of your feelings. Indicate how strongly each description applies to you. Click on "OK" to go to the next page"			
Challenged	Right now, I feel … [strongly underchallenged - strongly overstrained]	0 Strongly underchallenged 1 Somewhat underchallenged 2 challenged just right 3 Somewhat over-challenged 4 Strongly over-challenged	Task characteristic
Competence	Right now, I feel … [strongly incompetent - strongly competent]	0 Strongly incompetent 1 Somewhat incompetent 2 Neither 3 Somewhat competent 4 Strongly competent	Success expectancy
Excited	Right now, I feel excited.	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very strongly	Positive emotion
Interested	Right now, I feel interested.	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very strongly	Positive emotion
Frustrated	Right now, I feel frustrated.	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very strongly	Negative emotion
Bored	Right now, I feel bored.	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very strongly	Negative emotion
Anxious	Right now, I feel anxious.	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very strongly	Negative emotion
Stressed	Right now, I feel stressed.	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very strongly	Negative emotion
Relaxed	Right now, I feel relaxed	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very strongly	Positive emotion
Exam	If examined on what was just discussed, I would be able to do it…	0 not at all 1 rather not 2 Moderately well	Success Expectancy

		3 well 4 very well	
Explain	I could explain what was just discussed to a fellow student…	0 not at all 1 rather not 2 Moderately well 3 well 4 very well	Success Expectancy
IntrinsicV1	I liked doing what was just discussed	0 Not at all 1 Mostly didn't like 2 Liked it somewhat 3 Liked it quite a bit 4 Liked it very much	Intrinsic Values
AttainmentV1	What was just discussed is important for me to understand.	0 Not important at all 1 Mostly not important 2 Somewhat important 3 Quite important 4 Very important	Attainment Values
AttainmentV2	What was just discussed reinforces my belief that this course of study suits me as a person.	0 Does not reinforce my belief at all 1 Mostly does not reinforce my belief 2 Reinforces my belief somewhat 3 Reinforces my belief quite a bit 4 Strongly reinforces my belief	Attainment Values
AttainmentV3	What was just discussed is important for me personally.	0 Not important at all 1 Mostly not important 2 Somewhat important 3 Quite important 4 Very important	Attainment Values
UtilityV1	What was just discussed will be useful for my future career.	0 Not useful at all 1 Mostly not useful 2 Somewhat useful 3 Quite useful 4 Very useful	Utility Values
UtilityV2	What was just discussed will be useful to me later in my studies.	0 Not useful at all 1 Mostly not useful 2 Somewhat useful 3 Quite useful 4 Very useful	Utility Values
EmotionalC	What was just discussed gave me a bad feeling.	0 No bad feeling at all 1 Mostly no bad feeling 2 A somewhat bad feeling 3 A fairly bad feeling 4 A very bad feeling	Emotional Costs
OpportunityC	What was just discussed caused me to neglect other activities I would have preferred to engage in	0 Not at all 1 Rather not 2 To some extent 3 Quite a bit 4 To a great extent	Opportunity Costs
EffortC1	Mastering the topic just discussed would require…	0 No effort at all 1 Little effort 2 Some effort 3 Quite a lot of effort 4 Too much effortt	Effort Costs
EffortC2	I put effort into what was just discussed	0 Not at all 1 Barely 2 Somewhat 3 Quite a lot 4 Very much	Effort Costs

Appendix B

Table B-2

Overview of all figures and all tables in the Appendices

Research question	Labels	Description	Link to online materials
RQ1	**Figure 7 in manuscript; Tables G-1 to G-21 in Appendix G**	*Idiographic distributions of each item (1-21) across measurement time points for every individual (1-37)*: All 777 idiographic distribution plots of each item for every individual	https://osf.io/r6mfx/?view_only=bda6c52bb5ce48f0b2f72bd3ce106535
RQ2	**Figure 8 in manuscript**	*Between-person distributions of the idiographic within-person median of each item*	
RQ3	**Figure E-1 in Appendix E**	*All unimodal between-person distributions of intra-individual (Level 1) correlations among motivational ESM measures:* Unimodal correlation plots with the forest plots above	https://osf.io/qe2t6/?view_only=72b4661b91e64640b4244c2929987154
	Figure E-2 in Appendix E	*All bimodal between-person distributions of intra-individual (Level 1) correlations among motivational ESM measures:* Bimodal correlation plots with the forest plots above	https://osf.io/bhfy3/?view_only=8f862c42937a4223a822314580374062
	Figure E-3 in Appendix E	*All multimodal between-person distributions of intra-individual (Level 1) correlations among motivational ESM measures:* Bimodal correlation plots with the forest plots above	https://osf.io/umv8p/?view_only=5775a5a1b201452089673fb586bdb80e
RQ4 & RQ5	**Table C-1 in Appendix C**	*The nomothetic sample-level inter-correlations between SEVT and CVT items*: All within-and between person correlations between all SEVT and CVT items	https://osf.io/bk7fj/?view_only=2a79404926854c12ac50bb5bc7defadc
	Tables E-2 – E-38 in Appendix E	*The idiographic (person-specific) cross-construct inter-correlations between SEVT and CVT items*: A table for each participant with the correlation between all SEVT and CVT items	https://osf.io/wg64h/?view_only=8943b6160fe24b62a21a7a8e86ac5d26
	Figure 5 & 5 in manuscript	*Idiographic networks of within-person correlations among emotion (CVT)/motivation (SEVT) states (participants 1-37)*. A higher resolution of these figures can be found following the link on the right.	https://osf.io/vs2mn/?view_only=1a3f293b723048acb8c11ba946f38fc3
	Table I-1 in Appendix I	*Sensitivity Analyses: Does the number of ESM surveys per person predict within-person correlation effect size?*	https://osf.io/a6htj/?view_only=72dc670ea1e04426818811261b161c43

Appendix C

Table C-1: Nomothetic, sample-level within-person (Level 1) and between-person (Level 2) inter-correlations among all CVT facets (academic emotions) and all SEVT facets (motivation components):

To be found here: https://osf.io/bk7fj/?view_only=2a79404926854c12ac50bb5bc7defadc

Note. These are the estimated correlations that result from a multilevel analysis with ESM measurement time points (Level 1) nested in individuals (Level 2)

Appendix D

Table D-1

Between-person distributions of the different items across all 37 individuals

	Variance within-person (L1)	*Variance* between-person (L2)	*ICC*	*M (SD)* between-person (L2)	*Median* between-person (L2)	*overall sample Mode* (= most frequent value of all idiographic within-person modes)	*Kurtosis*[b]	*Skewness*[b]
excited	0.40	0.39	0.49	1.42	1	1	2.56	0.20
interested	0.46	0.37	0.45	1.82	2	2	2.71	-0.13
relaxed	0.45	0.33	0.42	1.63	2	1	2.48	0.14
frustrated	0.51	0.50	0.50	1.30	1	1	2.62	0.43
stressed	0.55	0.58	0.51	1.45	1.5	1/2	2.51	0.30
bored	0.59	0.46	0.44	1.40	1	1	2.56	0.34
anxious	0.43	0.36	0.46	1.74	1	0	3.88	1.06
competent	0.55	0.21	0.28	1.99	2	2	2.56	-0.32
challenged	0.33	0.13	0.28	2.27	2	2	3.11	0.03
Attainment Value 1	0.34	0.30	0.47	2.71	3	3	3.63	-0.60
Attainment Value 2	0.36	0.54	0.60	1.34	1	1	2.78	0.34
Attainment Value 3	0.39	0.83	0.68	1.75	2	2	2.26	0.15
Utility Value 1	0.42	0.53	0.56	2.10	2	2	2.49	-0.00
Utility Value 2	0.32	0.25	0.44	2.82	3	3	2.86	-0.32
Intrinsic Value 1	0.38	0.45	0.54	1.52	1.5	1	2.81	0.18
Effort Costs 1	0.34	0.41	0.55	2.23	2	2	3.26	-0.03
Effort Costs 2	0.39	0.40	0.51	2.02	2	2	2.97	-0.10
Emotional Costs	0.56	0.50	0.47	1.35	1	1	2.70	0.48
Opportunity Costs	0.56	0.90	0.62	1.67	2	1	2.02	0.23
Exam	0.46	0.36	0.44	1.68	2	2	2.44	-0.02
Explain	0.53	0.39	0.43	1.70	2	2	2.69	0.01

Note. [a] = between-person median of all within-person medians; [b] = Calculated by ignoring the nested multilevel structure of the data.

Appendix E

Table E-1

Overview and bottom-up generalization of within-person correlations: Which intra-individual correlations were found how often?

Item pair		% individuals with no corr.	% of individuals with negative correlations			% of individuals with positive correlations			Uni-, bi- or multimodal distribution between-person?
			weak	moderate	strong	weak	moderate	strong	
1. Competent	Excited	10.81	5.41	0.00	2.70	29.73	27.03	24.32	unimodal
2. Competent	Interested	18.92	5.41	0.00	0.00	24.32	40.54	10.81	unimodal
3. Competent	Relaxed	8.11	2.70	0.00	0.00	27.03	54.05	8.11	unimodal
4. Competent	Bored	19.44	25.00	16.67	8.33	27.78	0.00	2.78	unimodal
5. Competent	Stressed	13.89	38.89	19.44	22.22	5.56	0.00	0.00	unimodal
6. Competent	Frustrated	10.81	18.92	43.24	24.32	2.70	0.00	0.00	unimodal
7. Competent	Anxious	14.71	17.65	38.24	14.71	11.76	2.94	0.00	unimodal
8. Competent	Challenged	2.70	24.32	27.03	43.24	2.70	0.00	0.00	unimodal
9. Competent	Explain	0.00	5.41	0.00	0.00	10.81	18.92	64.86	bimodal
10. Competent	Exam	5.41	2.70	0.00	0.00	10.81	35.14	45.95	unimodal
11. Competent	IntrinsicV1	24.32	2.70	0.00	0.00	21.62	29.73	21.62	unimodal
12. Competent	AttainmentV1	28.57	17.14	2.86	0.00	31.43	20.00	0.00	unimodal
13. Competent	AttainmentV2	14.71	2.94	5.88	0.00	26.47	35.29	14.71	unimodal
14. Competent	AttainmentV3	16.22	16.22	5.41	2.70	37.84	18.92	2.70	unimodal
15. Competent	UtilityV1	28.57	14.29	2.86	0.00	31.43	22.86	0.00	unimodal
16. Competent	UtilityV2	22.22	11.11	5.56	0.00	47.22	11.11	2.78	unimodal
17. Competent	EffortC1	10.81	21.62	35.14	27.03	2.70	2.70	0.00	unimodal
18. Competent	EffortC2	21.62	24.32	16.22	0.00	27.03	8.11	2.70	bimodal
19. Competent	EmotionalC	8.33	25.00	19.44	47.22	0.00	0.00	0.00	bimodal
20. Competent	OpportunityC	22.86	40.00	22.86	5.71	5.71	2.86	0.00	multimodal
21. Excited	Interested	0.00	0.00	0.00	0.00	8.11	54.05	37.84	unimodal
22. Excited	Relaxed	35.14	5.41	0.00	0.00	24.32	21.62	13.51	bimodal
23. Excited	Bored	19.44	33.33	19.44	19.44	5.56	2.78	0.00	unimodal
24. Excited	Stressed	25.00	38.89	19.44	8.33	5.56	2.78	0.00	unimodal
25. Excited	Frustrated	21.62	29.73	21.62	13.51	8.11	2.70	2.70	unimodal
26. Excited	Anxious	8.82	26.47	23.53	2.94	29.41	8.82	0.00	bimodal
27. Excited	Challenged	16.22	29.73	21.62	18.92	8.11	2.70	2.70	unimodal
28. Excited	Explain	21.62	8.11	5.41	0.00	16.22	27.03	21.62	unimodal
29. Excited	Exam	24.32	8.11	2.70	0.00	29.73	16.22	18.92	unimodal
30. Excited	IntrinsicV1	8.11	0.00	0.00	0.00	32.43	16.22	43.24	bimodal
31. Excited	AttainmentV1	31.43	2.86	2.86	0.00	34.29	20.00	8.57	unimodal
32. Excited	AttainmentV2	20.59	5.88	0.00	0.00	20.59	32.35	20.59	unimodal
33. Excited	AttainmentV3	18.92	2.70	0.00	0.00	32.43	43.24	2.70	bimodal
34. Excited	UtilityV1	25.71	5.71	2.86	0.00	40.00	22.86	2.86	unimodal
35. Excited	UtilityV2	38.89	11.11	2.78	0.00	33.33	11.11	2.78	bimodal
36. Excited	EffortC1	21.62	35.14	18.92	8.11	13.51	2.70	0.00	unimodal
37. Excited	EffortC2	18.92	27.03	5.41	0.00	24.32	21.62	2.70	bimodal
38. Excited	EmotionalC	25.00	25.00	19.44	22.22	5.56	2.78	0.00	unimodal
39. Excited	OppportunityC	14.29	34.29	25.71	14.29	8.57	2.86	0.00	unimodal
40. Interested	Relaxed	13.51	5.41	8.11	0.00	40.54	27.03	5.41	bimodal
41. Interested	Bored	11.11	30.56	22.22	27.78	5.56	2.78	0.00	bimodal
42. Interested	Stressed	19.44	36.11	22.22	5.56	11.11	5.56	0.00	unimodal
43. Interested	Frustrated	21.62	21.62	29.73	8.11	16.22	2.70	0.00	unimodal
44. Interested	Anxious	29.41	26.47	17.65	5.88	14.71	0.00	5.88	bimodal
45. Interested	Challenged	16.22	32.43	27.03	13.51	5.41	5.41	0.00	unimodal
46. Interested	Explain	21.62	5.41	2.70	0.00	24.32	21.62	24.32	bimodal
47. Interested	Exam	24.32	8.11	0.00	0.00	32.43	24.32	10.81	unimodal
48. Interested	IntrinsicV1	16.22	0.00	0.00	0.00	13.51	29.73	40.54	bimodal
49. Interested	AttainmentV1	17.14	8.57	0.00	2.86	34.29	28.57	8.57	multimodal
50. Interested	AttainmentV2	14.71	2.94	0.00	0.00	38.24	23.53	20.59	unimodal
51. Interested	AttainmentV3	18.92	8.11	0.00	0.00	29.73	29.73	13.51	unimodal
52. Interested	UtilityV1	22.86	5.71	0.00	0.00	48.57	20.00	2.86	bimodal
53. Interested	UtilityV2	25.00	19.44	0.00	0.00	38.89	11.11	5.56	unimodal
54. Interested	EffortC1	32.43	29.73	16.22	5.41	8.11	8.11	0.00	unimodal
55. Interested	EffortC2	21.62	10.81	5.41	0.00	40.54	10.81	10.81	unimodal
56. Interested	EmotionalC	19.44	27.78	27.78	16.67	5.56	2.78	0.00	unimodal
57. Interested	OpportunityC	17.14	28.57	31.43	11.43	8.57	0.00	2.86	bimodal
58. Relaxed	Bored	19.44	22.22	13.89	2.78	19.44	19.44	2.78	unimodal
59. Relaxed	Stressed	11.11	22.22	27.78	38.89	0.00	0.00	0.00	unimodal
60. Relaxed	Frustrated	13.51	27.03	35.14	18.92	5.41	0.00	0.00	unimodal

61. Relaxed	Anxious	17.65	20.59	35.29	17.65	5.88	2.94	0.00	unimodal
62. Relaxed	Challenged	8.11	29.73	40.54	18.92	2.70	0.00	0.00	bimodal
63. Relaxed	Explain	13.51	2.70	0.00	0.00	24.32	45.95	13.51	unimodal
64. Relaxed	Exam	16.22	10.81	0.00	0.00	27.03	37.84	8.11	unimodal
65. Relaxed	IntrinsicV1	24.32	0.00	0.00	0.00	37.84	27.03	10.81	unimodal
66. Relaxed	AttainmentV1	34.29	11.43	5.71	2.86	34.29	8.57	2.86	unimodal
67. Relaxed	AttainmentV2	14.71	20.59	5.88	0.00	29.41	17.65	11.76	bimodal
68. Relaxed	AttainmentV3	29.73	16.22	0.00	2.70	35.14	16.22	0.00	unimodal
69. Relaxed	UtilityV1	22.86	14.29	11.43	0.00	31.43	20.00	0.00	unimodal
70. Relaxed	UtilityV2	33.33	13.89	19.44	0.00	22.22	11.11	0.00	unimodal
71. Relaxed	EffortC1	10.81	27.03	40.54	10.81	8.11	2.70	0.00	bimodal
72. Relaxed	EffortC2	24.32	27.03	18.92	5.41	13.51	5.41	5.41	unimodal
73. Relaxed	EmotionalC	11.11	19.44	36.11	27.78	2.78	2.78	0.00	unimodal
74. Relaxed	OpportunityC	20.00	31.43	20.00	8.57	8.57	11.43	0.00	bimodal
75. Bored	Stressed	25.71	28.57	5.71	0.00	14.29	17.14	8.57	unimodal
76. Bored	Frustrated	19.44	19.44	2.78	2.78	25.00	22.22	8.33	unimodal
77. Bored	Anxious	21.21	24.24	3.03	3.03	27.27	15.15	6.06	unimodal
78. Bored	Challenged	19.44	16.67	11.11	5.56	30.56	13.89	2.78	unimodal
79. Bored	Explain	19.44	30.56	11.11	5.56	13.89	16.67	2.78	unimodal
80. Bored	Exam	22.22	27.78	8.33	8.33	19.44	11.11	2.78	bimodal
81. Bored	IntrinsicV1	19.44	11.11	30.56	16.67	11.11	11.11	0.00	bimodal
82. Bored	AttainmentV1	32.35	23.53	17.65	8.82	11.76	5.88	0.00	unimodal
83. Bored	AttainmentV2	18.18	33.33	21.21	6.06	18.18	3.03	0.00	unimodal
84. Bored	AttainmentV3	27.78	33.33	22.22	8.33	5.56	2.78	0.00	unimodal
85. Bored	UtilityV1	26.47	35.29	20.59	0.00	14.71	2.94	0.00	bimodal
86. Bored	UtilityV2	22.86	45.71	14.29	2.86	5.71	8.57	0.00	bimodal
87. Bored	EffortC1	33.33	19.44	13.89	0.00	22.22	8.33	2.78	unimodal
88. Bored	EffortC2	33.33	13.89	38.89	2.78	8.33	2.78	0.00	bimodal
89. Bored	EmotionalC	11.43	22.86	11.43	0.00	31.43	11.43	11.43	bimodal
90. Bored	OpportunityC	25.71	2.86	2.86	2.86	37.14	20.00	8.57	bimodal
91. Stressed	Frustrated	16.67	2.78	0.00	0.00	33.33	22.22	25.00	unimodal
92. Stressed	Anxious	8.82	11.76	0.00	0.00	14.71	20.59	44.12	unimodal
93. Stressed	Challenged	11.11	8.33	2.78	0.00	16.67	30.56	30.56	unimodal
94. Stressed	Explain	5.56	25.00	47.22	11.11	11.11	0.00	0.00	bimodal
95. Stressed	Exam	13.89	44.44	25.00	11.11	5.56	0.00	0.00	unimodal
96. Stressed	IntrinsicV1	19.44	41.67	16.67	13.89	8.33	0.00	0.00	unimodal
97. Stressed	AttainmentV1	47.06	20.59	8.82	5.88	8.82	5.88	2.94	bimodal
98. Stressed	AttainmentV2	12.12	24.24	36.36	0.00	21.21	6.06	0.00	bimodal
99. Stressed	AttainmentV3	33.33	30.56	19.44	0.00	8.33	8.33	0.00	unimodal
100.Stressed	UtilityV1	23.53	20.59	26.47	2.94	17.65	8.82	0.00	bimodal
101.Stressed	UtilityV2	34.29	34.29	11.43	2.86	8.57	8.57	0.00	unimodal
102.Stressed	EffortC1	16.67	5.56	0.00	0.00	41.67	25.00	11.11	unimodal
103.Stressed	EffortC2	30.56	19.44	8.33	0.00	25.00	16.67	0.00	unimodal
104.Stressed	EmotionalC	5.56	2.78	0.00	0.00	27.78	38.89	25.00	unimodal
105.Stressed	OpportunityC	23.53	11.76	5.88	0.00	23.53	29.41	5.88	bimodal
106.Frustrated	Anxious	20.59	5.88	2.94	0.00	11.76	20.59	38.24	unimodal
107.Frustrated	Challenged	2.70	5.41	0.00	0.00	10.81	40.54	40.54	bimodal
108.Frustrated	Explain	10.81	24.32	40.54	24.32	0.00	0.00	0.00	unimodal
109.Frustrated	Exam	13.51	29.73	29.73	27.03	0.00	0.00	0.00	bimodal
110.Frustrated	IntrinsicV1	13.51	16.22	40.54	18.92	10.81	0.00	0.00	bimodal
111.Frustrated	AttainmentV1	31.43	37.14	17.14	2.86	8.57	0.00	2.86	unimodal
112.Frustrated	AttainmentV2	26.47	23.53	17.65	17.65	11.76	2.94	0.00	bimodal
113.Frustrated	AttainmentV3	18.92	35.14	16.22	2.70	18.92	5.41	2.70	unimodal
114.Frustrated	UtilityV1	25.71	37.14	20.00	0.00	14.29	0.00	2.86	unimodal
115.Frustrated	UtilityV2	19.44	30.56	19.44	8.33	13.89	5.56	2.78	unimodal
116.Frustrated	EffortC1	8.11	2.70	2.70	0.00	27.03	32.43	27.03	unimodal
117.Frustrated	EffortC2	35.14	10.81	8.11	0.00	27.03	16.22	2.70	unimodal
118.Frustrated	EmotionalC	5.56	0.00	0.00	0.00	13.89	33.33	47.22	multimodal
119.Frustrated	OpportunityC	14.29	8.57	5.71	0.00	37.14	20.00	14.29	unimodal
120.Anxious	Challenged	14.71	14.71	5.88	0.00	20.59	17.65	26.47	unimodal
121.Anxious	Explain	20.59	11.76	32.35	20.59	5.88	5.88	2.94	unimodal
122.Anxious	Exam	14.71	32.35	29.41	11.76	5.88	5.88	0.00	unimodal
123.Anxious	IntrinsicV1	26.47	23.53	17.65	14.71	17.65	0.00	0.00	bimodal
124.Anxious	AttainmentV1	21.88	31.25	21.88	3.13	12.50	6.25	3.13	unimodal
125.Anxious	AttainmentV2	19.35	35.48	19.35	6.45	16.13	3.23	0.00	unimodal
126.Anxious	AttainmentV3	32.35	23.53	11.76	2.94	17.65	8.82	2.94	unimodal
127.Anxious	UtilityV1	9.38	46.88	15.63	3.13	15.63	9.38	0.00	bimodal
128.Anxious	UtilityV2	21.21	30.30	9.09	12.12	15.15	12.12	0.00	unimodal
129.Anxious	EffortC1	20.59	11.76	8.82	0.00	14.71	26.47	17.65	unimodal
130.Anxious	EffortC2	26.47	20.59	5.88	0.00	44.12	0.00	2.94	unimodal
131.Anxious	EmotionalC	5.88	8.82	0.00	2.94	23.53	26.47	32.35	unimodal
132.Anxious	OpportunityC	31.25	18.75	0.00	0.00	18.75	9.38	21.88	bimodal
133.Challenged	Exam	0.00	10.81	32.43	54.05	2.70	0.00	0.00	bimodal
134.Challenged	Explain	5.41	21.62	35.14	37.84	0.00	0.00	0.00	unimodal
135.Challenged	IntrinsicV1	10.81	27.03	35.14	18.92	5.41	2.70	0.00	unimodal
136.Challenged	AttainmentV1	37.14	28.57	17.14	0.00	17.14	0.00	0.00	bimodal
137.Challenged	AttainmentV2	20.59	26.47	32.35	8.82	11.76	0.00	0.00	unimodal

138.Challenged	AttainmentV3	24.32	43.24	13.51	0.00	16.22	2.70	0.00	unimodal
139.Challenged	UtilityV1	31.43	28.57	17.14	2.86	14.29	5.71	0.00	unimodal
140.Challenged	UtilityV2	47.22	19.44	13.89	0.00	11.11	8.33	0.00	bimodal
141.Challenged	EffortC1	10.81	2.70	0.00	0.00	32.43	40.54	13.51	unimodal
142.Challenged	EffortC2	13.51	24.32	2.70	5.41	29.73	24.32	0.00	bimodal
143.Challenged	EmotionalC	5.56	2.78	0.00	0.00	19.44	36.11	36.11	unimodal
144.Challenged	OpportunityC	31.43	11.43	0.00	0.00	37.14	17.14	2.86	unimodal
145.Explain	Exam	0.00	0.00	0.00	0.00	10.81	21.62	67.57	unimodal
146.Explain	IntrinsicV1	2.70	2.70	2.70	0.00	24.32	27.03	40.54	unimodal
147.Explain	AttainmentV1	28.57	5.71	2.86	0.00	45.71	17.14	0.00	unimodal
148.Explain	AttainmentV2	17.65	5.88	0.00	0.00	32.35	23.53	20.59	unimodal
149.Explain	AttainmentV3	27.03	13.51	2.70	0.00	18.92	32.43	5.41	bimodal
150.Explain	UtilityV1	20.00	14.29	8.57	0.00	34.29	14.29	8.57	unimodal
151.Explain	UtilityV2	27.78	13.89	8.33	0.00	25.00	16.67	8.33	unimodal
152.Explain	EffortC1	8.11	21.62	32.43	37.84	0.00	0.00	0.00	unimodal
153.Explain	EffortC2	24.32	27.03	5.41	10.81	21.62	8.11	2.70	unimodal
154.Explain	EmotionalC	2.78	22.22	30.56	41.67	2.78	0.00	0.00	unimodal
155.Explain	OpportunityC	20.00	31.43	22.86	14.29	8.57	2.86	0.00	unimodal
156.Exam	IntrinsicV1	5.41	10.81	0.00	0.00	24.32	35.14	24.32	bimodal
157.Exam	AttainmentV1	20.00	14.29	2.86	0.00	40.00	17.14	5.71	bimodal
158.Exam	AttainmentV2	11.76	14.71	0.00	0.00	35.29	26.47	11.76	unimodal
159.Exam	AttainmentV3	32.43	21.62	2.70	0.00	21.62	18.92	2.70	unimodal
160.Exam	UtilityV1	28.57	11.43	5.71	0.00	22.86	25.71	5.71	unimodal
161.Exam	UtilityV2	25.00	19.44	2.78	0.00	36.11	13.89	2.78	unimodal
162.Exam	EffortC1	5.41	18.92	37.84	32.43	5.41	0.00	0.00	unimodal
163.Exam	EffortC2	18.92	24.32	10.81	0.00	35.14	8.11	2.70	bimodal
164.Exam	EmotionalC	13.89	19.44	30.56	30.56	5.56	0.00	0.00	unimodal
165.Exam	OpportunityC	51.43	11.43	22.86	8.57	2.86	2.86	0.00	multimodal
166.IntrinsicV1	AttainmentV1	20.00	14.29	2.86	0.00	37.14	22.86	2.86	bimodal
167.IntrinsicV1	AttainmentV2	20.59	2.94	0.00	0.00	26.47	26.47	23.53	unimodal
168.IntrinsicV1	AttainmentV3	21.62	2.70	0.00	0.00	29.73	35.14	10.81	unimodal
169.IntrinsicV1	UtilityV1	40.00	5.71	2.86	0.00	28.57	17.14	5.71	unimodal
170.IntrinsicV1	UtilityV2	33.33	8.33	5.56	0.00	30.56	19.44	2.78	bimodal
171.IntrinsicV1	EffortC1	10.81	40.54	18.92	21.62	8.11	0.00	0.00	unimodal
172.IntrinsicV1	EffortC2	29.73	21.62	5.41	2.70	21.62	10.81	8.11	unimodal
173.IntrinsicV1	EmotionalC	11.11	19.44	33.33	25.00	11.11	0.00	0.00	unimodal
174.IntrinsicV1	OpportunityC	17.14	31.43	17.14	25.71	5.71	2.86	0.00	unimodal
175.AttainmentV1	AttainmentV2	21.88	9.38	0.00	0.00	25.00	34.38	9.38	unimodal
176.AttainmentV1	AttainmentV3	22.86	5.71	5.71	0.00	31.43	25.71	8.57	unimodal
177.AttainmentV1	UtilityV1	11.76	5.88	2.94	0.00	29.41	35.29	14.71	bimodal
178.Attainment1	UtilityV2	14.29	8.57	2.86	0.00	28.57	25.71	20.00	unimodal
179.AttainmentV1	EffortC1	34.29	28.57	14.29	0.00	17.14	5.71	0.00	unimodal
180.AttainmentV1	EffortC2	22.86	28.57	5.71	0.00	20.00	22.86	0.00	unimodal
181.AttainmentV1	EmotionalC	35.29	32.35	11.76	5.88	11.76	2.94	0.00	unimodal
182.AttainmentV1	OpportunityC	48.48	30.30	9.09	9.09	3.03	0.00	0.00	unimodal
183.AttainmentV2	AttainmentV3	20.59	14.71	2.94	0.00	17.65	14.71	29.41	bimodal
184.AttainmentV2	UtilityV1	30.30	6.06	0.00	0.00	21.21	39.39	3.03	bimodal
185.AttainmentV2	UtilityV2	15.15	15.15	0.00	3.03	39.39	21.21	6.06	bimodal
186.AttainmentV2	EffortC1	14.71	41.18	11.76	17.65	14.71	0.00	0.00	unimodal
187.AttainmentV2	EffortC2	20.59	32.35	17.65	2.94	8.82	17.65	0.00	bimodal
188.AttainmentV2	EmotionalC	15.15	18.18	24.24	24.24	12.12	6.06	0.00	unimodal
189.AttainmentV2	OpportunityC	25.00	31.25	25.00	9.38	6.25	3.13	0.00	unimodal
190.AttainmentV3	UtilityV1	20.00	8.57	0.00	0.00	40.00	28.57	2.86	unimodal
191.AttainmentV3	UtilityV2	25.00	8.33	0.00	0.00	33.33	25.00	8.33	unimodal
192.AttainmentV3	EffortC1	21.62	27.03	18.92	2.70	21.62	8.11	0.00	bimodal
193.AttainmentV3	EffortC2	37.84	21.62	2.70	0.00	13.51	18.92	5.41	unimodal
194.AttainmentV3	EmotionalC	30.56	22.22	27.78	2.78	11.11	2.78	2.78	bimodal
195.AttainmentV3	OpportunityC	17.14	37.14	22.86	2.86	11.43	2.86	5.71	bimodal
196.UtilityV1	UtilityV2	5.71	0.00	0.00	0.00	37.14	20.00	37.14	bimodal
197.UtilityV1	EffortC1	31.43	25.71	14.29	5.71	20.00	2.86	0.00	unimodal
198.UtilityV1	EffortC2	31.43	20.00	11.43	2.86	28.57	5.71	0.00	unimodal
199.UtilityV1	EmotionalC	32.35	26.47	20.59	5.88	8.82	5.88	0.00	multimodal
200.UtilityV1	OpportunityC	24.24	24.24	33.33	6.06	9.09	0.00	3.03	bimodal
201.UtilityV2	EffortC1	36.11	33.33	8.33	0.00	19.44	0.00	2.78	bimodal
202.UtilityV2	EffortC2	38.89	22.22	8.33	2.78	25.00	2.78	0.00	unimodal
203.UtilityV2	EmotionalC	40.00	17.14	20.00	2.86	20.00	0.00	0.00	unimodal
204.UtilityV2	OpportunityC	26.47	32.35	8.82	14.71	11.76	5.88	0.00	unimodal
205.EffortC1	EffortC2	27.03	16.22	2.70	0.00	13.51	21.62	18.92	unimodal
206.EffortC1	EmotionalC	8.33	2.78	2.78	0.00	22.22	30.56	33.33	unimodal
207.EffortC1	OpportunityC	17.14	20.00	0.00	0.00	34.29	20.00	8.57	unimodal
208.EffortC2	EmotionalC	33.33	19.44	2.78	0.00	22.22	19.44	2.78	unimodal
209.EffortC2	OpportunityC	25.71	22.86	17.14	2.86	14.29	17.14	0.00	bimodal
210.EmotionalC	OpportunityC	17.65	5.88	0.00	0.00	26.47	32.35	17.65	unimodal

Note. We used Cohen's (1988) convention to classify the correlations: "no correlation": -0.1 < Tau < 0.1; "small correlation": 0.1 <= Tau < 0.3; "moderate correlation": 0.3 <= Tau < 0.5; "large correlation": Tau >= 0.5.

Tables E-2 – E-38: Idiographic, person-specific inter-correlations among all items

All 37 idiographic within-person correlation tables can be found here:
https://osf.io/wg64h/?view_only=8943b6160fe24b62a21a7a8e86ac5d26

Figure E-1: Forest plots with density curves for the *unimodal* within-person correlations

To be found here: https://osf.io/qe2t6/?view_only=72b4661b91e64640b4244c2929987154

Figure E-2: Forest plots with density curves for the *bimodal* within-person correlations

To be found here: https://osf.io/bhfy3/?view_only=8f862c42937a4223a822314580374062

Figure E-3: Forest plots with density curves for the *multimodal* within-person correlations

To be found here: https://osf.io/umv8p/?view_only=5775a5a1b201452089673fb586bdb80e

Appendix F

Modality Tests: Results, encountered complications and methodological decisions made when categorizing the modalities

We used three different tests to determine the modality of all idiographic within-person distributions of all items (777 distributions for 37 individuals * 21 items): (1) Silverman's test, (2) Hartigan's dip test, and (3) the bimodality coefficient. The complete overview of all results of all idiographic within-person distributions of all items, along with the modality test results for every item and person, can be found in Tables G-1 to G-21 in Appendix G.

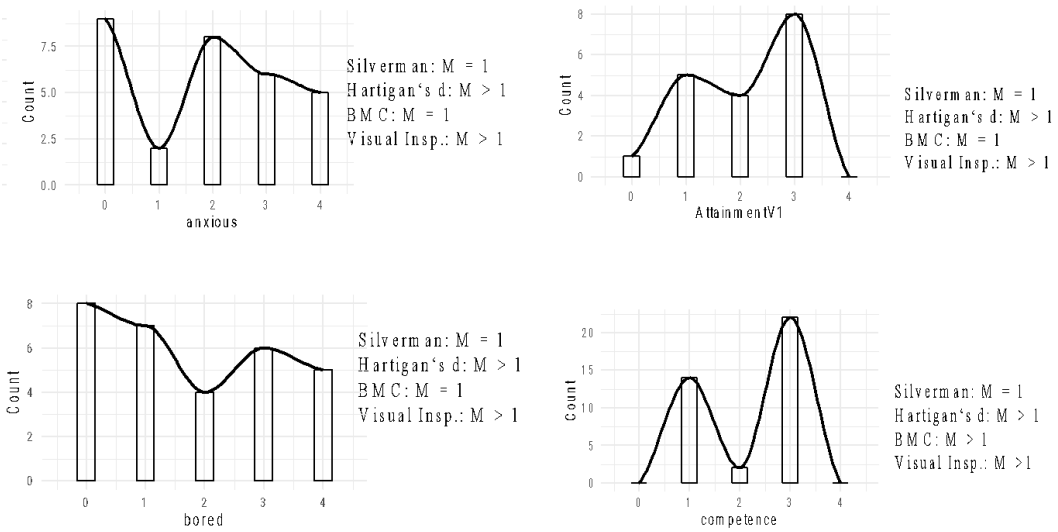
For Hartigan's dip test, we used the R package ‘diptest’ and the function ‘dip.test’, while for the bimodality coefficient we used the R package ‘mousetrap’ and the function ‘bimodality.coefficient’. Given Haslbeck et al.’s (2023) claim that the Silverman test works exceptionally well, we used two different methods for its calculation. First, we used the code proposed by Haslbeck et al. (2023) and then the R package ‘multimode’. Despite Haslbeck et al.'s (2023) paper highlighting the strong performance of Silverman's test, our results were less consistent, even when calculated using both Haslbeck et al.'s proposed code and our code using the “multimode” package (see Figure F-1 for examples). As already mentioned by Haslbeck et al. (2023) the Bimodality coefficient is very sensitive to skewed plots, which we also found in our results (see Figure F-2 for examples and Haslbeck et al., 2023).

Among the three applied tests, Hartigan's dip test performed the best in our study. However, neither method came reliably to the same conclusions as a visual inspection did: Some plots were either clearly unimodal, when Hartigan’s dip test proposed bimodality, or the other way around (see Figure F-3). Consequently, we determined that the tests were not consistent or robust enough, prompting us to rely on visual inspection for a more accurate determination of the modality. Two raters classified the modality for (1) the idiographic within-person distributions of each item in each person (see Table 3, step 1 and Appendix H-

4), and (2) the between-person distributions of the within-person medians of all items (see Table 3; step 2 and Figure 3), and (3) the between-person distributions of the within-person correlations among all item pairs (see Table 3, step 3 and Table E-1).

Figure F-1

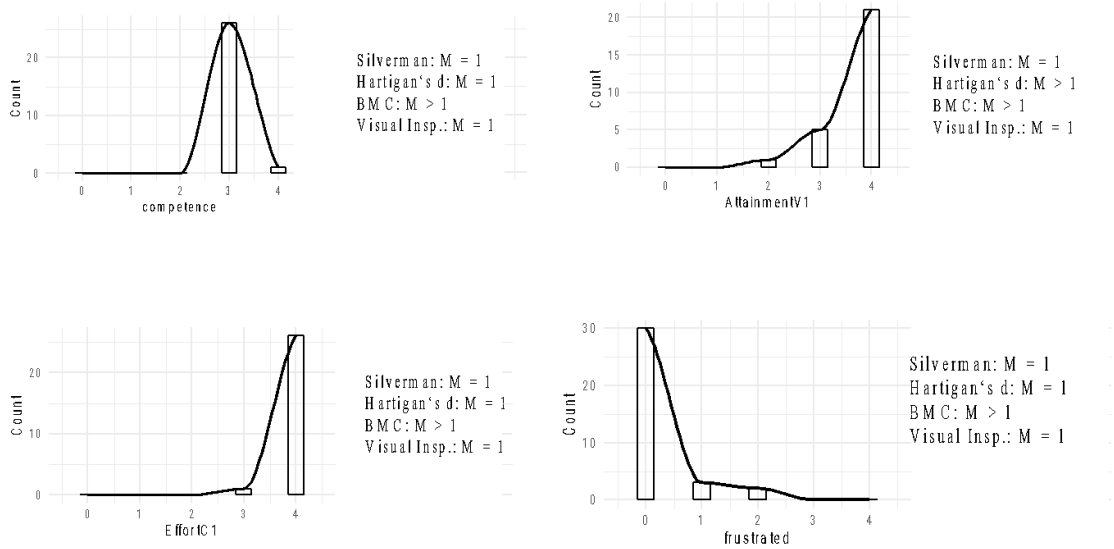
Examples for a Non-Trustworthy Performance of the Silverman's Method (multimodal distributions despite of Silverman's Method suggesting one mode)



Note. For the complete overview of all modality test results, see Appendices G-1 to G-21.

Figure F-2

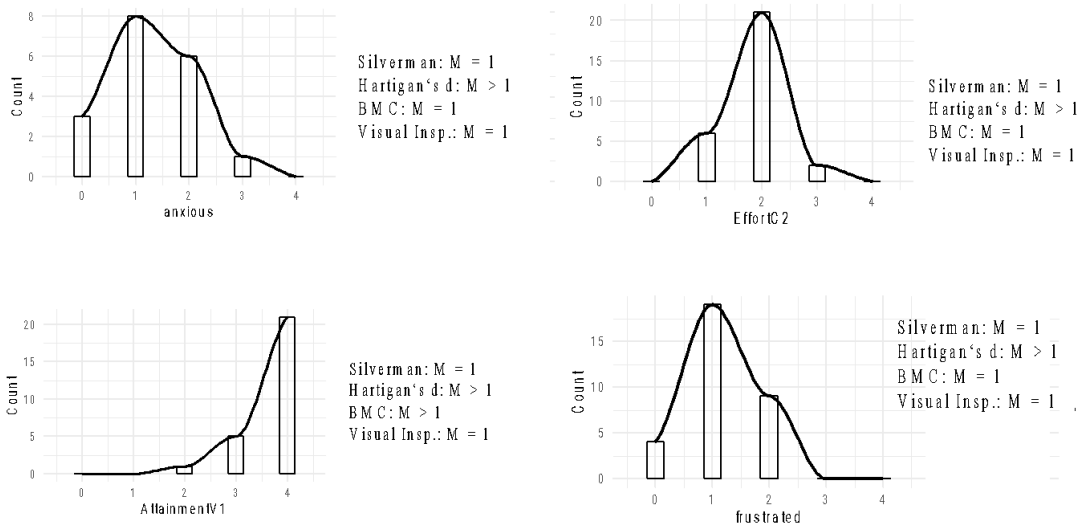
Examples for a Non-Trustworthy Performance of the Bimodality Coefficient (unimodality, despite of the BMC suggesting more than one mode)



Note. For the complete overview of all modality test results, see Appendices G-1 to G-21.

Figure F-3

Examples for a Non-Trustworthy Performance of the Hartigan's Dip Test (unimodality, despite of Hartigan's dip test suggesting more than one mode)



Note. For the complete overview of all modality test results, see Appendices G-1 to G-21.

Appendix G

Tables G-1-G-21:

Idiographic within-person distributions of each item (1-21) for every individual (1-37)

All 777 idiographic within-person distribution plots of each item for every individual can be found here: https://osf.io/r6mfx/?view_only=bda6c52bb5ce48f0b2f72bd3ce106535

Appendix H

Table H-1

R-Codes for all analyses and open data

Research Question analyzed with the R code	Description of the analyses	Referred to in the manuscript as	Link to the R Code
RQ1; Step 1	Within-person distributions of all items Modality tests	Appendix H-4	https://osf.io/aq2kc/?view_only=89a5924e14fc4225bfb5035193bb607a
RQ2; Step 2	Between-person distributions of the within-person medians of all items	Appendix H-5	https://osf.io/eu5cp/?view_only=02059a9719a34c7895a2df67db4d41be
RQ3; Step 3	Between-person distributions of within-person correlations	Appendix H-3	https://osf.io/8e2ru/?view_only=022b9e4251134eab805da4697ee0d4f0
RQ 4 (Step 4)	Multi-sample test	Appendix H-7	https://osf.io/9n2cg/?view_only=f7b45ab4ae014106a4f31f12d4aaf1cc
RQ4 & RQ5; Step 4	Idiographic networks Heatmaps	Appendix H-2	https://osf.io/7exhf/?view_only=8588fb64f31f433287b481729b3a1565
RQ4 & RQ5; Step 4	Nomothetic networks (multilevel analyses) Heatmaps nomothetic	Appendix H-1	https://osf.io/q85ja/?view_only=1140b1b4eb78414f860b8eef87901465
Open Dataset	All ESM data of 37 individuals and 21 items	Appendix H-6	https://osf.io/npt69/?view_only=243b035fbcbb4be8b5b61bfb3b386e00

Appendix I

Sensitivity Analysis

Table I-1

Robustness Checks: Between-person correlations (Kendall's τ) between the individuals' number of completed ESM surveys and the effect size of their within-person correlation for each item pair.

To be found here: https://osf.io/a6htj/?view_only=72dc670ea1e04426818811261b161c43

Appendix J

Procedure of smoothing the between-person distributions of the idiographic medians in order to avoid artificial multimodality

The median can be an integral number, or a number between two integers (i.e., 0.5; 1.5; 2.5; or 3.5 on a scale from 0 to 4). Medians ending in .5 are much rarer than the integral numbers, because it rarely happened that the split-half position lies between two numbers (e.g., between 1 and 2). This discrepancy in the frequency of integer medians and those ending in .5 complicates the categorization of modality. The rarity of medians ending in .5 causes dips in the density curve between the more common integer medians, resulting in artificial multimodality due to methodological artifacts.

To avoid that, we split the value for the appropriate medians evenly between the adjacent medians. For instance, if two within-person (idiographic) medians of 1.5 were observed for an item, we re-assigned one of them to the frequency of the medians of 1.0 and the other to the frequency of medians of 2.0. If only one median ending in .5 was observed (e.g., one median of 1.5), we added one count of 0.5 each to the next higher and the next lower integral and deleted the 0.5 value.

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A word on standardization in longitudinal studies: don't

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Keywords: standardization, ipsatization, longitudinal data, profiles, experience sampling method (ESM)

> *"Typically, you don't want to do a full z-score standardization of each variable, because then you lose the covariance metric that is needed for the SEM procedures, and you lose any information about mean-level changes over time."*
>
> *(Little, 2013, p. 18)*

This article discusses the risks of standardization and ipsatization in longitudinal studies. First, it summarizes some common purposes of standardization in psychological studies. Second, it explains why and when standardization and ipsatization are problematic in the analysis of longitudinal data and profiles. Third, it shows alternative ways to achieve similar purposes while avoiding the risks.

Definitions and Purposes of Z-standardization and Ipsatization

Z-standardization and ipsatization are procedures to transform absolute values, or ratings (e.g., 1 = *don't agree at all* to 7 = *totally agree*) to relative scores that reflect each answer's rank in comparison to the ranks of all responses in that sample. In z-standardization, the sample mean score is subtracted from each single observation, and this difference is then divided by the sample's standard deviation. The result is a scale where a score of 0 means that this observation was at the sample's mean level, and a z-score of 1 reflects an observation one standard deviation above the sample mean. Ipsatization also converts absolute ratings into relative ranks, but relates each answer to the individual's own mean, not the sample mean. An individual's responses are ipsatized by subtracting the individual's mean score from each response the individual gave in a questionnaire. A positive ipsatized score means that the individual rated this item higher (which often means: affirmed more strongly), than the average of other items in that questionnaire.

Standardization and ipsatization are applied for the following purposes:

(1) Standardization is used to bring variables with different response scales (e.g., a scale from 1 = *don't agree at all* to 7 = *totally agree* and another from 0 = *don't agree at all* to 10 = *totally agree)* to a comparable metric.
(2) Z-standardized scores are displayed in graphs to accentuate the mean-level differences between groups or profiles of observations.
(3) Ipsatization is used to account for uniform response biases, such as acquiescence (=tendency to affirm all items). For instance, in cross-cultural comparisons, items are often ipsatized to account for culture-specific response biases (Tweed and DeLongis, 2006). For the same purpose, within-person standardization is applied in intensive longitudinal studies with many observations per person across short time spans (e.g., experience sampling method, see Csikszentmihalyi and Schneider, 2000). While ipsatization refers to the individual's mean score across all variables, intra-individual standardization refers to the individual's mean of one variable across multiple observations. The resulting ipsatized and intra-individual z-scores reflect whether a response was "high" compared to other responses of the same individual. In

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regression analyses, the predictor variable is often ipsatized at the mean of the sample or group in order to make the intercept meaningfully interpretable (="centering," see Enders and Tofighi, 2007).

Problems Arising through Standardization and Ipsatization

While standardization and ipsatization are easy and widely accepted, there are many constellations in which these procedures are not useful or misleading. For cross-sectional studies, these issues have long been discussed (e.g., Fischer and Milfont, 2010), but additional problems arise when longitudinal data or profiles are examined. The reasons for the additional problems of standardization in longitudinal and nested data are the many additional possibilities to relate ratings to different reference frames and distributions. In longitudinal studies, the questions arise: Shall we standardize within time points, or across them? Standardize within or across individuals? Standardize within or across age groups/cohorts? The resulting problems are:

(1) Standardizing repeated measures within individuals impedes examining mean-level differences between individuals, because each individual's mean score becomes zero. The standardized means don't inform whether the individuals differed in their original experiences.
(2) Standardization across individuals within measurement time points impedes examining mean level changes from one time point to another, because all means at all time points become zero, whereas the raw-score means might have shown a decrease in the measured variable, such as interest (see e.g., Denissen et al., 2007).
(3) Standardization across individuals across time points obfuscates the information about the relative rank of an individual at given time points, and impedes disentangling rank-order and mean-level stability. For instance, Anna might have had relatively high interest in grade one and grade three, compared to others at the same time point. However, since interest often decreases with time, Anna's absolute interest was much lower in grade 3 than grade 1, as was everybody else's. With standardization across time points and individuals, the information about the time-point-specific relative rank-order gets mixed with the mean-level change, and it will look like Anna had high interest at time one but somewhat low or medium interest at time two.
(4) Standardization across individuals within age groups/cohorts impedes studying age differences at given time points. For instance, in a study that examined three cohorts (6th, 8th, and 10th grade) in 3 years (1992, 1995, 1997; see Csikszentmihalyi and Schneider, 2000), standardizing across individuals within each cohort or year impedes examining whether a mean score changed from the 6th to 8th grade, or from 1992 to 1995, because they all become zero.
(5) Misinterpretation of differences between profiles and groups is likely when z-standardized scores are used to compare these profiles, particularly if the variables differed in their means and variances prior to the transformation. Two problems complicate interpreting group differences based on z-scores: First, the z-scores represent ranks in relation to other individuals, but not the degree to which an item was affirmed by a given individual. If an item had a low sample mean score, then a "high" z-score above 0 (above the sample mean) can represent a "rather not" statement below the midpoint of the original response scale (see Moeller et al., in press). Second, plotting group differences using z-scores often makes eventually small differences look big, compared to a graph displaying the complete original response scale and raw scores. The reason is that z-score-based graphs often show only the part of the distribution where profiles differed, instead of the complete range of possible answers (e.g., Tuominen-Soini et al., 2011). This is similar to a graph with a truncated y-axis, which is considered misleading (e.g., Rovezzi Carroll and Carroll, 2002).
(6) Standardization across individuals should not be done with ipsatized scores, because that entangles the intra-individual frame of reference (ipsatization) and the inter-individual frame (standardization) and is hard to interpret.
(7) Ipsatization changes the covariance matrix in a way that makes the data unsuitable for correlational techniques like exploratory and confirmatory factor analysis, structural equation modeling, and multivariate techniques like multiple regression and multivariate analysis of variance (Cornwell and Dunlap, 1994; Closs, 1996; Chan, 2003).

Due to the complexity of longitudinal data and analyses, the above-described problems often co-occur. For instance, standardizing situation-specific repeated measures across individuals increases the risk of misinterpreting mean differences of z-scores between situation-level profiles of state measures, because the z-standardized situation-specific measures are at the same time determined by the intra-individual distribution of these variables (see problem no. 5), and the inter-individual distribution of these variables (see problems 2–5). This makes it almost impossible to interpret whether a relatively high rank (z-score) represents a variable that was rated as "high" on the original response scale by a specific person in a specific situation. For an example of intertwined standardization problems, see Denissen et al. (2007), who applied two different standardization strategies (within individuals across measurement time points; and across individuals within time points), and then compared within-time-point profiles of the standardized variables. This strategy includes three risks: those related to standardization within individuals (problem 1), those related to standardization within time points (problems 3 and 4), and those related to misinterpretation of profile mean scores (problem 5).

Alternatives to Standardization and Ipsatization

For bringing differently measured items to the same metric, there are several easy alternative monotonous scale transformations available, which, unlike standardization, do not change the multivariate distribution and covariance matrix of the

transformed variables. One solution is the proportion of maximum scaling ("POMS") method (Little, 2013), which transforms each scale to a metric from 0 (=minimal possible) to 1 (=maximum possible), by first making the scale range from 0 to the highest value, and then dividing the scores by the highest value.

$$\text{POMS} = [(observed - minimum)/(maximum - minimum)]$$

For instance, for a scale that originally ranged from 1 to 7, first the value 1 is subtracted from each observation to make the scale go from 0 to 6, and then each score is divided by 6 to make the scale go from 0 to 1. Contrary to standardization, this maintains the proportions of the absolute distances between the observed response options.

Another possibility is the percent of maximum possible ("POMP") method (Cohen et al., 1999), which makes each scale range from 0 (=minimal possible) to 100 (=maximum possible) by multiplying the result of the POMS transformation by 100. The resulting POMP-transformed scores can be interpreted as percentages of the possible maximum score. SPSS syntaxes for these transformations can be downloaded freely (Moeller, 2015).

For examining mean-level differences between profiles and groups, raw scores or scales transformed with the POMS or POMP method can be used. This has the advantages that the scores reflect the individual's degree of affirmation/rejection of the items, and that group differences are displayed in the correct proportions. For a discussion of further advantages and alternative transformations, see Little (2013), and Cohen et al. (1999).

To account for uniform response bias such as acquiescence, a common-method factor can be modeled in structural equation models (Billiet and McClendon, 2000; Geiser and Lockhart, 2012). For instance, a latent variable with similar factor loadings on all observed answers in the questionnaire can account for the response tendency that all observed answers had in common. The advantage over ipsatization is that the covariance metric remains useful for all exploratory and confirmatory factor analyses (EFA, CFA), and structural equation modeling (SEM). The procedure can be adapted to account for non-uniform response biases (e.g., combined affirmation bias to positively perceived questions, and negation bias to negatively perceived questions, as relevant in symptom validity assessment). If instead ipsatization is used with EFA, CFA, or SEM, then the ipsatization procedure must be modified (see Chan and Bentler, 1993; Cheung and Chan, 2002). To account for response biases in the analysis of profiles, method factors and raw- or POMS-scores can be combined in factor-mixture models (Lubke and Muthén, 2005; Leite and Cooper, 2010).

With both ipsatization and method factors, it remains difficult to disentangle biased response styles from genuine experiences. For instance, some individuals really are interested in a broad variety of topics (=affirm all interest items) and do not show are clear interest profile with high interest in some and low interest in other topics (Rounds and Tracey, 1993). To disentangle scale usage from genuine experiences, it helps including contradicting items and constructs in the questionnaire, or using validity scales, e.g., to assess the tendency to generally affirm items disregarding their content, or assessing social desirability. The research on "scale usage heterogeneity" provides further tools for this purpose (Rossi et al., 2001, 2005).

Summary

Z-standardization is a widely used procedure, applied for getting rid of acquiescence and other response biases, bringing variables of different metrics to the same metric, and emphasizing differences between groups in graphs.

In longitudinal data and analyses of subgroups of observations, z-standardization leads to a number of problems. It changes in often undesirable ways the distances between observations, and the multivariate distributions of cross-sectional and longitudinal data. The psychological literature is rich in examples of misinterpreted z-scores, some of which were described in this article. While many pitfalls are known for cross-sectional studies, longitudinal studies add further problems, due to confounded frames of reference (the original response scale, the intra-individual distribution, the inter-individual distribution within given time points, the inter-individual distribution across different time points, the variation within vs. between cohorts, and any combinations of these). Generally, it is not insightful to first standardize variables within units (individuals, cohorts, states, organizations) and then compare mean scores across these units that gave the reference frame for standardization. This should be trivial, but can often be observed in the current research, and is easily overseen or mishandled the more units and reference frames are added to the data structure.

Modeling common-method factors is a useful alternative to account for response biases while avoiding the downsides of ipsatization. Alternative easy monotonous scale transformations are available to get items with different response scales to the same metric (Cohen et al., 1999; Little, 2013). Given the ease and wide acceptance of standardization in the psychological literature, it seems necessary to emphasize the risks and possible misinterpretations during the methodological training, writing and review processes in psychology. As Little (2013) pointed out, it seems wise to avoid standardization in longitudinal data analyses and person-oriented analyses, unless the researcher is fully aware of and able to avoid undesirable consequences. There are many good uses for these procedures, but also many risks.

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So You Want to Do ESM? 10 Essential Topics for Implementing the Experience-Sampling Method

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Abstract

The experience-sampling method (ESM) captures psychological experiences over time and in everyday contexts, thereby offering exciting potential for collecting more temporally fine-grained and ecologically valid data for psychological research. Given that rapid methodological developments make it increasingly difficult for novice ESM researchers to be well informed about standards of ESM research and to identify resources that can serve as useful starting points, we here provide a primer on 10 essential design and implementation considerations for ESM studies. Specifically, we (a) compare ESM with cross-sectional, panel, and cohort approaches and discuss considerations regarding (b) item content and phrasing; (c) choosing and formulating response options; (d) timescale (sampling scheme, sampling frequency, survey length, and study duration); (e) change properties and stationarity; (f) power and effect sizes; (g) missingness, attrition, and compliance; (h) data assessment and administration; (i) reliability; and (j) replicability and generalizability. For all 10 topics, we discuss challenges and—if available—potential solutions and provide literature that can serve as starting points for more in-depth readings. We also share access to a living, web-based resources library with a more extensive catalogue of literature to facilitate further learning about the design and implementation of ESM. Finally, we list topics that although beyond the scope of our article, can be relevant for the success of ESM studies. Taken together, our article highlights the most essential design and implementation considerations for ESM studies, aids the identification of relevant in-depth readings, and can thereby support the quality of future ESM studies.

Keywords

experience-sampling method, ESM, ecological momentary assessment, EMA, ambulatory assessment, AA, daily life methods, intensive longitudinal data, ILD

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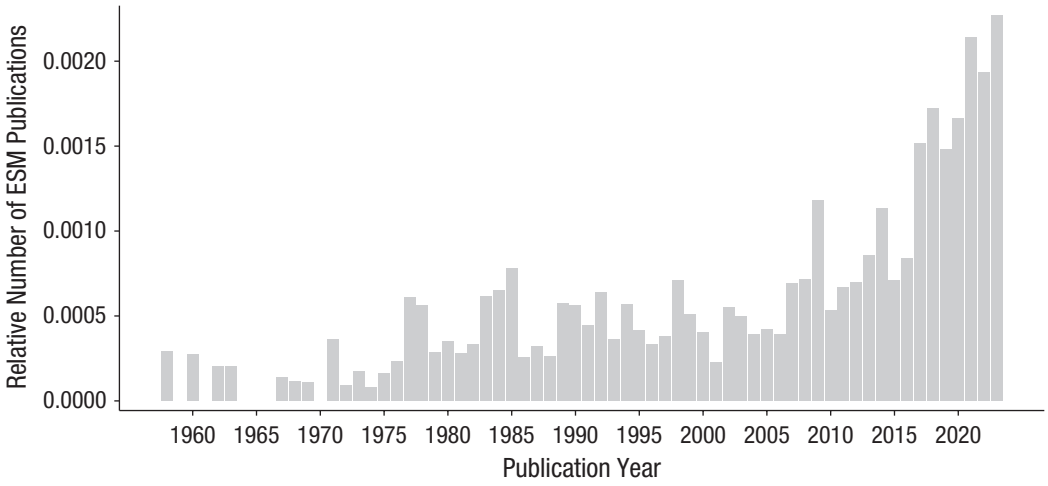

Fig. 1. The relative number of experience-sampling-method (ESM) publications on Web of Science (WoS). This figure shows the number of ESM publications on WoS per year since 1958 compared against the number of all psychological publications on WoS per year since 1958. For the absolute numbers of ESM and of all psychological publications on WoS, see Appendix B.

The experience-sampling method (ESM) is "a structured diary method in which subjects are asked in normal daily life to report their thoughts, feelings and symptoms, and also the context" (Myin-Germeys et al., 2009, p. 1533). "The longitudinal character of the ESM enables the study of dynamic patterns over time, thus capturing the film rather than a snapshot of daily life reality" (Myin-Germeys et al., 2009, p. 1539). ESM has also been referred to by other terms, including "ecological momentary assessment," "ambulatory assessment," "real-time data capture," "intensive longitudinal," and "time-series" research (Trull & Ebner-Priemer, 2009). Although some authors have used these terms interchangeably, others have suggested that the terms are associated with slightly different assessment techniques (e.g., ambulatory assessment is often used when physiological data are collected; Trull & Ebner-Priemer, 2009). Throughout, we use the term "ESM" to describe these methods.

Psychological research has been criticized for its limited ecological validity—a type of validity that denotes the degree to which research results can be generalized to the everyday context (Moeller, Dietrich, et al., 2023). This is an understandable criticism given that (quantitative) psychological research is usually either limited in the number of observations (e.g., observational studies with cross-sectional, panel, or cohort designs[1]), the setting (e.g., tightly controlled experimental lab studies), or both. Studies with limited observations or settings do not necessarily capture the dynamic and context-dependent nature of the human experience and can therefore not automatically be generalized to the everyday context. As a method that can effectively capture psychological experiences over time and in various everyday contexts, ESM offers exciting potential to mitigate this limitation (Trull & Ebner-Priemer, 2020).

In recent years, there has been a noteworthy surge in ESM publications, which seems to surpass a mere general increase in publications (see Fig. 1): More than a quarter of all ESM articles listed on Web of Science (372 of 1,269; 29.31%) were published in only the past 3 years (for details of our literature search strategy, see Appendix A). The growing use of ESM in psychological research is often attributed to the simultaneous advances in digital technologies that facilitate ESM research (e.g., Kuppens et al., 2022). By using smartphones, smartwatches, or other wearables, researchers can assess people's behaviors, thoughts, and feelings with an increasing ease and temporal resolution (e.g., collecting subjective self-report or objective biophysiological or mobile-sensing data[2]; e.g., Kuppens et al., 2022). These developments provide researchers with unique opportunities to investigate complex psychological processes that evolve dynamically both within a single individual and at large scale in populations.

Because technological advancements of the past years moved ahead at a rapid pace, it is unsurprising that the ESM field continuously progressed into new generations of tools and methods. However, these developments make it increasingly difficult for novice ESM researchers to be well informed about standards of ESM research and to identify resources that can serve as useful starting points for the design and implementation of ESM studies. Therefore, we provide a primer (foremostly) for novice ESM researchers on essential design and implementation considerations. This article originated from a panel discussion[3] with experienced ESM researchers, during

which we put a special focus on ESM design and implementation topics that are often not included in peer-reviewed articles (e.g., because of their perceived tangential nature regarding the content focus of the article or because of word counts). Most of the topics that we discuss in this article are to some degree similar for studies that are more limited in the number of observations (e.g., cross-sectional, panel, or cohort studies) and/or the setting (e.g., lab experiments). However, because of the high sampling frequency and the emphasis on ecological validity and temporal processes, some design and conduction aspects are unique to ESM (e.g., ESM-specific sampling schemes), and many require tailoring to the ESM context (e.g., power, reliability, and replicability).

We direct the reader elsewhere for (a) reporting guidelines for ESM studies (e.g., Liao et al., 2016; Stone & Shiffman, 2002; Trull & Ebner-Priemer, 2020), (b) reviews of methods and practices for ESM research (e.g., Bolger et al., 2003; Trull & Ebner-Priemer, 2020), (c) discussions on current and future applications of ESM (e.g., Mestdagh & Dejonckheere, 2021; Myin-Germeys et al., 2009, 2018), (d) analytical considerations (e.g., Hamaker & Wichers, 2017), (e) insights into conducting ESM studies in specific psychological subdisciplines (e.g., affective sciences: Kuppens et al., 2022; personality psychology: Kaurin et al., 2023), and (f) in-depth handbooks on how to conduct ESM research (e.g., Bolger & Laurenceau, 2013; Conner & Mehl, 2012; Hektner et al., 2007; Myin-Germeys & Kuppens, 2021). Instead, we provide a brief overview of the following 10 topics that we believe are essential for the quality of ESM design and implementation, discuss current challenges, and—if available—highlight potential solutions:

1. study type and research questions;
2. item content and phrasing;
3. choosing and formulating response options;
4. timescale (sampling scheme, sampling frequency, survey length, and study duration);
5. change properties and stationarity;
6. power and effect sizes;
7. missingness, attrition, and compliance;
8. data assessment and administration;
9. reliability;
10. replicability and generalizability.

For each of these topics, we additionally provide a selection of literature as starting points for more in-depth readings. Moreover, we have compiled a living, web-based resources library[4] with a more extensive catalogue of literature to facilitate further learning about the design and implementation of ESM studies. We hope that the resources library will be continually edited and improved by the vibrant ESM community so that it provides an ever more extensive and up-to-date overview of the available literature.

Study Type and Research Questions

In ESM studies, data are typically collected at least once per day in the participants' everyday lives (Trull & Ebner-Priemer, 2014). Over several days, weeks, or months, this results in so-called intensive longitudinal data (Hamaker & Wichers, 2017). ESM enables researchers to distinguish between within-subjects and between-subjects effects (e.g., Voelkle et al., 2014). To explain the differences between these effects, we use the relationship between rumination and anxiety as a running example.

The between-subjects approach compares multiple people against one another, thereby examining effects based on *interindividual* variance (Schuurman, 2023). This approach answers research questions such as "Do people who tend to ruminate more than others tend to also have higher levels of anxiety than others?" and necessarily requires data from multiple participants but can be applied to data from just one time point (e.g., cross-sectional data). For the between-subjects approach, researchers can therefore rely on data from either (a) many participants on one time point (e.g., cross-sectional data; although this comes with notable assumptions because cross-sectional data are technically not pure measures of between-persons associations; for details, see Hamaker, 2023), (b) many participants over a few time points (e.g., prospective panel or cohort data; Hamaker & Wichers, 2017), or (c) many participants over many time points (i.e., intensive longitudinal data).

The within-subjects approach typically[5] compares one person or multiple people against themselves over time, thereby examining *intraindividual* variance over time. This approach answers research questions such as "Does the average person tend to ruminate more than usual when she has a higher level of anxiety than usual?" and does not necessarily require data from multiple participants but requires data from multiple time points (for exceptions, see Note 5 and Schuurman, 2023). For the within-subjects approach, researchers can rely on data from either (a) many participants over several time points (e.g., prospective panel or cohort data; Hamaker & Wichers, 2017), (b) many participants over many time points (i.e., intensive longitudinal data), or (c) a few participants over many time points (i.e., multiple sets of single-case time-series data or intensive longitudinal data).

A specific case of the within-subjects research approach is when only one participant is assessed over many time points, which is also referred to as "*N*-of-1"

Table 1. Comparing Features of Cross-Sectional, Panel, Cohort, and ESM Studies

Study type	Cross-sectional	Panel or cohort[a]	ESM
Dynamic		(x)	x
Between-subjects research questions	(x)	x	x
Within-subjects research questions	(x)	x	x
N-of-1 research questions			x

Note: Parentheses "(x)" indicate the presence of noteworthy assumptions. ESM = experience sampling method.
[a]As the terms panel study and (single or accelerated) cohort study are inconsistently used and defined in the literature, we note that we specifically refer to longitudinal designs that assess a group of participants at several (mostly 2 to 10) time points (as opposed to assessing different groups of people at several time points, such as in repeated cross-sectional designs; Lugtig & Smith, 2019).

approach (or single-case, idiographic, or person-specific approach). Imagine our participant of interest is named Beth; this approach would answer research questions such as "Does Beth tend to ruminate more than usual when she has a higher level of anxiety than usual?" The *N*-of-1 version of the within-subjects approach is particularly fruitful when one either cannot recruit multiple participants, for example, because the construct or context of interest is rare (e.g., Wichers & Groot, 2016), or when the data are supposed to aid person-specific interventions (e.g., personalized feedback in clinical care; Bos et al., 2022; Bringmann et al., 2021; Piccirillo et al., 2019; Wright & Woods, 2020; Zuidersma et al., 2020).[6] In sum, because ESM studies collect intensive longitudinal data, they can more effectively capture the dynamics (i.e., temporal properties) of psychological experiences than cross-sectional, panel, and cohort studies and can also answer person-specific research questions (see also Table 1).

Last but not least, ESM studies are—by virtue of temporally fine-grained assessments—uniquely suited to answer research questions regarding the timescales of psychological experiences in real life and real time. For example, ESM studies are uniquely suited to answer questions such as "How strongly are past rumination levels (an hour, a few hours, or a day ago) associated with current anxiety levels?"; "How long does it generally take for people to return to their usual level of anxiety?"; "How stable are rumination levels (over an hour, a few hours, or a day)?"; "In which order do anxiety and rumination levels change over time?"; and many more.

Taken together, the foremost challenge for the researcher is to determine whether the question of interest is indeed best addressed with an ESM study. One critical consideration therein is that because of the need for frequent and time-sensitive assessments, ESM studies can be more burdensome for the participants than cross-sectional, panel, or cohort studies. This is especially the case for *N*-of-1 approaches because those depend solely on the number of observations collected for that one specific person (in other words, while in panel, cohort, and multiple-participant ESM analyses statistical power can be achieved through both the number of participants and the number of time points, in fully idiographic *N*-of-1 analyses, power can be achieved solely through the number of time points at which the respective person under examination has provided data; e.g., Mansueto et al., 2023).

Starting Points for In-Depth Readings

- Hamaker, E. L. (2012). Why researchers should think "within-person": A paradigmatic rationale. In M. R. Mehl & T. S. Conner (Eds.), *Handbook of research methods for studying daily life* (pp. 43–61). The Guilford Press.
- Molenaar, P. C. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement, 2*(4), 201–218. https://doi.org/10.1207/s15366359mea0204_1
- Schuurman, N. K. (2023). *A "within/ between problem" primer: About (not) separating within-person variance and between-person variance in psychology*. PsyArXiv. https://doi.org/10.31234/osf.io/7zgkx
- Wright, A. G. C., & Woods, W. C. (2020). Personalized models of psychopathology. *Annual Review of Clinical Psychology, 16*(1), 49–74. https://doi.org/10.1146/annurev-clinpsy-102419-125032

Item Content and Phrasing

After having determined that a research question is best addressed using ESM, the next step is to choose the relevant items or assessment sources to capture the variables of interest (e.g., subjective self-report or other-report items, objective data such as heart rate or smartphone sensor data). In this article, we specifically focus on self-report (or other-report) items. Given the dearth of measures developed specifically for ESM, researchers frequently create their own ESM items—either by adapting items from existing retrospective self-report questionnaires or by generating entirely new items—to assess the construct of interest. To keep ESM questionnaires short, the construct of interest is often assessed with a single item (Horstmann & Ziegler, 2020; J. Song et al., 2023; Wright & Zimmermann, 2019). Together with the lack of expert consensus regarding how best to determine the psychometric properties of

ESM items (e.g., reliability and validity; Dejonckheere et al., 2022; see also the Reliability section), this means that psychometrically validated ESM questionnaires are a rarity (Wright & Zimmermann, 2019). A number of sources (e.g., Eisele et al., 2021; Myin-Germeys et al., 2018; Palmier-Claus et al., 2011, 2019) provide guidance on item design, but there are no empirically supported "gold-standard" criteria for developing good ESM items. Currently, such "gold-standard" criteria are being developed (Eisele et al., 2024) as part of the ESM Item Repository (Kirtley et al., 2024; www.esmitemrepository.com), an open-science initiative that aims to increase the transparency and validity of ESM items. However, to date, ESM researchers are left in the uncomfortable position of making decisions about item content and phrasing with little empirical evidence to support those decisions (e.g., Eisele et al., 2021).

As described above, there are numerous unanswered questions regarding ESM items. Typical questions regarding item design relate to (a) item comprehensibility and psychometric properties, (b) the number and selection of items, and (c) the time frame that the items are assessing. Often, the first challenge is to ensure participants understand the items because confusing or illogical wording may increase participant burden and decrease compliance, thus reducing data quality (Eisele et al., 2022). The second challenge is to determine an appropriate length for the questionnaire. Shorter ESM questionnaires can sometimes result in better compliance and data quality (Eisele et al., 2022). However, researchers must ensure that developing short questionnaires does not come at the expense of validity. Focus groups or cocreation sessions with the target population can help with the operationalization of the construct of interest, such as with choosing the relevant content and an appropriate number of items (Soyster & Fisher, 2019; Vogt et al., 2004). Moreover, piloting the chosen or developed ESM items is recommended, ideally with the inclusion of a qualitative component (e.g., a "think-aloud study"; Ericsson & Simon, 1998), to guide further item refinement. Both those endeavors are especially necessary when new items or measures are being developed. A thorough briefing of participants before the commencement of the study creates the opportunity for aiding participants' understanding of items and is useful for improving adherence to the sampling protocol (Rintala et al., 2021)—this endeavor is relevant for all ESM studies regardless of using new or already established items.

Another important consideration when designing or selecting ESM items is whether the time frame of reference for the item (e.g., "since the last beep" or "right now") is appropriate (Deakin et al., 2022). Simply adding "right now" to an item that has been adapted from a retrospective trait-type questionnaire does not make the item suitable for ESM studies (Myin-Germeys et al., 2018). Researchers should also be mindful that although it may be desirable to capture certain behaviors, such as self-harm, "in the moment," this may realistically not be possible. Therefore, either asking about these behaviors "since the last beep" or assessing them only contingent to when they occur (i.e., event-contingent assessments that are initiated by the participant) may be a better option. Alternatively, if it is essential for the design of the study to assess equally spaced time windows, one could ask about these behaviors "in the last day," or when real-time anchoring is at the core of the study, one could ask for the behaviors "in the last hour" (see e.g., Kuppens et al., 2022; Medland et al., 2020).

In sum, unknowns regarding the construction of ESM items will be resolved only when the field devotes substantive empirical research to this topic. More knowledge and better guidance will not happen overnight but will be expedited by researchers increasing transparency about how and why items or measures have been selected, modified, or created—an essential step that all ESM researchers can adopt in their research workflow today. Moreover, to further facilitate the availability and refinement of validated ESM items, researchers are encouraged to submit their validated items to the ESM Item Repository (Kirtley et al., 2024; www.esmitemrepository.com).

Starting Points for In-Depth Readings

- Eisele, G., Kasanova, Z., & Houben, M. (2021). Questionnaire design and evaluation. In I. MyinGermeys & P. Kuppens (Eds.), *The open handbook of experience sampling methodology: A step-by-step guide to designing, conducting, and analyzing ESM studies* (pp. 71–89). KU Leuven. https://www.kuleuven.be/samenwerking/real/real-book
- Kirtley, O. J., Eisele, G., Kunkels, Y. K., Hiekkaranta, A. P., Van Heck, L., Pihlajamäki, M., Kunc, B., Schoefs, S., Kemme, N. D. F., Biesemans, T., & Myin-Germeys, I. (2024, July 2). *The experience sampling method (ESM) item repository*. PsyArXiv. https://doi.org/10.17605/OSF.IO/KG376

Choosing and Formulating Response Options

Once the items have been selected, the next step is to choose and formulate response options. ESM researchers typically use either a Likert scale or a visual-analogue scale (VAS). A Likert scale consists of multiple response options that, for example, reflect the extent of agreement,

frequency, or severity (e.g., 1 = *not at all*, 4 = *moderate*, 7 = *very much*). The number of response options can vary but often ranges from three to 10 and generally assesses data that are seen as discrete (Eisele et al., 2021). In contrast, a VAS often ranges from 0 to 100 (e.g., 0 = *not at all*, 50 = *moderate*, 100 = *very much*) and assesses data that are seen as continuous (Eisele et al., 2021). However, this distinction is not clear-cut. For example, when a Likert scale has seven or more answer options, it often starts to resemble a continuous scale, and in a similar vein, a VAS can be used in a rather discrete way when participants use only some parts of the scale (e.g., only the outer ends and middle part of the scale) or when researchers discretize the scale post hoc into ranges. ESM researchers sometimes also use binary (or dichotomous) response scales, which is a specific case of discrete data that have only two response options (e.g., 1 = *yes* and 0 = *no*). Finally, ESM researchers can choose to use open-text boxes as answer format so that participants can note down their individual answer in a nonstructured or semi-structured way.

For ESM researchers who want to proceed quantitatively, the first challenge is that they are left with the dilemma of having to choose between a binary, a Likert, and a VAS response format, which all come with noteworthy caveats. The upside of binary-response scales is that they are rather simple and efficient and can nonetheless capture relevant information for psychological constructs for which a more fine-grained assessment does not automatically result in more informative findings (e.g., Fisher, 2023; for an example on presence/absence vs. the severity of adverse experiences, see Schlechter et al., 2021). The downside of binary-response scales is, however, that the resulting data per definition do not follow a normal distribution, and not all analytic techniques for intensive longitudinal data can deal with this. Therefore, Likert scales or VASs are often preferred over binary scales. When deciding for a Likert scale, one generally risks the chance of overdiscretizing the answer options, which can lead to an insufficient variance. When deciding for a VAS, it is not always clear how to interpret response values and small differences therein. For example, researchers do not have a good idea of what a score of 76 really means or what the participant thinks it means. Moreover, a difference of 73 and 76 could be a meaningful difference but could equally just be measurement error because of the width of the participant's finger (Boring et al., 2012; Le et al., 2018; Mayer et al., 2018). That said, researchers can, however, also not be sure that all participants interpret the response options on Likert scales in the same way, particularly not if some response options have no label. From other fields of research, there are some indications that VASs might lead to more endorsement of the end points of the rating scale (i.e., the extreme responses) than Likert scales (Studer, 2012), and generally speaking, Likert scales have been suggested to be most user-friendly (Couper et al., 2006; Eisele et al., 2021). Therefore, although data assessed with VASs tend to be continuous—which is often considered advantageous for statistical analyses—VASs do not seem to provide a clear advantage to Likert response scales (e.g., Haslbeck et al., 2023; Müssig et al., 2022; Simms et al., 2019).

A second challenge that ESM researchers face, potentially regardless of using Likert scales or VASs, is that the number and location of anchor points may well matter. For example, evidence suggests that changes in scale anchors can alter response distributions (Matejka et al., 2016). And a third challenge is that the participants' interpretation of the response options may change over time, suggesting (no or) low longitudinal measurement invariance, which limits the validity of the items.

Even though not much is known yet about what an optimal response scale looks like for ESM items, the general advice is to keep the response scale consistent throughout a (construct-specific) set of ESM items (Eisele et al., 2021). Moreover, we recommend to use at least one of the following three strategies, which likely result in a more homogeneous and reliable use of the response-scale format: (a) piloting the chosen response-scale format with the target population (e.g., in form of a "think-aloud study"; Ericsson & Simon, 1998), (b) briefing participants before the commencement of the study regarding the response-scale format (e.g., in form of written or video instructions), and (c) examining the response distributions of the items to retrospectively increase understanding of how the scale has been used. It may also be useful to include an open-text box alongside quantitative response options so that participants can additionally record individual answers. Using additional open-ended answer formats can be particularly useful when research questions or items are exploratory (van Roekel et al., 2019) and when nonvalidated items or response options are used and researchers want to better understand response distributions and response shifts.

Starting Points for In-Depth Readings

- Couper, M. P., Tourangeau, R., Conrad, F. G., & Singer, E. (2006). Evaluating the effectiveness of visual analog scales: A web experiment. *Social Science Computer Review, 24*(2), 227–245. https://doi.org/10.1177/0894439305281503
- Haslbeck, J. M. B., Ryan, O., & Dablander, F. (2022). *Multimodality and skewness in emotion time series*. PsyArXiv. https://doi.org/10.31234/osf.io/qudr6

- Matejka, J., Glueck, M., Grossman, T., & Fitzmaurice, G. (2016). The effect of visual appearance on the performance of continuous sliders and visual analogue scales. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (pp. 5421–5432). Association for Computing Machinery. https://doi.org/10.1145/2858036.2858063
- Schwartz, C. E., Sprangers, M. A., Carey, A., & Reed, G. (2004). Exploring response shift in longitudinal data. *Psychology & Health, 19*(1), 51–69. https://doi.org/10.1080/08870440310001184560

Timescale (Sampling Scheme, Sampling Frequency, Survey Length, and Study Duration)

Inherent to designing ESM items and response scales are timescale decisions. Timescale decisions usually comprise (a) the sampling scheme (i.e., a fixed, random, semirandom, or event-contingent scheme), (b) the sampling frequency (i.e., the number of assessments per day, also sometimes referred to as "sampling density"), (c) the prompt depth and survey length (i.e., the number of items and the time it takes to assess them), and (d) the study duration (Dejonckheere & Erbas, 2021; Kaurin et al., 2023). Regarding the sampling scheme, fixed schemes consist of preselected assessment time points, whereas random schemes consist of assessments on randomly selected time points. Semirandom schemes consist of assessments that occur at a random time within a predetermined time frame, to ensure that each day has an equal number of assessments (and, on average, equal time windows between assessments; Janssens et al., 2018). Fixed sampling schemes are sometimes also referred to as "interval-contingent schemes," and (semi) random schemes are also referred to as "signal-contingent schemes." In these schemes, assessments are initiated by prompts (Himmelstein et al., 2019). In contrast, event-contingent schemes consist of assessments that take place whenever the event of interest happens (e.g., social interactions; Dawood et al., 2020; Himmelstein et al., 2019; Stadel et al., 2024), that is, assessments are typically initiated by the study participants.

Besides the sampling scheme, timescale decisions regarding sampling density, prompt depth, and study duration—also referred to as "3Ds" (see Kaurin et al., 2023)—vary widely across ESM studies because timescale decisions should align with the theory and research question of interest (Bolger & Laurenceau, 2013; Collins, 2006). Kaurin and colleagues (2023) showed in their meta-analysis that less dense samplings protocols have usually deeper prompts (i.e., longer surveys) and that more dense sampling protocols have usually a shorter study duration. Hence, the 3Ds are related to each other across studies because ESM researchers tend to decrease one when increasing another. Further discussions on timescale decisions can be found in Conner and Lehman (2012), Dejonckheere and Erbas (2021), Himmelstein and colleagues (2019), Kaurin and colleagues (2023), Leertouwer and colleagues (2021), Shiffman and colleagues (2008), Stadel and colleagues (2024), and van Roekel and colleagues (2019).

A first challenge regarding timescales in ESM studies is to choose a sampling scheme. Most sampling-scheme choices have both advantages and disadvantages. Consider social interactions as an example: Fixed or (semi) random schemes may be advantageous when beginning and end points of an event are not clearly identifiable or when events happen (nearly) continuously (e.g., Himmelstein et al., 2019), such as when participants work in settings where they interact with other people close to nonstop (e.g., preschool teachers, hairdressers, general practitioners), so that fixed or (semi)random sampling can help to ensure both a manageable and a sufficient number of assessments. On the other hand, event-contingent schemes may be advantageous for more infrequent events, such as social interactions in representative samples of college students, so that the event-contingent sampling can help record social interactions at the actual frequency and time at which they occur, increasing the ecological validity (e.g., Himmelstein et al., 2019). Although widespread lore or intuition suggests that event-contingent schemes require less retrospection than fixed or (semi)random schemes, the little empirical data that exist on this topic suggest that this might not (always) be the case (e.g., Himmelstein et al., 2019; Stadel et al., 2024).

Unless one chooses an event-contingent scheme, a second major challenge is to determine an appropriate sampling frequency for the construct of interest, because the eventual results depend in large part on the timing of the assessments (see Collins & Graham, 2002; Haslbeck & Ryan, 2022). For example, the strength of temporal (lagged) associations has been demonstrated to differ as a function of the length of the time interval between assessments (also referred to as the "lag problem"; Bolger & Laurenceau, 2013; Collins & Graham, 2002; Dormann & Griffin, 2015; Gollob & Reichardt, 1987; Hamaker & Wichers, 2017; Jacobson et al., 2019; Kuppens et al., 2022; Shiffman et al., 2008; Stone & Shiffman, 2002). Thus, the frequency of assessments needs to be based on the assumed fluctuation speed of

the construct of interest (or on the speed of its interaction with other relevant constructs of interest). A third and related challenge is that the timing of assessments should be representative (Conner & Lehman, 2012; Dejonckheere & Erbas, 2021; Stone & Shiffman, 2002), which is particularly challenging when contextual factors influence the assessment (e.g., assessing tiredness right after lunch) or when the assessment time is associated with the measured outcome (e.g., assessing tiredness in a fixed design every day during the late evening).

A fourth challenge is that even though ESM designs reduce recall bias, this bias may still occur to some extent when using retrospective questions (e.g., affect during the past day) in contrast to concurrent questions (e.g., affect right now; Neubauer, Scott, et al., 2020). Designs with retrospective questions are therefore sometimes referred to as "ecological retrospective assessment" (Leertouwer et al., 2021). Likewise, recall bias may increase when the construct of interest is assessed over a time span during which it is likely to fluctuate meaningfully (e.g., appetite level during the past day; Conner & Mehl, 2012; Kuppens et al., 2022; Shiffman et al., 2008). To address these issues, ESM researchers can increase the sampling frequency of fixed or (semi)random assessments, implement periods of measurement bursts into these assessment schemes (for more details on this strategy, see Ram et al., 2014), or use event-contingent schemes. Event-contingent sampling is especially useful when the timescale of the construct of interest is both unknown and relevant (e.g., assessing the amount of alcohol consumed right after the event as opposed to at the end of the day).

A fifth challenge is that the depth of the prompts should be sufficiently comprehensive to capture the construct of interest validly without having a completion time that burdens the participants and decreases their compliance. Here, ESM researchers could consider conducting a pilot study to qualitatively assess study burden and to quantitatively assess compliance. Moreover, in some situations, planned-missingness designs can help to systematically reduce the number of items per prompt in such a way that eventually enough data for all items are available to be able to infer the relevant estimates (for more details, see the section on missingness, attrition, and compliance; or Silvia et al., 2014). Finally, a sixth challenge is that the duration of the study will need to be attuned to (a) a period (or periods) with sufficient occurrences of the construct of interest (e.g., Helmich et al., 2021), (b) the required statistical power, (c) an acceptable level of participant burden, (d) financial resources and researcher availability, and if applicable, (e) the cyclicity of the construct of interest (e.g., week vs. weekend, seasonal, periodic, or event-related effects, such as stress among college students likely being higher around exam times than before or after exam times; Fritz et al., 2021).

Starting Points for In-Depth Readings

- Collins, L. M., & Graham, J. W. (2002). The effect of the timing and spacing of observations in longitudinal studies of tobacco and other drug use: Temporal design considerations. *Drug and Alcohol Dependence, 68*, 85–96. https://doi.org/10.1016/S0376-8716(02)00217-X
- Dejonckheere, E., & Erbas, Y. (2021). Designing an experience sampling study. In I. Myin-Germeys & P. Kuppens (Eds.), *The open handbook of experience sampling methodology: A step-by-step guide to designing, conducting, and analyzing ESM studies* (pp. 71–89). KU Leuven. https://www.kuleuven.be/samenwerking/real/real-book
- Shiffman, S., Stone, A. A., & Hufford, M. R. (2008). Ecological momentary assessment. *Annual Review of Clinical Psychology, 4*, 1–32. https://doi.org/10.1146/annurev.clinpsy.3.022806.091415

Change Properties and Stationarity

When designing the timescale of an ESM study, one aspect that is often overlooked is that the properties of the construct of interest can change systematically—which we hereafter refer to as "change." Change differs from momentary (i.e., timepoint-specific) fluctuations—which we hereafter refer to as "fluctuations," "variations," or "deviations." Change can occur in many ways. First, the mean level can change—for instance, a participant can become systematically sadder or happier (or a combination of both). Such changes are not just fluctuations, variations, or deviations from the participant's mean, but are structural changes of the construct's mean on average over time. Second, the variance of a construct can change—for example, a participant's sadness or happiness can become systematically more variable over time. Third, the association between variables can change systematically—for example, sadness and happiness can be highly related at the beginning of the study but may have no association at the end of the study. Fourth, how well an ESM item can explain itself—also known as "autocorrelation" or "inertia"—can change systematically; for example, the degree to which sadness

at one time point can explain sadness at the next time point can change over the course of the study (Bringmann et al., 2017; Koval & Kuppens, 2012).[7] And to make matters even more complex, over time, participants can begin to give a different meaning to the construct or its response options, which erroneously gives the appearance of observed change in the statistics of the construct (e.g., in the mean level; e.g., McNeish et al., 2021). Data in which the statistical properties of a process, such as the mean, the variance, or the autocorrelation, do not change over time—or in other words, data with no (or very little) change—are also called "stationary data" (Haslbeck & Ryan, 2022; Molenaar, 2004).

So what does this mean for the design of an ESM study? The first and foremost challenge is to decide whether one is interested in change and to then make appropriate choices regarding the timescale and the analytic method (Bringmann et al., 2022). If one is not interested in measuring change itself, it is critical to reduce external influences that may induce change, such as psychosocial treatment, exams, or holidays. Therefore, one could incorporate a prescreener into the study protocol to check before the study starts whether participants go on holidays or are very likely to encounter influential experiences (e.g., a wedding when the participant takes part in a study on mood). Collecting high-frequency assessments during a brief period of time may also increase the chance of capturing approximately stationary data. However, collecting truly stationary psychological data can be complex, given the dynamic nature of many psychological constructs (Bringmann et al., 2022). If one is interested in assessing change itself, or in other words, "nonstationary ESM data" (i.e., means, variances, associations, or autocorrelations change systematically over time), a longer study duration covering a period during which participants are assumed to change would be advantageous (or necessary; e.g., Helmich et al., 2021).

Planning for change is particularly important for choosing the analytic approach. Currently, many of the commonly used analytic approaches (e.g., linear multilevel models, dynamic structural equation models, or vector-autoregressive models) assume stationarity (Ariens et al., 2020). Given that there are only a few approaches that can deal well with nonstationarity, modeling nonstationary data represents a much needed (yet complex) area of future study. Note also that it matters what kind of change one would like to model. For example, modeling approaches used for detecting change in the autocorrelation require more observations than approaches used for detecting change in the mean (Bringmann et al., 2017).

Starting Points for In-Depth Readings

- Ariens, S., Ceulemans, E., & Adolf, J. K. (2020). Time series analysis of intensive longitudinal data in psychosomatic research: A methodological overview. *Journal of Psychosomatic Research, 137*, Article 110191. https://doi.org/10.1016/j.jpsychores.2020.110191
- Bringmann, L. F., Albers, C., Bockting, C., Borsboom, D., Ceulemans, E., Cramer, A., Epskamp, S., Eronen, M. I., Hamaker, E., Kuppens, P., Lutz, W., McNally, R. J., Molenaar, P., Tio, P., Voelkle, M. C., & Wichers, M. (2022). Psychopathological networks: Theory, methods and practice. *Behaviour Research and Therapy, 149*, Article 104011. https://doi.org/10.1016/j.brat.2021.104011
- Collins, L. M. (2006). Analysis of longitudinal data: The integration of theoretical model, temporal design, and statistical model. *Annual Review of Psychology, 57*(1), 505–528. https://doi.org/10.1146/annurev.psych.57.102904.190146

Power and Effect Sizes

Alongside carefully planning design features related to items and timescale, it is important to ensure adequate statistical power, that is, to ensure that sufficient observations are collected that can be manipulated via the number of time points and/or the number of participants.[8] Generally speaking, power is defined as the probability of rejecting a statistical null hypothesis when it is truly false or more colloquially, detecting an effect as statistically significant that is "truly there." Power is an important and often underestimated parameter for empirical research. Studies with low power that fail to detect true effects are problematic for determining the validity of a given construct. Some simulation studies and resulting recommendations for sample sizes and number of time points for ESM studies exist (see e.g., Lane & Gates, 2017; Mansueto et al., 2023; Schultzberg & Muthén, 2017). However, because these recommendations are highly context specific (e.g., type and complexity of the analytic model, timescale of the relevant constructs, and their interactions), we believe that a power analysis conducted for the specific analytic model and content at hand would provide much more decisive guidance. There are simulation-based tools available that can help ESM researchers to estimate the number of

participants and/or the number of time points needed to obtain sufficient power (e.g., Bolger et al., 2012; Lafit et al., 2021, 2022; Schultzberg & Muthén, 2017). However, we want to stress that calculating the statistical power for ESM studies is rather challenging (given the complexity of the time-series models) and sometimes even impossible. Many of the software resources that are available do not account for temporal dependencies, and those that do are so far available for only a limited number of analytic approaches (Lafit et al., 2021). Moreover, conducting a power analysis requires researchers to think clearly about the anticipated or postulated effect size, because an appropriate effect-size expectation is crucial to ensuring that the power calculation is accurate. Therefore, measures of effect sizes play a pivotal role in the advancement of ESM research.

The first challenge that needs to be overcome regarding power considerations is a shift in how ESM researchers approach statistical power. ESM researchers frequently rely on design decisions of prior work (e.g., "We've always recruited 80 participants and administered surveys for 14 days"), which were often determined by pragmatic considerations and not by using power analyses. Pragmatic considerations, such as the participant burden and the amount of financial resources available, are, of course, crucial, and it should be the norm to report them. Yet pragmatic considerations should not deter researchers from conducting and reporting power analyses (even if the power analysis resulted in an insignificant target effect).

A second notable challenge is that ESM researchers may not have a set expectation about the effect size, which is required when calculating statistical power. Here, we recommend that researchers either choose the smallest effect size that they still consider meaningful and want to be able to detect, or that they search the literature for similar studies and base expectations of effects on prior studies. That said, turning to the literature may leave ESM researchers disappointed because effect sizes are often not reported in a standardized metric, which is problematic for conducting power analyses. For instance, researchers might use a Monte Carlo simulation approach (e.g., using the tools developed by Lafit and colleagues, 2021, or Muthén & Muthén, 2017) and estimate the power for a regression coefficient of .10. But if this regression coefficient is unstandardized, a regression coefficient of .10 can be anything from a tiny to a huge standardized effect depending on the scale of the independent and dependent variable. To date, resources to compute and interpret standardized effect sizes are available (e.g., Bulteel et al., 2016; Jaeger et al., 2017; Rights & Sterba, 2019; Schuurman et al., 2016) but are not consistently used. Therefore, if ESM researchers would start to routinely report standardized effects, the field could together move the quality of ESM research forward. And last but not least, if a suitable power analysis seems truly implausible, we recommend using guidance from existing simulation studies (where applicable) or to at least openly report about the alternative (pragmatic) considerations taken and their potential implications.

Starting Points for In-Depth Readings

- Bolger, N., Stadler, G., & Laurenceau, J.-P. (2012). Power analysis for intensive longitudinal studies. In M. R. Mehl & T. S. Conner (Eds.), *Handbook of research methods for studying daily life* (pp. 285–301). The Guilford Press.
- Jaeger, B. C., Edwards, L. J., Das, K., & Sen, P. K. (2017). An R^2 statistic for fixed effects in the generalized linear mixed model. *Journal of Applied Statistics, 44*(6), 1086–1105. https://doi.org10.1080/02664763.2016.1193725
- Lafit, G., Adolf, J. K., Dejonckheere, E., Myin-Germeys, I., Viechtbauer, W., & Ceulemans, E. (2021). Selection of the number of participants in intensive longitudinal studies: A user-friendly shiny app and tutorial for performing power analysis in multilevel regression models that account for temporal dependencies. *Advances in Methods and Practices in Psychological Science, 4*(1). https://doi.org/10.1177/2515245920978738
- Rights, J. D., & Sterba, S. K. (2019). Quantifying explained variance in multilevel models: An integrative framework for defining R-squared measures. *Psychological Methods, 24*(3), 309–338. https://doi.org/10.1037/met0000184
- Schultzberg, M., & Muthén, B. (2017). Number of subjects and time points needed for multilevel time-series analysis: A simulation study of dynamic structural equation modeling. *Structural Equation Modeling: A Multidisciplinary Journal, 25*(4), 495–515. https://doi.org/10.1080/10705511.2017.1392862

Missingness, Attrition, and Compliance

A topic that is closely related to statistical power is how to address missing data. "The best solution to the missing data problem is not to have any," as the saying goes (Allison, 2001, p. vi). In ESM studies, this goal is usually unattainable because assessing participants throughout

their everyday activities almost assuredly means that participants will miss some of the many ESM prompts. In some situations, the participants may simply be unable (e.g., a bus driver at work) and in other situations unwilling (e.g., during a cinema visit) to respond to a prompt. Typically, researchers administer ESM prompts during certain time windows, after which they expire, to make sure that responses are recorded at the intended time and context—meaning that participants miss prompts if they are not able to complete the survey within the appropriate window. If prompts did not expire, following prompts would pile up on previously unanswered ones, thus making it increasingly unlikely for researchers to receive valid responses. Even though many ESM studies are much more intensive for participants than cross-sectional, panel, or cohort studies, reviews of the ESM literature found average compliance rates of 70% to 80% (e.g., de Vries et al., 2021; Jones et al., 2019; Wen et al., 2017; Williams et al., 2021; Wrzus & Neubauer, 2023). However, recent reviews also revealed that about 30% to 50% of ESM studies do not report compliance rates (de Vries et al., 2021; Wrzus & Neubauer, 2023), suggesting that true rates of compliance may be lower than the reported average rates of 70% to 80%. Thus, the key message is that it is unavoidable to plan for missing prompts when designing an ESM study. When planning for missing prompts, it is important to differentiate between randomly missed prompts (e.g., prompts not being received because of technological failures) and systematically missed prompts that relate to constructs of interest (e.g., patients with depressive symptoms who tend to oversleep in the morning could systematically miss early morning prompts), because systematic missingness in particular can bias the data (see Schreuder et al., 2022).

When designing an ESM study, there are generally speaking four approaches that can address missingness. First, researchers should develop a plan to increase compliance. Second, researchers can implement planned-missingness designs. Third, researchers can plan the analysis of missing-data patterns. And fourth, researchers can plan the statistical handling of missing responses based on observed responses. Regarding the first approach, researchers should set out a plan for increasing participant compliance when designing the ESM study. However, besides some mixed evidence for the suggestion that longer questionnaires are significantly associated with reduced compliance (Eisele et al., 2022; Hasselhorn et al., 2021; Williams et al., 2021), there is so far only limited empirical evidence regarding the impact of study design on compliance (see Reiter & Schoedel, 2024; van Berkel et al., 2020). The majority of the advice for increasing compliance in ESM studies is anecdotal. For example, ESM researchers attempt to increase compliance by (a) training the participants regarding the study protocol, (b) compensating based on compliance (e.g., paying participants per response or offering a "bonus" for reaching a certain compliance threshold), (c) implementing suspension windows (i.e., participants can register times at which they are unavailable to prevent getting prompts during those times), (d) making ESM studies overall as user-friendly as possible, and (e) troubleshooting directly with participants if compliance appears problematic (e.g., Burke et al., 2017; Trull & Ebner-Priemer, 2020).

The second approach—planned missing-data designs—is an approach that is implemented before the conduction of the study. It allows researchers to administer all items but to systematically reduce the number of items per prompt in such a way that eventually enough data for all items are available to be able to infer the relevant estimates (Silvia et al., 2014). This is possible because such missing data are missing completely at random and thus unrelated to the construct at hand and to other constructs (Little & Rubin, 2002; Silvia et al., 2014). However, although such a design reduces the assessment burden, unintended additional missing data are likely. Moreover, this approach seems predominantly useful when multiple items are used to assess one construct, and it requires the usage of specific analytic procedures (e.g., latent variable models) to compensate for the intended missing data, which are not (yet) available for all time-series analyses (see e.g., Silvia et al., 2014).

The third approach—planning the examination of missing-data patterns—can include analyses that help to disentangle whether missingness is completely at random (MCAR) or missing at random (MAR; some variables are associated with the missingness; e.g., Enders, 2010; Little, 1988). Although tests to identify MCAR in cross-sectional data are readily available (Enders, 2010; Little, 1988; Rubin, 2018), we were unsuccessful in finding similar methods for ESM data. Moreover, MAR pattern can technically be examined with correlational analyses that enable the identification of factors associated with missingness patterns; however, because of the wealth of temporally dependent data, even this seemingly simple endeavor can quickly prove challenging. Thus, although knowing whether missingness is MCAR, MAR, or missing not at random informs about the implications of the missing data, obtaining such knowledge is not straightforward for ESM data. It may, however, be worthwhile to implement event-related suspension windows and/or open-text boxes that can help to gather information on why responses have been missed.

For the fourth approach—handling missing prompts statistically—researchers need to be aware of two caveats. The first caveat is of a theoretical nature because one needs to decide whether it makes sense to impute

situational states (i.e., impute values for missing prompts) with statistical techniques. Among experts, this seems still to be an open debate. According to some ESM researchers, ESM prompts are highly context-specific and fluctuating in nature, which is why they argue that inferring (or estimating) values for missing prompts based on values from earlier or later prompts would be illogical. Other ESM researchers do, however, recommend to use techniques such as missing data imputation, full information maximum likelihood (FIML) estimation, or Kalman filtering (Ji et al., 2018; McNeish & Hamaker, 2020). The second caveat regarding treating missing data is of a statistical nature, because the relevant techniques (e.g., multiple imputation, FIML, or Kalman filtering; Ji et al., 2018; McNeish & Hamaker, 2020), although readily available, are not necessarily straightforward for ESM data.

Starting Points for In-Depth Readings

- Ji, L., Chow, S. M., Schermerhorn, A. C., Jacobson, N. C., & Cummings, E. M. (2018). Handling missing data in the modeling of intensive longitudinal data. *Structural Equation Modeling: A Multidisciplinary Journal*, *25*(5), 715–736. https://doi.org/10.1093/iwcomp/iwaa019
- Silvia, P. J., Kwapil, T. R., Walsh, M. A., & Myin-Germeys, I. (2014). Planned missing data designs in experience sampling research: Monte Carlo simulations of efficient designs for assessing within-person constructs. *Behavioral Research Methods, 46*(1), 41–54. https://doi.org/10.3758/s13428-013-0353-y
- Williams, M., Lewthwaite, H., Fraysse, F., Gajewska, A., Ignatavicius, J., & Ferrar, K. (2021). Compliance with mobile ecological momentary assessment of self-reported health-related behaviors and psychological constructs in adults: Systematic review and meta-analysis. *Journal of Medical Internet Research, 23*(3), Article e17023. https://www.jmir.org/2021/3/e17023
- Wrzus, C., & Neubauer, A. B. (2023). Ecological momentary assessment: A meta-analysis on designs, samples, and compliance across research fields. *Assessment, 30*(3), 825–846. https://doi.org/10.1177/10731911211067538

Data Assessment and Administration

After design decisions for an ESM study have been finalized, decisions regarding data assessment and administration need to be made. Because of the frequency with which participants are assessed, ESM data sets are often much larger than data sets from cross-sectional (panel or cohort) studies. Accordingly, participants have not only an identifying ID but also identifying time stamps that encode the time structure of the data. Given the wealth of rapidly incoming data, it is wise to carefully choose a data-collection software and to plan its implementation early on.

When deciding on an ESM software for the data collection, some initial questions should be considered, such as: What is the budget? Does the software need to be compatible for both computer and mobile screens? Should the study use an internet- or (offline-capable) app-based software? Does the software need to be able to handle complex ESM designs (e.g., with item-order randomization)? Is the data storage in line with the data-protection regulations that apply to the study? Should notifications be sent as text messages, emails, or push notifications (or something alike)? Is a feedback module required? What level of technical support is needed (e.g., around-the-clock support)? Because there is a large and consistently evolving range of software that can be used for the assessment of ESM data, it would go beyond this article to provide an exhaustive discussion of the options. However, the interested reader can find an overview that compares various features of relevant software tools in form of a crowdsourced table, "ESM & Mobile Sensing Solutions Feature Table"[9] (this effort was initiated and is led by Arslan et al., 2018). Moreover, several articles provide guidance regarding choosing ESM software (e.g., Henry et al., 2024; Trull & Ebner-Priemer, 2020; Weermeijer et al., 2021), and some also provide insights into using wearables for the passive collection of physiological and movement data (see De Calheiros Velozo et al., 2024; Mehl et al., 2023; Vaid & Harari, 2019). Deciding on the ESM software is, however, only the initial challenge regarding data assessment.

The myriad of challenges that ESM researchers face when assessing and administering their data are beyond the scope of what could be covered in this article, but we want to briefly provide a few fundamental suggestions for (novice) ESM researchers. First, if participants change time zones, the assessment software should ideally be able to accommodate those changes, because trying to correct time zones retrospectively can not only be difficult, but can also, in the worst case, bias the data (note that similar problems can occur if the study is conducted in countries that switch winter and summer time on different days). Second, when planning the data assessment, it is helpful to plan for regular compliance checks to be able to intervene on time if necessary (see also the section above on missingness, attrition, and compliance). Third, open-text fields can be a useful

addition to the data-assessment form, because these can help during data collection to spot whether participants experience technical issues they might report there. And last but not least, given the wealth of data and the complexity of their structure, it is advisable to carefully plan how variables should be coded, stored, and (ideally) shared in a trusted (public or protective-access) repository.

Starting Points for In-Depth Readings

- Arslan, R. C., Tata, C. S., & Walther, M. P. (2018). *ESM & mobile sensing solutions: Feature table*. https://comparison-to.formr.org
- Weermeijer, J., Kiekens, G., & Wampers, M. (2021). Experience sampling platforms. In I. Myin-Germeys & P. Kuppens (Eds.), *The open handbook of experience sampling methodology: A step-by-step guide to designing, conducting, and analyzing ESM studies* (pp. 103–117). KU Leuven. https://www.kuleuven.be/samenwerking/real/real-book

Reliability

Many of the topics discussed above determine (part of) the validity of ESM measurements. Reliability is another aspect that affects validity. Reliability concerns the consistency of measurements: If the true value of a construct does not vary and researchers accordingly get (approximately) the same result each time that they replicate their assessments, then they have reliable measurements. If random noise is present in addition to the true value of the construct and researchers accordingly get a (substantially) different result each time that they replicate their assessments, then they have unreliable measurements. Importantly, reliability concerns only random errors because measurements with a high amount of consistent error could still be perfectly reliable (albeit clearly invalid; Lord & Novick, 1968; Revelle, 2023). In the context of psychology, the consensus is that measurements will never be perfectly reliable. If unaccounted for, an imperfect reliability will bias the results of one's analyses: the less reliable, the more bias (e.g., see Buonaccorsi, 2010; Schuurman & Hamaker, 2019). Hence, an important step in validating ESM items (or entire scales) is to evaluate their reliability. Low reliability should be accounted for in the analyses; however, given that statistical corrections are always imperfect, too, it would be best to develop as reliable ESM measures as possible (Yang et al., 2022).

Considering and accounting for reliability in ESM studies comes with notable challenges. Psychology has a strong history of evaluating the reliability of measurements in the context of between-subjects research, but (validation) results from between-subjects research cannot be expected to generalize to within-subjects research (see Hamaker, 2012; Molenaar, 2004; Schuurman, 2023). Moreover, it is so far an unresolved question whether reliability methods for between-subjects research (e.g., Cronbach's alpha coefficients) are suitable in the ESM context without adaptions. One reason why this seems unlikely is that between-subjects reliability methods usually treat within-subjects fluctuations as measurement error. Some reliability methods have, however, specifically been developed for or adapted to the ESM context. For example, reliability methods for multilevel models can be applied to intensive longitudinal data with multiple subjects (e.g., multilevel alpha and omega; Castro-Alvarez et al., 2024; Geldhof et al., 2014; Nezlek, 2017; Revelle & Condon, 2018). One potential downside of these approaches for ESM research is that many of them disregard the dependency (dynamics) among repeated measures. Another downside of many current reliability methods is that they rely on multiple-item measures, whereas ESM research often relies on single-item measures (see former section, Item Content and Phrasing). Recently, two test–retest reliability methods have been introduced for estimating reliabilities for single-item measures (the measurement errors multilevel autoregressive [MEAR] model approach by Schuurman & Hamaker, 2019, and the immediate test–retest method by Dejonckheere et al., 2022). Furthermore, ESM studies are often (to some extent) idiographic, and most of the current methods do not take into account that within-subjects reliability may differ from person to person (Dejonckheere et al., 2022; Lord & Novick, 1968; Schuurman & Hamaker, 2019).[10] Accordingly, person-specific reliability methods (e.g., dynamic extensions of omega that model dependency among the true scores over time, e.g., Castro-Alvarez et al., 2022; H. Song & Ferrer, 2012; or the immediate test–retest method and the MEAR model approach described above, Dejonckheere et al., 2022; Schuurman & Hamaker, 2019) are necessary for ESM studies that consider person-specific results. And finally, reliability may also change over time. Therefore, establishing measurement invariance across both persons and time is another key challenge in evaluating the reliability of ESM items (Adolf et al., 2014; McNeish et al., 2021; Vogelsmeier et al., 2023). Although tackling these challenges is not easy, a worthwhile step forward is to consistently study and report on the behavior of ESM measures (e.g., via the ESM item repository: Kirtley et al., 2024; www.esmitemrepository.com). That way, researchers can learn from each other and progressively develop the required measures and reliability methods for ESM research.

Starting Points for In-Depth Readings

- Dejonckheere, E., Demeyer, F., Geusens, B., Piot, M., Tuerlinckx, F., Verdonck, S., & Mestdagh, M. (2022). Assessing the reliability of single-item momentary affective measurements in experience sampling. *Psychological Assessment, 34*(12), 1138–1154. https://doi.org/10.1037/pas0001178
- Geldhof, G. J., Preacher, K. J., & Zyphur, M. J. (2014). Reliability estimation in a multilevel confirmatory factor analysis framework. *Psychological Methods, 19*(1), 72–91. https://doi.org/10.1037/a0032138
- Schuurman, N. K., & Hamaker, E. L. (2019). Measurement error and person-specific reliability in multilevel autoregressive modeling. *Psychological Methods, 24*(1), 70–91. https://doi.org/10.1037/met0000188

Replicability and Generalizability

The crisis of replicability and generalizability within psychological sciences (Ioannidis, 2012, 2022) extends to ESM research. Replicability refers broadly to finding similar results as an original study, when using the same methods to examine novel data in *the same context*. Generalizability refers to finding similar results as an original study, when using the same methods to examine novel data in *a new context*, such as in a different country, with a different therapy modality, or in a different school (see Moeller, Dietrich, et al., 2023). Because research on the replication crisis has revealed that many psychological findings could not be replicated because of artifacts caused by flawed research methods, much effort has been invested to improve the trustworthiness of psychological science. Nevertheless, many challenges for replicability and generalizability remain, and some of them are unique or particularly aggravated in ESM research compared to cross-sectional, panel, or cohort research. For example, ESM studies have historically often (a) employed small sample sizes (typically because of a relatively higher participant burden in combination with limited research-related resources), (b) used items or measures that have not been validated for ESM research (typically because of the need of momentary and short surveys), and (c) been conducted with novel data-collection tools or data-analyses procedures, resulting in many studies being the first of their kind (Myin-Germeys et al., 2018). Moreover, ESM studies usually set out to examine constructs that are time- and context-specific and are likely to differ between individuals—a level of heterogeneity that cannot be fully solved by methodological innovations (Bryan et al., 2021). And finally, publications based on ESM data often lack important details on the design and conduction of the ESM study that would be necessary to replicate the study, such as participant instructions, incentivization, or data cleaning and preparation (Kaurin et al., 2023). Taken together, a combination of missing study details, design-related shortcomings, the time and context specificity of ESM studies, as well as rapid innovations impede future replication efforts and hamper generalization efforts.

Although the challenge of replicability and generalizability of ESM research should not be underestimated, the good news is that every ESM researcher can contribute to an enhanced replicability in simple but meaningful ways already at the designing (and implementation) stage. First, one can preregister the study protocol (see Kirtley et al., 2021, for an ESM registration template). Various topics that are necessary for such a preregistration have been discussed in this article. Although many topics that are reported in a preregistration are similar for cross-sectional, panel, and cohort data, some topics are specific to ESM studies or at least more pertinent for a registration of an ESM study, including (a) sampling scheme (i.e., fixed, random, or event-related), (b) sampling frequency, (c) survey depth (i.e., number of items and completion time), (d) timescale nature of the items (here specifically, time-variant vs. time-invariant variables), and (e) incentivization and increasing participant compliance (cf. Kirtley et al., 2021). There are two additional topics that were beyond the scope of our article but are relevant for a preregistration and should be considered at the designing stage of the ESM study: (a) the response window, that is, how much time participants get to respond to a prompt before it expires (sometimes also referred to as "time-out specification;" e.g., Deakin et al., 2022), and if applicable, (b) passive monitoring (e.g., wearable data) and its linkage with self-report data (see e.g., Myin-Germeys et al., 2009; Niemeijer et al., 2023). Second, ESM researchers can publish a protocol paper (Kaurin et al., 2023) to facilitate the replicability of the study. Several checklists exist for the reporting of design-related features of ESM studies that can aid both the fine-tuning of design choices and a more complete reporting of details relevant for replication efforts (e.g., Kaurin et al., 2023; Liao et al., 2016). Alternatively, researchers can, of course, when the study is completed, publish an article with a comprehensive and fully replicable methods section (if needed, with additional online resources that can be released on public or protective-access repositories, such as OSF: https://osf.io/). Another valuable way toward improving the replicability and generalizability of ESM studies would be to conduct collaborative data collections across different research labs and contexts (see e.g., Dora,

Piccirillo, Foster, Arbeau, et al., 2023)—a consideration that should ideally be made early at the designing stage of the study, to be able to increase the harmonization of procedures across labs and contexts as much as possible. The analyses of such context-diverse data sets could then be performed by many independent analysts (see e.g., Bastiaansen et al., 2020) in a theoretically replicable way, so that sources of variation can be teased apart and can inform on how the field can better ensure the replicability and generalizability of ESM studies in the future. And last, ESM researchers can contribute to the development of novel solutions for the replicability and generalizability of ESM research by joining efforts of relevant initiatives such as the ManyMoments project (https://bit.ly/ManyMoments; Moeller, 2023; Moeller, Bergmann, et al., 2023; Moeller, Dietrich, et al., 2023; Moeller, Langener, et al., 2023).

Starting Points for In-Depth Readings

- Bastiaansen, J. A., Kunkels, Y. K., Blaauw, F. J., Boker, S. M., Ceulemans, E., Chen, M., Chow, S.-M., de Jonge, P., Emerencia, A. C., Epskamp, S., Fisher, A. J., Hamaker, E. L., Kuppens, P., Lutz, W., Meyer, M. J., Moulder, R., Oravecz, Z., Riese, H., Rubel, J., . . . Bringmann, L. F. (2020). Time to get personal? The impact of researchers choices on the selection of treatment targets using the experience sampling methodology. *Journal of Psychosomatic Research, 137*, Article 110211. https://doi.org/10.1016/j.jpsychores.2020.110211
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Discussion and Concluding Remarks

This article was aimed at providing an overview of collective wisdom on designing and implementing ESM studies, current problems therein, and potential solutions. The reader may have noticed that although a lot can already be learned about how to best design an ESM study (for highlights, see Table 2), there are at least equally many design choices for which researchers currently do not have sufficient knowledge. For an overview

Table 2. Relevant Design Considerations for ESM Studies at a Glance

Study type and research questions
• The research question of interest needs to be both suitable for and best addressed with an ESM study design. • Because of frequent and time-sensitive assessments, ESM studies can be more burdensome for participants than cross-sectional, panel, or cohort studies (especially with *N*-of-1 designs).
Item content and phrasing
• Focus groups or cocreation sessions can inform the operationalization of the construct of interest (e.g., choosing content and number of items). • Shorter ESM questionnaires can sometimes result in better compliance and data quality, but that is not always the case and should not come at the expense of validity. • Piloting ESM questionnaires can prevent confusing or illogical wording as well as the inclusion of redundant items, and may therefore reduce participant burden, increase compliance, and improve data quality. • Training participants before the study can increase understanding of the items and may thereby reduce participant burden, increase compliance, and improve data quality. • The time frame for the item (i.e., fixed, e.g., “in the last day”; [semi]random, e.g., “right now” or “since the last beep”; or event-contingent, e.g., “right now”) should be aligned with the research aim.

(continued)

Table 2. *(continued)*

Choosing and formulating response options
• Likert or VAS response formats are often preferred over dichotomous response formats because not all analytic techniques for ESM can deal with binary responses. • VAS response formats do not seem to have a clear advantage to Likert response formats, which have been suggested to be most user-friendly. • Response formats should be consistent throughout a (construct-specific) set of ESM items. • If response formats are not validated, they should ideally be pilot-tested, participants should be briefed about the usage (via text or video) before the study, and response distributions and shifts should retrospectively be explored. • Additional open-text boxes can be useful to record person-specific answers and to identify reasons for shifts in response distributions or missingness.
Timescale (sampling scheme, sampling frequency, survey length, and study duration)
• The sampling scheme needs to match the research aim and should be suitable for the context at hand. • Sampling scheme and sampling frequency need to align with the (assumed) fluctuation speed of the construct of interest (or its interrelations) because this can affect the quality of the results. • The timing of assessments should be representative and should ideally not fully depend on the time of day or contextual factors (e.g., assessing tiredness only in the late evening). • Recall bias can be addressed through increasing the sampling frequency of fixed or (semi)random assessments, via measurement bursts, or by using event-contingent schemes. • A pilot study can inform about burden and compliance to enable a more beneficial choice of sampling frequency weighed against prompt depth (survey length and completion time) and study duration. • The duration of the study needs to balance sufficient assessments of the construct of interest (e.g., regarding power, variability, and cyclicity) with feasibility considerations (e.g., participant burden and finances).
Change properties and stationarity
• It should be carefully considered whether one is interested in measuring change itself (i.e., means, associations, or other statistical properties that change systematically over time, resulting in nonstationary data). o If yes, the study duration should (be long enough to) cover a period during which participants are expected to change. o If no, external influences that may induce change should be reduced (e.g., by prescreening for potential external influences and/or by keeping the study period brief). • It is useful to keep in mind that to date, only a few analytic approaches exist that can deal well with nonstationary ESM data.
Power and effect sizes
• Simulation studies are available that provide recommendations for sample size; however, such guidance is highly context specific and should be used cautiously. • Power should be calculated based on the smallest effect sizes considered meaningful or based on previous reports of standardized effect sizes. • Pragmatic considerations, such as participant burden and financial resources, are crucially important but should not deter researchers from conducting and reporting power analyses. • Results of power analyses should also be reported when the target effects are insignificant. • When a power analysis seems truly implausible, existing simulation studies may provide guidance, and alternative (pragmatic) considerations and their potential implications should be transparently reported.
Missingness, attrition, and compliance
• Strategies to increase compliance can for example include piloting ESM studies to improve the user experience, financial incentives, training participants in the study procedures, allowing for suspension windows, and motivation enhancement conversations with participants during periods of low compliance. • Planned missingness designs allow researchers to collect a subset of the items per prompt while eventually still being able to obtain relevant estimates for all items. • Suspension windows, open-text boxes, and the like can help to determine the type of missing-data patterns and may help to identify factors that are associated with missingness. • If one is content assuming that data of missed prompts can be statistically compensated for (e.g., via imputation, FIML, Kalman filtering), the use of relevant statistical techniques—which although readily available are not always straightforward—should be planned.

(continued)

Table 2. *(continued)*

Data assessment and administration
• The ESM software for data collection should be chosen carefully to ensure that it suits one's study requirements well (e.g., costs, data protection regulations, compatibility with mobile phones, offline capability, prompt notification options, feedback capabilities, around-the-clock support, time-zone changes, or other study-specific design choices). • Variables should be logically encoded, and a plan for storing, downloading, and sharing the temporally dense data should be set out.
Reliability
• Reliability methods from between-subjects research (e.g., Cronbach's alpha coefficients) are not necessarily suited for the ESM context; therefore, reliability methods should be used that are tailored to ESM data (e.g., multilevel alpha and omega). • Special attention to the choice of reliability methods is required when working with single-item measures (e.g., use the MEAR modeling approach or the immediate test–retest method) or person-specific analyses (e.g., use dynamic extensions of omega) because not all ESM-tailored reliability methods can be applied in these contexts.
Replicability and generalizability
To accelerate replicability, generalizability and meta-science, ESM researchers are encouraged to do the following: • report their methods transparently and in an as replicable way as possible (including sufficient detail on the sample, items, timescale, compliance, software, and analytic procedure), for example, via preregistrations, protocol papers, or thorough methods sections; • submit validated items to the ESM Item Repository (www.esmitemrepository.com); • report standardized effect sizes (to aid subsequent power analyses); • share their code and data (when possible) on trusted repositories; • engage in (cross-lab) collaborations for data collection, (meta-)analysis, and the interpretation of findings.

Note: ESM = experience-sampling method; VAS = visual analogue scale; FIML = full information maximum likelihood; MEAR = measurement errors multilevel autoregressive.

of exactly these issues, we refer the reader to the article "A Momentary Assessment on the Future of ESM Research" (Piccirillo et al., 2024), which proposes a research agenda of empirical work that could be conducted to advance the design of ESM studies.

There are several topics that were beyond the scope of our article. First, we did not report on theory development, including important considerations such as the relevance of context or testing meaningful hypotheses (read more on this in Kuppens et al., 2022). Second, we did not discuss challenges related to measurement reactivity, that is, when behavior, thought pattern, or mood change merely because of completing the ESM assessments (Myin-Germeys et al., 2018; Truijens et al., 2023). Third, we addressed neither analytic methods nor analytic caveats of intensive longitudinal data, such as overnight effects (i.e., gaps in the collected data between an evening and the next morning prompt) or cyclical effects (e.g., week vs. weekend, seasonal, periodic, or event-related effects). And fourth, we did not discuss common data-related biases of intensive longitudinal data that can be dealt with statistically, such as zero inflation if the construct of interest (e.g., an emotion or an event) does only rarely occur (Shao et al., 2023) or initial elevation bias when initial data have systematically higher values than successive data (see Kuppens et al., 2022; Shrout et al., 2018).

Although it is impossible to cover the full scope and complexity of designing an optimal ESM study in one article, we provide an overview of 10 crucial design- and implementation-related topics and a living, web-based resources library (see Note 5) with a more extensive list of relevant literature to facilitate further learning. Therefore, we believe that our work contributes to the quality of future ESM research through bringing areas of caution, essential consideration, and helpful readings to our readers' attention.

Appendix A

Material to reproduce Figure 1

Step 1a: search query for the experience-sampling method (ESM) publication records of Web of Science (WoS). "ESM" OR "experience sampling method" OR "EMA" OR "ecological momentary assessment" OR "IL" OR "intensive longitudinal" OR "time-series" OR "timeseries" OR "time series" OR "ambulatory assessment" OR "real time data capture" OR real-time data capture" OR "daily life methods" OR daily-life methods" (Title) AND Article OR Book OR Book Chapter OR Data Paper OR Discussion OR Early Access OR Editorial Material OR Letter OR Meeting Summary OR Proceedings Paper OR Review OR Software Review (Document Type) AND English OR Dutch

OR German (Language) AND Psychology (Web of Science Categories)

Step 1b (alternative for 1a): search link for the ESM publication records of WoS. https://www.webofscience.com/wos/woscc/summary/3cd90184-f89c-4cfc-8c56-d589a7bded1c-ab980ba7/relevance/1

Step 2: follow these steps on WoS to retrieve the data set for ESM publication records. (click) Analyze results → (select) Publication years → (select and insert) Showing the "maximum" number out of "all" entries → (click) Select all → (choose) All data rows → (click) Download table

Step 3a: search query for all psychological publication records of WoS. Article OR Book OR Book Chapter OR Data Paper OR Discussion OR Early Access OR Editorial Material OR Letter OR Meeting Summary OR Proceedings Paper OR Review OR Software Review (Document Type) AND English OR Dutch OR German (Language) AND Psychology (Web of Science Categories)

Step 3b (alternative for 3a): search link for all psychological publication records of WoS. https://www.webofscience.com/wos/woscc/summary/8d13299a-0d53-43d2-8c69-657cc4acf6aa-ab9903e2/relevance/1

Step 4: follow these steps on WoS to retrieve the data set for all psychological publication records. (click) Analyze results → (select) Publication years → (select and insert) Showing the "maximum" number out of "all" entries → (click) Select all → (choose) All data rows → (click) Download table

Step 5: code for the reproduction of the figures, to be used in R studio. Note that you need to insert file directories and file names where indicated:

****

```
#loading libraries
library(dplyr)
library(tidyr)
library(ggplot2)

#load in ESM publication records data, and convert the data into usable format
ESMyears <- read.delim("specify the file directroy & name", sep = "")
ESMyears$PubNumbers <- ESMyears$Years
ESMyears$PubYears <- ESMyears$Publication
ESMyears_ordered <- ESMyears[order(ESMyears$PubYears),]
ESMyears_ordered$PubYears <- as.numeric(ESMyears_ordered$PubYears)
ESMyears_ordered$type <- "ESM"

#plot the ESM publication records per year
ESM_plot <- ggplot(
  data = ESMyears_ordered,
  mapping = aes(x = PubYears, y = PubNumbers)
) +
  geom_col(
    size = 0.5,
    fill = "grey"
  ) +
  labs(
    x = "\npublication year",
    y = "number of ESM publications\n"
  ) +
  scale_x_continuous(
    breaks = seq(from = 1960, to = 2020, by = 5)
  ) +
  scale_y_continuous(
    expand = expansion(mult = 0.025)
  ) +
  theme_minimal() +
  theme(
    panel.grid.minor = element_blank(),
    panel.grid.major = element_blank(),
    axis.title = element_text(size = 14, family = "serif"),
    axis.text.x = element_text(size = 12, angle = 45, hjust = 1, family = "serif"),
    axis.text.y = element_text(size = 12, family = "serif"),
    axis.line = element_line(colour = "black", size = 0.5),
    axis.ticks = element_line(colour = "black", size = 0.5)
  )
ESM_plot
ggsave(plot = ESM_plot,
       filename = "choose a file directory & name for the plot",
       device = cairo_pdf,
       width = 8, height = 4, units = "in")
ggsave(plot = ESM_plot,
       filename = "choose a file directory & name for the plot",
       device = png, type = "cairo", dpi = 300,
       width = 8, height = 4, units = "in")

#load in all psychology publication records data, and convert the data into usable format
ESMyears_all <- read.delim("specify the file directroy & name ", sep = "")
ESMyears_all$PubNumbers <- ESMyears_all$Years
ESMyears_all$PubYears <- ESMyears_all$Publication
ESMyears_all_ordered <- ESMyears_all[order(ESMyears_all$PubYears),]
ESMyears_all_ordered$PubYears <- as.numeric(ESMyears_all_ordered$PubYears)
ESMyears_all_ordered <- ESMyears_all_ordered[(ESMyears_all_ordered$PubYears)>=1958,] # this line of codes removes all data before 1958, to cover the same time frame as available for the ESM data
ESMyears_all_ordered$type <- "ALL"
```

```
#bind the two datasets together
ESM_years_data <- bind_rows(
   ESMyears_all_ordered %>% select(PubYears,
    PubNumbers, type),
   ESMyears_ordered %>% select(PubYears, PubNumbers,
    type)
)
#plot the number of ESM publication records (black) in
    comparison to all psychological publication records (grey),
    per year
comparison_plot <- ggplot(
   data = ESM_years_data %>%
      mutate_at("type", factor, levels = c("ESM", "ALL"),
          ordered = TRUE),
   mapping = aes(x = PubYears, y = PubNumbers, fill = type)
) +
   geom_col(
      size = 0.5
   ) +
   labs(
      x = "\npublication year",
      y = "number of publications\n"
   ) +
   scale_x_continuous(
      breaks = seq(from = 1960, to = 2020, by = 5)
   ) +
   scale_y_continuous(
      expand = expansion(mult = 0.025)
   ) +
   scale_fill_manual(
      values = c("ALL" = "lightgrey", "ESM" = "black")
   ) +
   theme_minimal() +
   theme(
      panel.grid.minor = element_blank(),
      panel.grid.major = element_blank(),
      axis.title = element_text(size = 14, family = "serif"),
      axis.text.x = element_text(size = 12, angle = 45, hjust =
          1, family = "serif"),
      axis.text.y = element_text(size = 12, family = "serif"),
      axis.line = element_line(colour = "black", size = 0.5),
      axis.ticks = element_line(colour = "black", size = 0.5)
   )
comparison_plot
ggsave(plot = comparison_plot,
       filename = "choose file directory & name",
       device = cairo_pdf,
       width = 8, height = 4, units = "in")
ggsave(plot = comparison_plot,
       filename = "choose file directory & name",
       device = png, type = "cairo", dpi = 300,
       width = 8, height = 4, units = "in")

#convert the merged datasets into a wide format
ESM_years_data_wide <- pivot_wider(ESM_years_data,
    names_from = "type", values_from = "PubNumbers",
    id_cols = "PubYears")

#plot the relative number of ESM publication records, i.e.,
    ESM publication records divided by all psychological
    publication record
ESM_years_data_wide$PubNumbers_ratio <- ESM_years_
    data_wide$ESM / ESM_years_data_wide$ALL
ratio_plot <- ggplot(
   data = ESM_years_data_wide,
   mapping = aes(x = PubYears, y = PubNumbers_ratio)
) +
   geom_col(
      size = 0.5,
      fill = "grey"
   ) +
   labs(
      x = "\npublication year",
      y = "relative number of ESM publications"
   ) +
   scale_x_continuous(
      breaks = seq(from = 1960, to = 2020, by = 5)
   ) +
   scale_y_continuous(
      expand = expansion(mult = 0.025)
   ) +
   theme_minimal() +
   theme(
      panel.grid.minor = element_blank(),
      panel.grid.major = element_blank(),
      axis.title = element_text(size = 14, family = "serif"),
      axis.text.x = element_text(size = 12, angle = 45, hjust =
    1, family = "serif"),
      axis.text.y = element_text(size = 12, family = "serif"),
      axis.line = element_line(colour = "black", size = 0.5),
      axis.ticks = element_line(colour = "black", size = 0.5)
   )
ratio_plot
ggsave(plot = ratio_plot,
       filename = "choose file directory & name",
       device = cairo_pdf,
       width = 8, height = 4, units = "in")
ggsave(plot = ratio_plot,
       filename = "choose file directory & name",
       device = png, type = "cairo", dpi = 300,
       width = 8, height = 4, units = "in")

#calculate sum of all ESM publications since 1958
372/sum(ESMyears_ordered$PubNumbers) # = 0.2931442

**********************************************************************
```

Appendix B

Figures showing the absolute numbers of experience-sampling method (ESM) publications and of all psychological publications

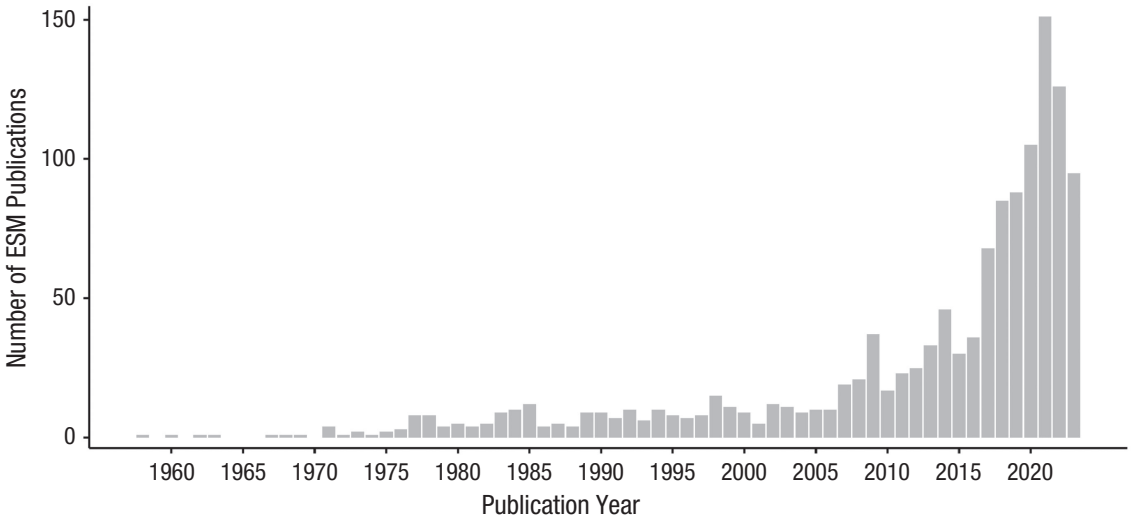

Fig. B1. The absolute number of experience-sampling-method (ESM) publication records on Web of Science (WoS) per year since 1958.

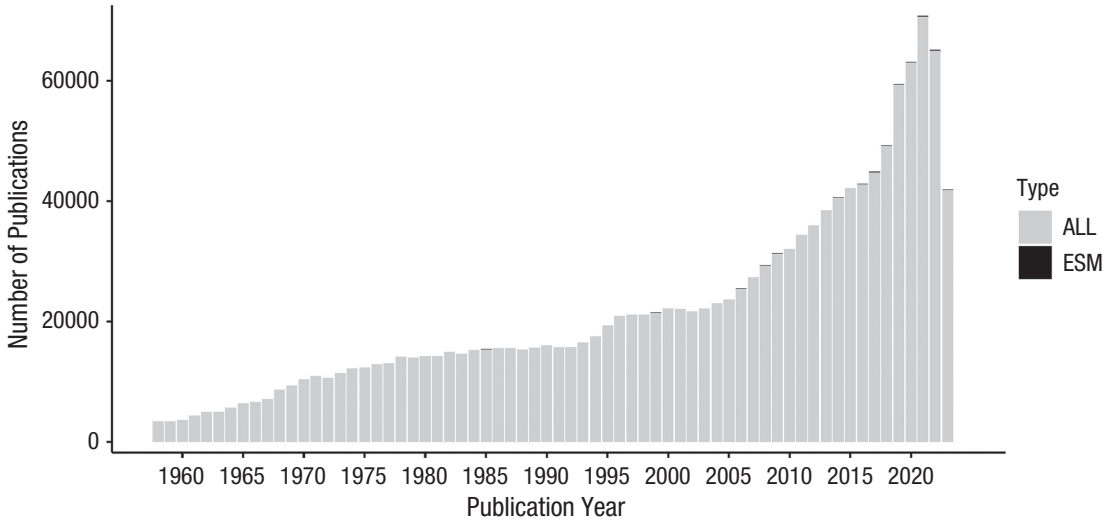

Fig. B2. The absolute number of both experience-sampling method (ESM) and all psychological publication records on Web of Science (WoS) per year since 1958. Black = ESM publication records; gray = all psychological publication records.

Transparency

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J. Fritz and M. L. Piccirillo share first authorship.

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Marilyn L. Piccirillo: Conceptualization; Investigation; Writing – original draft; Writing – review & editing.

Zachary D. Cohen: Investigation; Writing – review & editing.

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Supplemental Material

Additional supporting information can be found at http://journals.sagepub.com/doi/suppl/10.1177/25152459241267912

Notes

1. Because the terms "panel study" and (single or accelerated) "cohort study" are inconsistently used and defined in the literature, we note that we specifically refer to longitudinal designs that assess a group of participants at several (mostly two to 10) time points (as opposed to assessing different groups of people at several time points, such as in repeated cross-sectional designs; Lugtig & Smith, 2019).
2. In this article, we focus predominantly on subjective self-report assessments, but many of our recommendations will also apply to objective biophysiological (e.g., heart rate) or mobile-sensing (e.g., GPS tracking) data. For recommendations that are specific to passive sensing, we refer the reader elsewhere (e.g., Harari et al., 2017; Jacobson et al., 2020; Mehl et al., 2023; Niemeijer et al., 2023).
3. The public panel discussion, which we organized in response to the growing attention to ESM research, included seven speakers with long-standing experience in ESM research. The corresponding slides and videos are provided on our OSF page: https://osf.io/4a723/. The panel discussion was split in two events.
4. https://docs.google.com/document/d/1fAYQqIc4bUpO-FqjSvoGABEQfeZfMjib2I5UM9xgNRc/edit.
5. Note that under specific circumstances, within-subjects effects can also be based on assessments from just one time point. One way to do this is through assessing within-subjects focused questions (Eid et al., 1999). For example, instead of asking participants about their current level of anxiety, one can ask about their current level of anxiety compared with their usual level of anxiety, thereby directly assessing within-subjects effects. For further examples for estimating within-subjects effects based on data from just one time point, see Schuurman (2023).
6. Moreover, it is possible to apply hybrid within- and between-subjects approaches that first examine separate idiographic models for each person (i.e., within-subjects) and then compare the idiographic results across persons (i.e., between-persons) to identify patterns that generalize across people versus those that do not (e.g., see the group iterative multiple model estimation approach by Gates et al., 2023).
7. Note that there are several other statistical properties (e.g., the rank order) that can also change systematically, but this goes beyond the scope of what can be covered in this article.
8. For completeness, we want to stress that power can usually also be increased through reducing the complexity of the chosen analytic method or the number of variables included therein.
9. https://docs.google.com/spreadsheets/d/18R9x9Qbl9tADJGpJBJID_T9EWZeQ_4W3OFdn3iKRU7U/edit#gid=204277638.
10. Some do take into account that within-subjects reliability differs from person to person, but assume that error variances are equal from subject to subject and that only the true scores vary (e.g., Neubauer, Voelkle, et al., 2020).

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